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POSTER ABSTRACTS

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P300

# IMPROVING QUALITY AND SAFETY IN MATERNITY SERVICES: CAN WE IMPROVE THE PREVENTION, DETECTION AND MANAGEMENT OF CONGENITAL ABNORMALITIES IN PREGNANCY?

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**Background:** Congenital abnormalities account for one third of perinatal deaths. We aimed to determine whether there were 'quality gaps' in the provision of care, including prevention, diagnosis and treatment, which if closed may reduce the numbers of perinatal deaths.

**Method:** Perinatal deaths from congenital cardiovascular, central nervous system or chromosomal abnormality in 2010 were identified. Audit tools were developed to gather information from the clinical notes on the care and the quality of ultrasound provided.

**Results:** There were 137 perinatal deaths due to a congenital cardiovascular, central nervous system, or chromosomal abnormality. First contact with a health professional during pregnancy was predominantly with a primary care provider. First contact occurred within 14 weeks in 85% of pregnancies however there was often a delay before booking with the lead maternity carer. Twenty perinatal deaths from neural tube defects could potentially have been prevented through the use of pre-conceptual folate. Prenatal screening was offered to 75% of the women who presented prior to the cut off for screening. Of those offered screening, 84% undertook at least one of these tests. There did not appear to be significant delays in the pregnancy pathway. Review of ultrasound images found some abnormalities could have been detected earlier.

**Conclusions:** The importance of pre-conceptual folate needs emphasis. Failure to offer screening early or delay in booking was responsible for delay in diagnosing screen detectable abnormalities. We should evaluate the false negative rates and the cost benefit of using multiple algorithms to detect a wider variety of abnormalities.

P301

# CONGENITAL HYPOTHYROIDISM WITH SEIZURES: A CASE REPORT

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**Introduction:** Congenital hypothyroidism (CH) is defined as thyroid hormone deficiency, present at birth. It is seen in 1:4000 births, caused by an anatomical defect, known as thyroid dysgenesis, inborn errors of metabolism, genetic mutations or iodine deficiency. The former accounts for 80–85% of cases. If CH remains untreated, severe neurological impairment develops. Clinical manifestations, which are varied and generally standard for CH, are often subtle at birth, due to placental transfer of T4. Newborn screening programs have improved outcomes, through early diagnosis and treatment. Raised thyroid stimulating hormone (TSH) levels and low levels of T4, confirm this disorder. We describe the case of a baby with CH, who presented with neonatal seizures; a phenomenon reported only twice before in the literature.

**Case:** Our term male infant was admitted with a history of abdominal distension. Investigations revealed no signs of obstruction or sepsis. Mother was treated with thyroxine for a slightly elevated TSH. At day 4, our baby had apnoeas and bradycardias with seizure activity, confirmed on BrainZ monitoring. These passed without treatment. Breastfeeding ability was slow initially, as was weight gain. Neonatal screening results revealed TSH > 500 mU/L and T4 < 3.2 pmol/L, consistent with CH. Thyroxine was commenced immediately. MRI and EEG were normal but he referred bilaterally on the hearing screen.

**Conclusion:** Our case highlights the need to eliminate CH, as a cause of seizures, although neonatal screening will confirm the diagnosis. If thyroid function tests are done as part of the investigative process, treatment can be commenced even earlier.

P302

# ANTI-MÜLLERIAN HORMONE: A PREDICTOR OF ADVERSE PREGNANCY OUTCOME?

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**Background:** Anti-Müllerian hormone (AMH) is used to measure ovarian reserve and to predict responsiveness to assisted reproductive technologies. AMH decreases with age and is lower in women with infertility. We investigated the association between AMH and pregnancy outcomes ≥20 weeks gestation.

**Methods:** A retrospective cohort study of women who had AMH tested at 11–13 weeks of pregnancy at the Royal Hospital for Women, Sydney between October 2009 and April 2012. Maternal demographic and pregnancy outcome data was extracted from the hospital database (Obstetrix) and patient files. Pregnancy outcomes included pregnancy hypertension (pre-eclampsia/gestational hypertension), gestational diabetes, caesarean section, small for gestational age neonates, stillbirth, preterm birth, and neonatal intensive care admission. Multivariate analyses were undertaken to determine associations between AMH and pregnancy outcomes. As there is no agreed definition of low AMH, AMH < 10<sup>th</sup> centile was used to compare outcomes.

**Results:** 331 women had AMH tested and had known pregnancy outcomes of singleton fetuses ≥20 weeks gestation. The median AMH level was 9.7 ng/mL (Inter-quartile range (IQR) < 3.9–17.3). AMH levels were significantly lower in those women who developed pregnancy hypertension than those who did not (5.1 ng/mL, IQR 1.5–13.2, p = 0.03). After adjusting for age, parity and body mass index, women with an AMH < 10<sup>th</sup> centile (1.5 ng/mL), had a 3.3 fold (95% CI 1.2–8.7) increased risk of pregnancy hypertension. There was no significant association of low AMH levels with the other adverse pregnancy outcomes.

**Conclusions:** Low AMH (<10<sup>th</sup> centile) in pregnancy is associated with pregnancy hypertension ≥20 weeks gestation.

P303

# SILDENAFIL WEANING POST-DISCHARGE IN INFANTS WITH CONGENITAL DIAPHRAGMATIC HERNIA

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**Background:** Sildenafil is frequently used to treat pulmonary hypertension (PAH) in infants with congenital diaphragmatic hernia (CDH), and is often continued after hospital discharge. However there is limited data to guide sildenafil dosing and weaning rate in CDH. This is of particular concern in light of a recent dose-ranging study which observed increased mortality and side effects in children with non-CDH PAH receiving high-dose sildenafil over 2 year follow-up<sup>1</sup>.

**Method:** We conducted a retrospective cohort study of sildenafil usage, dosing and weaning in infants with CDH managed in our institution between September 2005 and September 2012. Participants were identified from our departmental database. Time to wean sildenafil dose to <1.5 mg/kg/day (considered lower risk for side-effects and mortality in recent study<sup>1</sup>) was determined.

**Results:** Of 122 infants with CDH, 19 (17%) were receiving sildenafil at discharge. Follow-up data on sildenafil doses was available for 17 infants (Table 1). Median dose at discharge was 8 (range 1–19) mg/kg/day. Mean rate of weaning was 0.1 (0.01–0.5) mg/kg/week. In infants who ceased therapy during the study period, median duration was 343 (105–671) days. At 1 year of age, 29% were still receiving a dose of sildenafil >1.5 mg/kg/day. One infant died due to severe PAH.

**Conclusions:** Sildenafil therapy at discharge is common in severe CDH. Variation in dosing and weaning rates highlight the need for standardised assessment and treatment of PAH after discharge, in order to optimise benefits and minimise potential adverse effects of sildenafil.

#### Reference

1. Barst *et al.* *Circulation* 2012;125(2):324–34.

**Table 1.** Demographics and sildenafil treatment data for CDH infants discharged on sildenafil

| N = 17 (9 female, 8 male)                                                     | Median | Range    |
|-------------------------------------------------------------------------------|--------|----------|
| Weight at discharge (Kg)                                                      | 4.6    | 2.9–5.8  |
| Length of NICU stay (days)                                                    | 71     | 32–132   |
| Sildenafil dose at discharge (mg/kg/d)                                        | 8      | 1–19     |
| Time to stopping sildenafil, (days), (N = 10)                                 | 257    | 105–671  |
| Time to halve sildenafil dose after discharge (days) (N = 15)                 | 209    | 88–577   |
| Time to achieve <1.5 mg/kg/dose (days) (N = 10)                               | 226    | 0–562    |
| Duration of sildenafil treatment (days) (N = 17, of which 10 stopped therapy) | 343    | 105–671  |
| Rate of reduction (mg/kg/week)                                                | 0.1    | 0.01–0.5 |

#### P304

##### THE HIGH PERI-PARTUM AND POST-NATAL MORBIDITY OF GASTROSCHISIS: A DIAGNOSIS TO BE RESPECTED

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**Background:** Diagnosis of gastroschisis is increasing. We reviewed its presentation with postnatal management and outcome over eleven years. **Method:** Data were collected from databases of two tertiary perinatal hospitals (RWH, MHW), a neonatal surgical unit (RCH) and NETS from January 2000–December 2010. Data included maternal, antenatal and neonatal characteristics, delivery management, and post-operative course until discharge. Descriptive statistics are reported as mean (median, range). **Results:** Of 93 infants identified: 53 (57%) were male. Data were incomplete for some infants. Maternal age at delivery was 23.5 (23, 16–40) years. GA at diagnosis was 19 (18.5, 12–39) weeks. GA at delivery was 34 (36, 16–40) weeks. 95% delivered in a tertiary perinatal centre. 42 (54%) delivered by Caesarean section. BW was 2304 g (2370, 658–3670). 41 (44%) infants were small for gestational age. Of 10 (10.7%) infants with other anomalies, one had Trisomy 13. 42/87 (48.3%) infants required resuscitation, with 32/87 (36.8%) requiring respiratory support; 21/87 (24%) were intubated. 65/77 (84%) required a mean of 26 (20, 0–110) ml/kg fluid resuscitation. Primary closure occurred in 66 (75%) patients, with staged closure in 22 (25%) patients, for whom time to closure was 5 (0, 0–93) days. Proven sepsis occurred in 21/93 (22.6%) patients. Another 14 (15%) had suspected/clinical sepsis. Age at full feeds was 30.2 (22, 11–123) days. Discharge occurred at 51 (32.5, 12–263) days. Survival was 97%. **Conclusions:** Outcomes are similar to those previously reported. Despite the high survival rate, the number requiring caesarian section, significant resuscitation at birth, and incidence of post-natal sepsis serves as a reminder that gastroschisis is a high risk birth defect not only peri-partum, but also in the postnatal period.

#### P305

##### THE GENETICS OF G6PD DEFICIENCY AND UGT1A1 MUTATION ASSOCIATED WITH JAUNDICE IN THE EARLY NEONATAL PERIOD

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**Background:** G6PD deficiency is a common X-linked inherited enzymopathy. Mutations in the G6PD and the UGT1A1 enzymes are important causes of neonatal jaundice. However, the genetics of these mutations and the clinical correlates have not been widely studied. Aim: To relate G6PD and UGT1A1 mutations with jaundice severity in G6PD deficient neonates. **Method:** Term healthy infants born in the UKM Medical Centre, diagnosed as G6PD deficient by cord blood fluorescent spot test (FST), and hospitalised for 5 days for monitoring of jaundice were enrolled. Their gender, ethnicity, peak bilirubin, reticulocyte, photo-light therapy, were related to the most common G6PD (n = 10) and UGT1A1 (n = 2) mutations.

**Results:** Over an 18-month period, n = 154 infants were enrolled; 125 (80%) males and 29 (20%) females. Almost three-quarters (n = 111) were Malays; n = 43 were Chinese. The mean peak bilirubin was 234 ± 66 micromol/L. One out of every two infants (53%) received phototherapy; none had exchange transfusion; mean reticulocyte was normal, and similar between photo-treated and non-treated infants (3.56% vs 3.95%, p = 0.54). The G6PD mutations, most commonly being c.871G>A (n = 50), c.487G>A (n = 17) and c.1388G>A (n = 13) were associated with similar peak bilirubin levels even in combined carriage with either the c.211G>A (n = 15) or c.-3279T>G (n = 90) UGT1A1 mutation. Only G6PD deficient males with homozygous c.-3279T>G had almost significantly lower peak bilirubin (231 vs. 251 micromol/L, p = 0.057).

**Conclusions:** G6PD mutation itself or in combination with UGT1A1 mutation did not predict jaundice severity, although early preventive phototherapy could have masked true genetic expression. In G6PD deficient males, being c.-3279T>G UGT1A1 homozygous may be protective against significant hyperbilirubinaemia.

#### P306

##### BEFORE AND AFTER PHOTO PRESENTATION – CONGENITAL CHYLOTHORAX.

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There are conditions when the appearance of the newborn baby can be scary to young parents. Congenital chylothorax is one of these conditions, being a rare cause of hydrops fetalis. We describe a baby where photos were taken soon after birth (a, grossly hydropic) and nearly 2 years later (happy little girl). Photos will be presented in the poster.

**Case:** A 25 year old, primiparous lady who is 32+3 weeks pregnant, presented with lethargy, shortness of breath and palpitation. Ultrasound showed severe hydrops fetalis with absent end diastolic flow in the umbilical artery.

A baby girl, delivered by caesarean section was born unwell, grossly hydropic with Apgar scores of 3 at 1<sup>st</sup>, 5<sup>th</sup> and 10 minutes. She was intubated at birth and 407 mls of serous fluid were drained from chest drains in the first day. Her initial ventilation improved but ventilation worsened due to increased pleural effusion and turbidity after expressed breast milk via gastric tube was started on day 4. Excessive lymphocytes in the effusion and lymphopenia confirmed a chylothorax. Her condition improved with monogen feeds and a 29 day course of Octreotide and she was extubated on day 27. She is currently 22 months old and her growth and development are age appropriate. The parents gave written informed consent for photos to be published.

**Conclusions:** Photos documenting the progress of conditions like hydrops could be a useful aid during counselling for parents and staff in the

antenatal and neonatal journey. We are building up a library of such photos and welcome any contribution from colleagues.

P307

#### SURVIVAL AGAINST THE ODDS: IS CYSTIC FIBROSIS IN EXTREME PREMATURITY COMPATIBLE WITH SURVIVAL?

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**Background:** Cystic fibrosis (CF) occurring in extreme prematurity is rare with the few published cases describing universal lethality<sup>1</sup>. Babies with CF may have a threefold higher risk of being born prematurely<sup>2</sup>.

**Methods:** We describe our experience of three preterm infants with CF and their management in the neonatal intensive care setting.

**Results:** 24 week female infant 510 g (DCDA twin 1) born by caesarean section to 23 years primigravida, due to Preterm labour and APH, ΔF508 on newborn screening. Received Dexamethasone, physiotherapy, Dornase-alfa, inhaled Tobramycin, hypertonic saline and prophylactic Flucloxacillin. PERT added to enteral nutrition. Early lung disease with multiple episodes of collapse consolidation. Respiratory support for 8 months, first year of life spent in hospital.

24 Week female infant 680 g (DCDA twin 2), sibling of above. Severe cystic lung disease and bilateral grade 4 IVH, subsequent NEC. Ductus required ligation on day 26. Succumbed to ventilator associated pneumonia resistant to all therapy on D40.

29 week male infant 1420 g born by caesarean section to 23 years G6P0 due to abnormal CTG. Discoloured abdomen at birth with worsening abdominal distension. Laparotomy; ischaemic bowel with meconium ileus. Newborn Screening confirmed homozygous ΔF508 on day 3. Prolonged parenteral nutrition with eventual PERT, received UDCA, amoxycillin/clavulanic acid prophylaxis, multiple antibiotics for eradication of MRSA. **Conclusion:** Care for Cystic fibrosis is complex especially when coupled with prematurity. Survival beyond 3 months for an extreme preterm infant with CF has not previously been described.

#### References

1. Lu KD *et al.* Journal of Perinatology 2011;31:504–506.
2. Festini F *et al.* The Journal of Pediatrics 2005;4:316–319.

P308

#### STILL GOT THE BLUES: A CASE OF CONGENITAL METHAEMOGLOBINAEMIA

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**Background:** Methaemoglobin occurs when Fe is oxidized from Fe<sup>2+</sup> to Fe<sup>3+</sup>, the ferric form (Fe<sup>3+</sup>) cannot transport oxygen. Methaemoglobin is chocolate brown in colour and gives a cyanosed hue when present in high levels. Normally NADH-Cytochrome B5 reductase will reduce methaemoglobin and keep levels near 1%. Congenital methaemoglobinaemia is an extremely rare cause of cyanosis in the newborn.

**Methods:** We describe a case of presumed recessive congenital methaemoglobinaemia (RCM) who presented with persistent cyanosis.

**Results:** A non-dysmorphic female infant was born at 39 + 4 weeks gestation by normal vaginal delivery following an uncomplicated pregnancy. Birth weight 2610 g (<10<sup>th</sup> percentile) with Apgars of 8, 9. No maternal medications, smoking, alcohol or drugs. Central cyanosis at birth with normal cardiorespiratory examination, saturations in mid 90's with no improvement in oxygen. Co-oximetry on day 3: total haemoglobin 210, oxyhaemoglobin 77%, Methaemoglobin 22.5%, G6PD normal. Skin biopsy for enzymology and mutation analysis is pending.

**Conclusion:** Congenital methaemoglobinaemia results from deficiency of Cytochrome B5 reductase, either isolated to erythrocytes (type I-benign)

or involving all cells (type II). Type II carries a far worse prognosis with neurodevelopmental sequelae. Therapies for methaemoglobinaemia (high dose ascorbic acid or methylene blue) will provide cosmetic benefit but do not alter prognosis for type II. Although rare, congenital methaemoglobin needs to be considered as a cause of cyanosis. Pulse oximetry is unreliable in this setting and co-oximetry will help with diagnosis. Methylene blue can precipitate haemolysis in G6PD deficient patients. Diagnosing the type of congenital methaemoglobinaemia has significant prognostic implications.

P309

#### DUAL CHAMBER PACEMAKER IMPLANTATION IN AN EXTREMELY LOW BIRTH WEIGHT INFANT WITH LONG-QT SYNDROME AND A PATHOLOGICAL C.1750G>C MUTATION FOR THE KCNH2 GENE

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**Background:** Congenital long-QT syndrome (LQTS) is a heterogeneous genetic channelopathy with variable penetrance which is associated with increased propensity for polymorphic ventricular tachyarrhythmias and sudden cardiac death. Neonatal presentation of the LQTS is an uncommon diagnosis that carries a high mortality risk.

Different therapeutic options exist for affected patients, ranging from pharmacologic management to more aggressive therapies including left cardiac sympathetic neural denervation, implantable pulse generators and cardiac defibrillators. Use of dual chamber pacemakers is generally reserved for larger infants due to the size of the generators and challenges of lead placement in premature infants. To date all reported patients fitted with dual chamber pacemakers have been 2.0 kg or above.

**Method:** Retrospective case report based on clinical records, genetic testing and cardiological investigations in a preterm twin with LQTS.

**Results:** This case study presents an extremely preterm infant born at 26<sup>+</sup>5 weeks weighing 796 grams who had recurrent episodes of ventricular tachycardia and second degree AV block. LQTS was diagnosed and a pathological c.1750G>C mutation of the KCNH2 gene identified. A Medtronic ADAPTA<sup>®</sup> ADDR51 dual chamber rate responsive pacemaker was successfully implanted on day 41 with the infant weighing 1100 grams. The pacemaker was set in DDD mode. Her twin sister also had LQTS managed with propranolol.

**Conclusion:** Implantation of a dual-chamber epicardial pacemaker in an extremely low birth weight premature neonate with a structurally normal heart is technically challenging but feasible. However, long term outcome, quality of life and cost effectiveness in this cohort remain to be established.

P310

#### INTELLECTUAL, ACADEMIC AND SOCIAL-EMOTIONAL OUTCOME IN SCHOOL-AGED CHILDREN WITH OESOPHAGEAL ATRESIA

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**Background:** The limited literature on long-term outcome in children born with oesophageal atresia (OA) has focused on mental health with higher than expected rates of internalizing symptomatology in parents and children. Only one known study has reported intellectual outcome in a group of school-aged children with OA. It found significantly reduced FSIQ and increased emotional and behavioural problems.



**Method:** Children born with OA in Victoria in 2003 and 2006 were invited to participate in a neurodevelopmental clinic. Of 36 eligible participants, 29 underwent assessment using the WPPSI-III, the WISC-IV, WRAT4 and BASC-2. Two children died during the neonatal period, and five were lost to follow-up.

**Results:** The mean age of the participants was  $6.7 \pm 1.6$  years, and the median length of hospital stay was 21 (IQR 12–35) days. Six children were identified as having VACTERL association and two had long-gap OA without a fistula. One had Autism Spectrum Disorder and one was syndromic. The mean FSIQ was  $92.8 \pm 18.9$  points. Thirty percent of the sample demonstrated impaired FSIQ ( $<85$  IQ points); significantly more than expected in the typically developing population. Similarly, academic skills were impaired in 18–41% of the sample. Anxiety was elevated in 34% on parental questionnaire. Two children could not be fully assessed due to selective mutism or significant behavioural problems.

**Conclusions:** A significant number of children with OA demonstrated intellectual and academic impairments as well as internalizing symptomatology. This finding may inform future clinical care, but further research into this high-risk group is vital.

### P311

#### NEONATAL HYPERTENSION – A LONG TERM FOLLOW-UP STUDY

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**Background:** Neonatal hypertension occurs in up to 3% of neonates, more commonly in those admitted to neonatal intensive care. The cause in the majority of cases is renal in origin. The aims of this study were to review renal function and renal volumes in children who had a history of neonatal hypertension.

**Methods:** Children with a history of neonatal hypertension from January 2001 to December 2008 were included in the study during 2011. Blood pressure, weight, height and BMI were recorded. Renal ultrasound with 3D volume, urine for electrolytes, albumin,  $\beta_2$  microglobulin, and blood for electrolytes, urea, creatinine, calcium, phosphate, renin and aldosterone were collected depending on parental consent.

**Results:** Of the 41 neonates with neonatal hypertension, 11 (27%) were included in the study (6 died; 24 moved interstate or declined involvement). Two patients (18%) had hypertension on follow-up visits. One child was still on antihypertensive medication and one was found to be hypertensive on review. This child had small volume kidneys and albuminuria. Three out of 9 renal volume measurements were low (33%) and 2 out of 11 had renal scarring (18%). The 6 available renin/aldosterone results were normal.

**Conclusions:** Neonatal hypertension appears to have implications for long term renal function and BP in later childhood. Ongoing surveillance of BP and renal function of children with a history of neonatal hypertension is recommended.

### P312

#### ECHOCARDIOGRAPHIC ASSESSMENT OF LEFT VENTRICULAR OUTPUT USING THREE SITES FOR OUTFLOW TRACT MEASUREMENTS IN PRETERM INFANTS ON RESPIRATORY SUPPORT

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**Introduction:** Echocardiographic cardiac output (CO) measurements provide useful clinical information about cardiac function in neonates. It is

important to standardise the measurement of LVO in preterm infants. The site of assessment of the left ventricular outflow tract diameter (LVD) is not consistent in the literature. Left ventricular output (LVO) and right ventricular output (RVO) should be equal when no shunts are present. We calculated LVO using three different 'standard' LVD measurements and compared each to RVO.

**Methods:** Written consent was obtained from parents of preterm infants treated with nasal continuous positive airway pressure (nCPAP). Infants with any significant extra- and intracardiac shunts were excluded. LVO was calculated with LVD measured at the aortic valve (AV), the aortic root (AR) and at the sinotubular junction (STJ). Limits of agreement and mean differences between RVO and LVO were assessed using Bland-Altman plots.

**Results:** 52 infants were included; mean (SD) postmenstrual age of 31 (28–34) weeks, and weight of 1.35 (0.68–2) kg. The mean difference between RVO and LVO was largest for LVD measured at the AV and AR, and smallest when measured at the STJ.

|                                | LVO based<br>on AV     | LVO based<br>on AR     | LVO based<br>on STJ     |
|--------------------------------|------------------------|------------------------|-------------------------|
| mean difference<br>(RVO – LVO) | 106 ml/kg/min          | –106 ml/kg/min         | 9 ml/kg/min             |
| limits of<br>agreement         | –31, 243 ml/<br>kg/min | –282, 69 ml/<br>kg/min | –102, 120 ml/<br>kg/min |

**Conclusion:** Our study demonstrates that LVO is underestimated when measuring the LV outflow tract at the AV and overestimated when measured at the AR. We found that measurement at the STJ is more precise and accurate than measurement at the aortic valve or the aortic root.

### P313

#### SELECTIVE HEAD COOLING IN NEONATAL ENCEPHALOPATHY – STUDY OF IMMEDIATE COMPLICATIONS AND OUTCOME IN NICU DUBAI HOSPITAL, UAE

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**Background:** Cooling (Selective head or total body) has been recognized as a standard modality to treat moderate to severe neonatal hypoxic ischemic encephalopathy. Dubai Hospital NICU has been using selective head cooling with Natus cool cap machine since last 2 years where the rectal temperature is maintained around 34.5 degree centigrade by applying cool caps on the scalp with a temperature fluctuating around 9–12 degree centigrade.

**Objective:** To look at the immediate complications of selective hypothermia both local and systemic during the process of cooling for 72 hours. Immediate outcome at the time of discharge was correlated with the MRI brain findings and the neurologic assessment.

**Methods:** Neonates  $>34$  weeks born in DH who fulfilled criteria for selective head cooling were admitted in NICU and Natus cool cap was applied within 6 hours of birth and continued for 72 hrs. Amplitude integrated EEG (aEEG) was recorded before the cooling and continued for the entire cooling period. During this period they were monitored for bradycardia, hypotension, bleeding episodes, seizures, oliguria. Septic screen was done to evaluate sepsis. Scalp skin changes and facial edema were also recorded. MRI brain after cooling was done and result was recorded. Detailed neurological exam was done by neonatologist at the time of discharge.

**Results:** 10 neonatal data were analyzed. Average Gestational age was 37.5 weeks, Average birth weight 2.87 kg. 4 had severe hypoxia and 6 of them had moderate hypoxia based on the clinical condition and aEEG criteria. 5 delivered by LSCS and 5 by NVD. One had cord prolapse. 9 (90%) of them needed ventilator support at the time of birth. Cool cap was initiated within 6 hrs mean of 3.9 hrs (3.5–5 hrs). 4 (40%) had abnormal PT/PTT and 3 (30%) had facial edema. 1 (10%) had blood culture

positive and received antibiotics. All of them had seizure activity and 2 (20%) had recurrent seizure needing multiple anticonvulsant medication. Bradycardia (heart rate <100) was noted in 5 cases (50%) during cooling but no ECG abnormality was noted. One (10%) died and 8 (80%) were neurologically assessed as normal at discharge. MRI Brain were also normal in 8 cases (80%) in these infants at discharge. One (10%) had grade 2 IVH. Mild scalp erythema was noted in all cases which resolved spontaneously after cooling was stopped. One (10%) had hypocalcemia and hypomagnesemia.

**Conclusion:** Selective head cooling is relatively safe therapy with minimal immediate complications like facial edema (30%) and coagulation abnormality (40%). Culture positive Sepsis was 10%. Normal neurological assessment with normal brain MRI at the time of discharge was noted in 80% of such cases. However, larger studies are needed to look for further complications and long term neurological outcome.

P314

# OUTCOMES FOR MODERATE TO SEVERE HYPOXIC ISCHAEMIC ENCEPHALOPATHY: A SINGLE INSTITUTION'S EXPERIENCE

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**Background:** Hypoxic ischaemic encephalopathy (HIE) remains a significant contributor to mortality in the neonatal period and potential ongoing neurological morbidity in survivors. There are numerous neonatal prognostic indicators used to predict outcome and guide counselling of families.

**Method:** Single centre retrospective clinical study analysing the Liverpool NICU data on infants ≥36 weeks gestation treated for HIE from 01/01/2005 until 01/03/2011. Medical records of infants were reviewed and follow-up outcomes at 8 and 24 months were recorded where available. HIE was classified as mild, moderate or severe using Sarnat staging. MRI, formal EEG and amplitude-integrated electroencephalogram (aEEG) data if performed, were collected.

**Results:** Over the six year period, 56 infants that were admitted to Liverpool NICU diagnosed with HIE were analysed. 61% were male, 39% were female with 14 (25%) neonatal deaths occurring. 5 infants (9%) had mild HIE, 32 (57%) had moderate HIE and 19 (34%) had severe HIE. 28 infants (50%) received cooling therapy. Follow-up data was available for 33 infants. 86% of infants with severe HIE died, 7% had neurodevelopmental disability and 7% had a normal outcome at follow-up. Of the infants diagnosed with moderate HIE, 47% had a normal outcome, 35% had neurodevelopmental disability and 18% died. The usefulness of additional aEEG in the first 72 hours of life, formal EEG and MRI will be presented.

**Conclusions:** Clinical severe HIE had the worst prognosis with death the most common outcome while moderate HIE continues to be the most ambiguous stage with the widest range of outcomes.

P315

# EFFECTIVENESS OF ENVIRONMENTAL ENRICHMENT INTERVENTIONS FOR IMPROVING THE MOTOR OUTCOMES OF INFANTS AT HIGH RISK OF CEREBRAL PALSY: A SYSTEMATIC REVIEW

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**Background:** Neuroplasticity evidence from animals favours an early enriched environment for promoting optimal brain injury recovery. In

infants, systematic reviews show environmental enrichment improves cognitive outcomes but the effect on motor skills is less understood with reviews suggesting traditional therapy has little effect on motor outcomes. The study aim was to appraise the effectiveness of environmental enrichment for improving motor outcomes of infants at high risk of cerebral palsy (CP).

**Method:** A systematic review was conducted. CENTRAL, CINAHL, ERIC, SocINDEX and PsychINFO databases were searched for literature meeting inclusion criteria: randomised controlled trials (RCTs); were high risk of/ diagnosis of CP; >25% participants ≤2 years; parent or infant interventions post discharge; and motor outcomes reported. Two reviewers independently extracted data using the Cochrane protocol regarding participants, intervention characteristics, and outcomes. Methodological quality was assessed using risk of bias assessment and GRADE.

**Results:** 226 studies were identified, after removing duplicates and unrelated studies, 16 full-text articles were reviewed, of which seven studies met inclusion criteria. The risk of bias varied between studies with the more recent studies demonstrating the lowest risk. Enrichment interventions varied in type and focus making comparisons difficult. A meta-analysis was conducted of studies that compared enrichment to standard care (n = 5), and totalled 150 infants. A small positive effect for enrichment was found; SMD 0.49 (95% CI 0.03 to 0.96; I<sup>2</sup> = 45%; p = 0.01).

**Conclusions:** Environmental enrichment looks promising for CP and therefore high quality clinical trials with well-defined environmental enrichment strategies are urgently required.

P316

# DEVELOPMENTAL AND NEUROSENSORY OUTCOMES FOR A COHORT OF EXTREMELY PRETERM OR EXTREMELY LOW BIRTH WEIGHT INFANTS FROM THE ROYAL WOMEN'S HOSPITAL, MELBOURNE

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**Background:** Rates of developmental delay for extremely preterm (EP) and extremely low birth weight (ELBW) infants are a useful benchmark for centres involved in their clinical care. This study aimed to determine the rates of follow-up and developmental delay for a single centre cohort.

**Method:** A retrospective review of two-year follow-up data for EP and/or ELBW infants born at the RWH in 2008 and 2009 was undertaken. Developmental outcomes were assessed using the Bayley-III (mild delay -2 to <-1 standard deviation (SD), moderate delay -3 to <-2 SD, severe delay <-3SD), along with rates of cerebral palsy, blindness and deafness at two years. Reasons for failure to follow-up were identified.

**Results:** The follow-up rate was 50% (109/219 survivors to discharge). The mean scores and rates of delay on the Bayley-III assessment for the cohort evaluated at two years are shown in the Table. The mean scores were all within 1 SD (15) of the Bayley-III mean of 100. Mild delay in any domain of development was the most prevalent problem. The rate of cerebral palsy was 6.4% and no children had blindness or deafness. The main reasons for loss to follow-up were failure to attend, lack of available appointments or no contact details.

| Variable                | Cognitive   | Language    | Motor       |
|-------------------------|-------------|-------------|-------------|
| Number of children      | 103         | 99          | 102         |
| Mean ± SD               | 93.5 ± 14.6 | 91.7 ± 18.1 | 95.0 ± 17.6 |
| Severe delay n (%)      | 2 (1.9)     | 2 (2.0)     | 5 (4.9)     |
| Moderate delay SD n (%) | 5 (4.9)     | 7 (7.0)     | 3 (2.9)     |
| Mild delay n (%)        | 12 (11.7)   | 20 (20.2)   | 11 (10.8)   |
| Normal range n (%)      | 84 (81.5)   | 70 (70.7)   | 83 (81.4)   |

**Conclusions:** Rates of developmental delay in EP and/or ELBW infants at RWH were similar to other published cohorts in Australia. Follow-up for these infants is an important part of clinical care and strategies need to be determined to improve rates of loss to follow-up.

## P317

## EXCESS PRENATAL CORTICOSTERONE EXPOSURE PROGRAMS HYPOTENSION IN AGED MALE OFFSPRING IN THE MOUSE

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**Background:** Exposure to synthetic glucocorticoids *in utero* has been associated with the development of hypertension in later life for both humans and animal models. The effects of natural glucocorticoids have been less well characterized. We used a mouse model of short-term exposure to excess corticosterone (CORT) to investigate the long-term cardiovascular and renal outcomes.

**Method:** CORT was delivered between E12.5-E15. Untreated counterparts were used as controls (UNTR). Nephron endowment was estimated at PN30. At 12 months of age the blood pressure of the offspring was measured by radiotelemetry in both basal and stressed conditions. Urine was collected and analysed using a Cobas Integra® 400 plus analyzer.

**Results:** Male CORT offspring had significantly ( $P < 0.05$ ) lower basal mean arterial, systolic, and diastolic blood pressures compared to UNTR. In response to a 15 min restraint stress the increase in heart rate was attenuated in the CORT males. Male CORT offspring had increased albuminuria ( $4.18 \pm 0.39$  vs.  $2.32 \pm 0.39$  ng/kg BW;  $P < 0.05$ ) as well as increased excreted levels of sodium, potassium and chlorine ions in their urine. CORT males had 33% decrease in nephron number. Male CORT were lighter at 12 months of age ( $33.3 \pm 1.0$  vs.  $36.7 \pm 1.1$  g;  $P < 0.05$ ), and although they tended to have larger hearts and kidneys this was not statistically significant. No differences were observed in female offspring.

**Conclusions:** Prenatal exposure to excess CORT leads to a nephron deficit, albuminuria, electrolyte imbalance and hypotension in aged male offspring.

## P318

## CHILDREN AT RISK OF NEONATAL HYPOGLYCAEMIA HAVE IMPAIRED WORKING MEMORY AT TWO YEARS

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**Background:** Neonatal hypoglycaemia is associated with adverse neurodevelopmental outcomes, including impaired executive function. The aim of this study was to assess one aspect of executive function; working memory (WM) in children born at risk of hypoglycaemia, and to compare WM in children with different risk factors.

**Method:** Children were born at risk of neonatal hypoglycaemia (infant of a diabetic mother (IDM), late preterm, small (birth weight  $< 2500$  g or  $< 10^{\text{th}}$  centile) or large (birth weight  $> 4500$  g or  $> 90^{\text{th}}$  centile). At 24 months parents completed the Behaviour Rating Inventory of Executive Function – Preschool Version (BRIEF-P), and children were assessed using the Multi-search-multi-location test (MSML). Outcomes were mean scores on the WM subscale of the BRIEF-P (mean = 50, SD = 10), where higher BRIEF-P scores indicate poorer performance, and proportions achieving success in two stages of the MSML; training and pre-switch, and overall success. Groups were compared using ANOVA and Chi-square.

**Results:** Mean BRIEF-P WM scores were approximately 1 SD above the mean in all groups (Table). In the MSML task 87.9% of children completed training and achieved pre-switch success, and 84.6% achieved success in all stages. There were no significant differences between risk groups.

**Conclusions:** Children born at risk of neonatal hypoglycaemia are reported by parents to have problems with WM at 24 months, independent of the reason for neonatal risk. The MSML task may provide additional information about the development of WM processes in two-year-old children.

Table. Assessments of WM in children from different risk groups for neonatal hypoglycaemia

|                                                                               | IDM<br>N = 66 | Large<br>N = 16 | Late Preterm<br>N = 54 | Small<br>N = 40 | Total<br>N = 176 |
|-------------------------------------------------------------------------------|---------------|-----------------|------------------------|-----------------|------------------|
| BRIEF-P WM Mean (SD)                                                          | 60.7 (13.0)   | 58.1 (8.6)      | 59.5 (12.0)            | 64.1 (13.3)     | 61.1 (12.5)      |
| MSML                                                                          | N = 120       | N = 25          | N = 88                 | N = 47          | N = 280          |
| Pre Switch Success (completed training; 3 consecutive correct searches) N (%) | 101 (84.2)    | 24 (96.0)       | 79 (89.8)              | 42 (89.4)       | 246 (87.9)       |
| Overall Success N (%)                                                         | 95 (79.2)     | 23 (92.0)       | 77 (87.5)              | 42 (89.4)       | 237 (84.6)       |

## P319

## GLOMERULAR CAPILLARY LENGTH AND SURFACE AREA IN THE PRETERM LAMB KIDNEY

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**Background:** Preterm neonates are born with developmentally immature kidneys, hence vascularisation of the immature glomeruli is likely to be ongoing after birth. The aim of this study was to determine the effect of preterm birth on glomerular capillary growth and surface area in a lamb model of preterm birth.

**Method:** Moderate preterm lambs (n = 7) were delivered at 130 d gestation (term = 147 d), and ventilated for 3 days after birth. Gestational fetal age-matched controls were euthanised at 133 d gestation (n = 5). Term

lambs were delivered naturally, and they were either non-ventilated or ventilated for 3 days after birth (n = 16). At necropsy, the left kidney from each animal was perfusion-fixed. In epon-araldite embedded sections of inner, mid, and outer renal cortex, stereological techniques were used to measure glomerular capillary surface area, length and total renal filtration surface area (TRSFA).

**Results:** There was a significant increase in capillary length per glomerulus and TRSFA in preterm lambs, compared to age matched-fetal controls. Preterm lambs showed no difference in capillary length, capillary surface area and TRSFA compared to terms. Term lambs ventilated after birth exhibited a significantly reduced capillary length and capillary surface area per glomerulus compared to term non-ventilated control lambs.

**Conclusions:** These findings suggest that in response to being born there is marked growth of the glomerular capillaries in the preterm kidney. This does not appear to be the result of ventilation after birth, since growth of the glomerular capillaries in the term kidney was reduced following ventilation.



P320

# EARLY UNDERNUTRITION ALTERS ABUNDANCE OF FACTORS REGULATING MYOGENESIS AND PROTEIN SYNTHESIS IN SINGLETONS AND TWINS IN SKELETAL MUSCLE OF THE SHEEP FETUS

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Maternal undernutrition around conception results in a decrease in myofiber number in the sheep fetus. The impact of undernutrition during the periconceptual (PCUN) or preimplantation (PIUN) period on the abundance of factors regulating myogenesis and protein synthesis is unknown. The effects of PCUN (70% energy requirement (ER), from 45 d before to 6 d after conception) or PIUN (70% ER, 0 d–6 d) compared to controls (100% ER, –45 d to post mortem at 136–138 d gestation) on the abundance of myogenic factors (myostatin, follistatin, MyoD, MyoG and Myf5) and factors regulating protein synthesis (mTOR, pmTOR, P70S6K, pP70S6K, RPS6, pRPS6, p4EBP1, and eIF4E) in fetal skeletal muscle was determined using Western Blotting in singletons and twins. The abundance of myostatin and Myf5 was lower, while follistatin and p4EBP1 was higher in both PCUN and PIUN groups in singletons and twins. mTOR abundance was lower in the PCUN group in singletons, not in twins. pP70S6K abundance was not different in singletons, but higher in the PIUN group in twins. pRPS6 abundance was higher in singletons, but lower in twins in the PIUN group. The abundance of MyoD, MyoG, P70S6K, RPS6 or eIF4E was not different between groups. PCUN and PIUN acts to decrease the abundance of the myogenic inhibitor, myostatin. However the decrease in Myf5 abundance may result in a decrease in myofiber in singletons and twins. PCUN and PIUN may increase protein translation initiation in singletons and twins, however ribosomal protein translation may be increased in singletons, but decreased in twins in the PIUN group.

P321

# INCREASED PROLYL HYDROXYLASE (PHD) 2 AND 3 EXPRESSION IS ASSOCIATED WITH DECREASED SURFACTANT PROTEIN IN THE LUNG OF THE GROWTH RESTRICTED SHEEP FETUS

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**Background:** Antenatal glucocorticoid (GC) treatment reduces the risk of respiratory distress in normally grown preterm infants, but there is conflicting evidence about lung outcomes in intrauterine growth restricted (IUGR) infants. Experimental placental restriction in fetal sheep results in IUGR, chronic hypoxemia, increased cortisol and decreased lung surfactant protein (SP) production. The mechanisms responsible for decreased SP production are unknown but may involve decreased GC action or changes in hypoxia signalling.

**Method:** Endometrial caruncles were removed from nonpregnant ewes to induce placental restriction. Lungs were collected from Control and IUGR fetuses at 130–135 (n = 19) and 139–145 (n = 28) days gestation. qRT-PCR and Western blotting were used respectively to quantify lung mRNA and

protein expression of molecules that regulate GC or hypoxia inducible factor (HIF)1 $\alpha$  action or HIF1 $\alpha$  target genes.

**Results:** SP-A, -B and -C mRNA expression decreased in IUGR fetuses as previously determined, but SP-D mRNA expression was unchanged. There was no change in mRNA expression of glucocorticoid receptor or 11 $\beta$ HSD-1 and -2, which control GC availability. PHD2 mRNA and protein and PHD3 mRNA abundance increased concomitantly with HIF1 $\alpha$  and 1 $\beta$  mRNA expression. There was no increase in expression of the hypoxia-responsive gene adrenomedullin.

**Conclusions:** Despite increased circulating cortisol in IUGR fetuses, there is no evidence for increased intracellular GC action. While acute hypoxia normally inactivates PHDs, chronic hypoxemia in the IUGR fetus increases PHD abundance which normally prevents HIF signalling. This may represent a mechanism by which chronic hypoxemia results in a decrease in SP production in the IUGR fetal lung.

P322

# STANDARDISATION OF INSERTION OF PERIPHERALLY INSERTED CENTRAL CATHETERS (PICCS) USING PROFESSIONAL VIDEO RECORDING AS A TEACHING AID: A PRODUCT OF STATE-WIDE COLLABORATION

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**Background:** Central line associated blood stream infections (CLABSIs) are a well-documented complication of neonatal central lines in newborn care units in particular PICC insertions. The elimination of line associated sepsis is a complex problem. Use of a specially trained and dedicated team for catheter insertion has proven effective in reducing CLABSI rates. Quality Improvement Collaboratives have become popular internationally and have demonstrated efficacy in reducing CLABSI rates. Common among these programmes is an attempt to standardize the delivery of best practices so that they are carried out consistently among all participating centres.

**Method:** The insertion of a PICC line into an infant by a neonatal fellow in a NICU was filmed by a professional video crew. This took place under the direct supervision of a clinical nurse consultant who was instrumental in the development of the central line bundle of the NICU. The video was then professionally edited in a studio. The script was produced by the clinical nurse consultant and the author. Voice over was performed by a senior neonatologist with experience in electronic teaching aids.

**Results:** A high quality professional video demonstrating a standardized technique for the insertion of PICC lines in neonates was produced. The video can be used in the medical orientation package for registrars and residents in NICUs statewide.

**Conclusions:** We have developed a successful model of collaboration between tertiary NICU's in NSW. This has resulted in the production of a professionally produced teaching aid that will standardize a common practice state-wide and contribute to the improvement of the quality of care to preterm neonates in reducing CLABSI rates.

P323

# SAFE SLEEP PRACTICES IN A SPECIAL CARE BABY UNIT – HAVE WE IMPROVED?

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**Background:** Although the 'safe sleeping' public health campaign of 1990 significantly reduced the number of sudden and unexpected deaths of



infants during sleep, this remains a leading cause of preventable deaths. Neonatal nurses have an important role in modelling sleeping standards of care to parents. Safe sleeping practices were audited within the Special Care Nursery at Women's and Children's Hospital, Adelaide.

**Methods:** The first audit was undertaken in 2005 and repeated in 2012. All infants over a 6 week period were observed across all shifts to determine compliance with supine position, flat cot, face uncovered, use of bolsters and feet at end of cot.

**Results:** 106 infants were observed for supine infant position in 2005 and 102 in 2012. For the remainder of categories 50 infants were observed in 2005 and 102 in 2012 (Table). The comparison between the proportion complying in 2005 and 2012 are presented in the Table.

**Table.** Compliance with safe sleeping 2005 and 2012

|                    | 2005<br>n = 50 | 2012<br>n = 102 | P       |
|--------------------|----------------|-----------------|---------|
| Supine position    | 33 (31%)*      | 92 (90%)        | <0.0001 |
| Flat cot           | 41 (82%)       | 87 (85%)        | 0.64    |
| Face uncovered     | 43 (86%)       | 91 (89%)        | 0.60    |
| Toys near face     | 2 (4%)         | 1 (1%)          | 0.25    |
| Bolsters           | 7 (14%)        | 1 (1%)          | 0.002   |
| Feet at end of cot | 0              | 37 (36%)        | <0.0001 |

\*n = 102 for supine position only, n = 50 infants audited for remainder of indices in 2005.

**Conclusions:** Although there was a significant improvement in compliance with positioning infants' supine with feet at end of the cot, there is still need for improvement.

#### P324

##### FROM NOVICE TO EXPERT – GRADUATE NURSE EDUCATION IN NICU

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**Background:** The Canberra Neonatal Intensive Care Unit (NICU) has designed a Neonatal Nursing Foundation Program (NNFP) to attract and retain graduate nurses. Specialty areas across the Canberra Hospital (CH) are also using this model to accelerate skill development for newly registered nurses. Research has discussed a positive correlation between retention rates of new and graduate nurses and the completion of the NNFP.

**Method:** The NNFP commenced for the first time in 2012. The NNFP focused primarily on the orientation, education and support of nurses while working in the Special Care and High Dependency areas of the NICU. The course consists of didactic teaching sessions combined with self-directed packages and workshops. Clinical support is facilitated by Clinical Development Nurses. A comprehensive curriculum has been developed; it is comprised of four modules: Module 1 – Basic Skills; Module 2 – Special Care; Module 3 – High Dependency and Module 4 – Introduction to Intensive Care.

**Results:** The NNFP has improved the education and support of staff commencing in the specialty of Neonatal Nursing. All of the 2012 participants successfully completed the program and are working in the NICU. They have all enrolled in the Post-graduate Neonatal Nursing Certificate course in 2013 and have been given credit points towards a Graduate Certificate in Neonatal Nursing since completing the NNFP.

**Conclusions:** The NNFP has attracted and retained four course participants and assisted with their journey from 'novice to expert'. We are currently recruiting for the 2013 NNFP.

#### P325

##### ASSESSING WELL BABIES – A MIDWIVES PROGRAM

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**Background:** The Staff Development Unit at Canberra Hospital has been offering a well baby assessment program to assist with midwives assessing the baby prior to discharge. This approach ensures holistic care of the mother and baby. The assessment is performed on babies born >37 weeks gestation and weighing >2.5 kg. The principle of the assessment is to detect significant abnormalities, birth injuries and cardio respiratory disorders that may compromise a successful adaptation to extra uterine life.

**Method:** The well baby workshop has been running at Canberra Hospital since 1998. The program is a full day program which provides a systematic approach to assessing the newborn. The program consists of didactic lectures followed by an interactive session which includes listening to heart sounds and assessing hips. This is followed by an interactive video on detecting abnormalities of the newborn. The midwife must undertake a competency assessment after the completion of the program and is then deemed competent to commence assessment on newborn babies.

**Results:** Over 50% of the midwives working at Canberra Hospital have completed the program and are now assessing babies prior to discharge. If the babies are assessed as requiring further treatment they receive prompt and appropriate referral to the medical staff. Finally midwifery assessment of the newborn baby results in timely discharge for the mother and baby.

**Conclusions:** All term newborn babies receive appropriate and timely assessments and examination. Any baby that requires further assessment or treatment is referred straight to the medical officer.

#### P326

##### DEVELOPMENT OF A NURSING CLINICAL PRACTICE STANDARD FOR COMPLEX NEONATAL RESUSCITATION

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**Background:** A neonatal resuscitation nurse is an advanced nursing role. Changing patterns within the NICU environment have resulted in higher numbers of part-time staff, impacting upon their exposure to complex neonatal resuscitations. We sought to develop a practice standard for those nursing staff attending neonatal resuscitations.

**Method:** Information was sought from all Australasian NICU's regarding specific requirements for nursing staff attending neonatal resuscitations. A consultation process was undertaken with all NICU trained staff within our institution and the results used to develop a clinical practice workbook for neonatal resuscitation.

**Results:** Clinical practice information was obtained from 50% of Australasian units. While no unit had a specific clinical practice guideline outlining the minimum training/experience for nurses attending neonatal resuscitations, all respondents required staff to have a Neonatal Intensive Care Certificate and complete basic life support. Newborn resuscitation programme certification (NRP) was not mandatory. In our nursery 31 staff (36% of permanent staffing) currently attend neonatal resuscitations with 22 (70%) employed <1.0 FTE. All hold a Neonatal Intensive Care Certificate. Input into requirements for the role were received from 17 (46%) and highlighted 6 monthly scenario training, NRP accreditation every 3 years, formalised debriefing sessions following complex resuscitations as desirable.

**Conclusions:** The need for a minimum clinical practice standard for staff participating in complex resuscitations is recognised by those staff involved. NRP accreditation and exposure to clinical scenario training appears critical to establishing a team with the expertise, experience and enthusiasm to make them effective in supporting the resuscitation of the sick newborn.

P327

**WHICH MUSIC BEST IMPROVES SYNCHRONIZATION OF RESPIRATORY SUPPORT DURING SIMULATED CARDIO-PULMONARY RESUSCITATION OF NEONATES?**Roehr CC<sup>1,2</sup>, Schmöler GM<sup>2,3,4,5,6</sup>, Dawson JA<sup>2,4,6</sup>, Wong C<sup>4</sup>, Dold SK<sup>1</sup>, Davis PG<sup>2,4,6</sup><sup>1</sup>Department of Neonatology, Charité University Medical Centre, Berlin, Germany, <sup>2</sup>Department of Newborn Research, The Royal Women's Hospital, Melbourne, Australia, <sup>3</sup>Department of Neonatology, Royal Alexandra Hospital, Edmonton, Canada, <sup>4</sup>Critical Care Stream, Murdoch Children Research Institute, Melbourne, Australia, <sup>5</sup>Division of Neonatology, Department of Paediatrics, Medical University Graz, Austria, <sup>6</sup>Department of Obstetrics & Gynaecology, The University of Melbourne, Australia Email: charles.roehr@thewomens.org.au**Introduction:** The 3:1 coordination of chest compressions (CC) to manual inflations (MI) is mandated for effective neonatal cardiopulmonary resuscitation (CPR). Manikin studies in adults showed that listening to music improved the coordination of CC and MI, suggesting 'Stayin' alive' (105 beats per minute (bpm)) to adequately pace adult CPR. We speculated that a faster tempo was needed for neonatal CPR.**Aim:** To identify the best tune to pace neonatal CPR.**Methods:** Thirty individuals were studied during simulated 3:1 neonatal CPR. The order of tunes played was randomized. Five tunes of: 'Stayin' alive' (The Bee Gees: 105 bpm), 'Radetzky March' (J. Strauss: 110 bpm), 'Jingle Bells' (J. Pierpont: 115 bpm), 'I will survive' (G. Gaynor: 120 bpm), and 'S.O.S.' (ABBA: 125 bpm), were played in random order. Delivery of CC and MI were measured at baseline (participants counting '1,2,3 - breathe') and during each tune, using a medical data acquisition system.**Results:***Chest compressions*

Baseline median (SD) CC/min = 80 (6) compared to 77 (7) 'Stayin' alive' (p = 0.458), 80 (6) 'Radetzky March' (p = 1.0), 84 (9) 'Jingle Bells' (p = 1.0), 83 (6) 'I will survive' (p = 1.0), and 86 (7) 'S.O.S.' (p = 0.046).

*Manual inflations*

Baseline mean (SD) MI/min = 28 (2), compared to 28 (2) 'I will survive' (p = 1.0), 27 (2) 'Radetzky March' (p = 1.0), 28 (2) 'Jingle Bells' (p = 0.931), 26 (2) 'Stayin' alive' (p = 0.350), and 29 (2) 'S.O.S.' (p = 0.040).

**Conclusion:** Musical tunes have an impact on the coordination of MI to CC during neonatal CPR. 'S.O.S.' significantly raised the rate of CC and MI from baseline and best improved the 3:1 coordination.

P328

**EXPERIENTIAL LEARNING IN OBSTETRICS: SKILLS GAINED WITH THE USE OF PLACENTAE TOGETHER WITH SIMULATION**

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**Background:** The study set out to explore the insights of medical students and their clinical supervisors of the preparation for clinical placement in obstetrics after prior analysis revealed students would appreciate specific practice in examination skills. To incorporate this finding, examination of donated human placentae was added to one of the preparation skill days where the mechanisms of labour and assessment of a newborn infant at birth (Apgar) using simulation models was also undertaken.**Method:** In-depth semi-structured interviews were conducted with participating students shortly after completion of the eight week obstetrics and gynecology clinical rotation. Interviews were held with the clinical supervisors after all the students had completed their rotation. Thematic analysis was used to identify and document the emergent themes.**Results:** Student perceptions overall reported feeling adequately prepared in the added aspects with clearer links between theory and practice. The use of placentae, with the opportunity to spend time discussing the findings in the teaching space environment, assisted students to feel more confident in the clinical environment. Applying the mechanisms of labour, with the use of the simulation model, allowed students to demonstrate

their knowledge in the controlled setting prior to entering the birthing suite.

**Conclusions:** The findings provide an increased understanding of the factors a cohort of medical students and their clinical supervisors consider essential for preparation in obstetrics.

P329

**HUMAN MILK FORTIFIER AND NECROTIZING ENTEROCOLITIS: A RETROSPECTIVE CASE CONTROLLED STUDY**

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**Background:** Necrotizing enterocolitis (NEC) is the most common gastrointestinal emergency in preterm babies. Previous case controlled studies have identified several risk factors, with prematurity being the common variable. Although Human milk fortifier (HMF) is widely used to meet the caloric requirements of preterms, data on its relationship to NEC is lacking. The aim of this study was to detect if addition of HMF to breast milk increased the risk of NEC in preterm babies.**Method:** Data from medical records were used to identify all 21 cases diagnosed with NEC based on modified Bells criteria (stage II or greater) over a 10-year period (01/2000–12/2009). They were matched with 2 controls/case by gestational age, birth weight, volume and type of feed. Demographic and clinical data previously shown to predispose to NEC were recorded for cases and controls. Stratified Mantel-Haenszel Chi-square test was used to calculate the odds ratio for categorical variables and paired t-tests for comparing the continuous variables.**Results:** Incidence of NEC at <34 weeks gestation was 1%. The odds ratio for HMF use was 9 (p = 0.1, 95% CI = 0.86–94.3). There were also trends towards increased risk of NEC with prolonged antibiotic use, recent blood transfusion and recent administration of dilating eye drops.**Conclusions:** Addition of HMF to breast milk is associated with increased incidence of NEC in preterm infants <34 weeks gestation. Future case-control studies should match cases and controls for multiple attributes as described, to delineate the real effect of HMF.

P330

**ASSOCIATION OF INHIBITORS OF GASTRIC ACID SECRETION AND HIGHER INCIDENCE OF NECROTIZING ENTEROCOLITIS IN PRETERM VERY LOW BIRTH WEIGHT INFANTS**More K<sup>1,2</sup>, Athalye-Jape G<sup>1,2</sup>, Rao S<sup>1,3</sup>, Patole SK<sup>2,3</sup><sup>1</sup>Department of Neonatal Paediatrics, Princess Margaret Hospital, Perth, Australia, <sup>2</sup>Department of Neonatal Paediatrics, KEM Hospital for Women, Perth, Australia, <sup>3</sup>Centre for Neonatal Research and Education, University of Western Australia, Perth, Australia Email: drkiranmore@yahoo.com**Background:** Inhibitors of gastric acid (IGA) are used for upper gastrointestinal bleeding or gastro oesophageal reflux in preterm infants. The resultant increase in gastric pH may enhance the growth of pathogens and increase the risk of necrotizing enterocolitis (NEC). NEC is a potentially disastrous gastrointestinal emergency in preterm infants with significant mortality and morbidity. Our systematic review examined the association between IGA and NEC in preterm infants.**Methods:** Standard methodology of systematic reviews was followed. PubMed, Embase, Cochrane and CINAHL databases were searched in August 2012. Two reviewers independently collected data and assessed the quality of the studies based on Newcastle Ottawa Scale. Using fixed effect model meta-analysis was performed and odds ratio and 95% confidence interval were calculated.**Results:** One case control and one prospective cohort study (N = 11,346); both evaluating H2-blockers as IGA, were included. Meta analysis showed a significant association between NEC and IGA (OR: 1.78, 95% CI: 1.4,

2.27,  $p < 0.00001$ ). The prospective cohort study found higher incidence of infection (sepsis, pneumonia, urinary tract infection) with IGA (37.4% v 9.8%, OR: 5.5, 95% CI: 2.9–10.4,  $p < 0.001$ ).

**Conclusions:** Exposure to H2 receptor antagonists may be associated with increased risk of NEC and infections in preterm infants.

P331

# PROLONGED INITIAL EXPOSURE TO ANTIBIOTICS AND HIGHER INCIDENCE OF NECROTIZING ENTEROCOLITIS IN PRETERM INFANTS – A SYSTEMATIC REVIEW

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**Background:** Necrotizing enterocolitis (NEC) is a potentially life threatening illness affecting about 4–6% preterm very low birth weight (VLBW) infants. Aberrant microbial colonization of the gut is an important risk factor in the pathogenesis of NEC. Prolonged antibiotic exposure might disrupt normal colonization of the developing gut and increase the risk of NEC. We conducted a systematic review to study the association between prolonged initial antibiotics (PIA) and NEC.

**Methods:** The standard methodology of neonatal systematic reviews was followed. PubMed, Embase, Cochrane and CINAHL databases were searched in October 2012. Quality assessment was done by Health Technology Assessment (HTA) for non-randomised studies, and by the 'Risk of Bias' assessment tool (Cochrane Handbook) for randomised controlled trials (RCTs). Meta-analyses were performed by using random effect model, and odds ratio and 95% confidence interval were calculated.

**Results:** Four non-randomised studies (N = 4,983) and one RCT (N = 140) with subgroup of 53 preterm infants were included. Meta analysis of data from these studies showed a significant association between PIA and NEC [OR: 1.21 (95% CI: 1.01–1.49);  $p = 0.04$ ]. PIA also significantly increased the odds of death [OR: 1.43; (95% CI: 1.26, 1.62);  $p < 0.00001$ ], and the composite outcome of NEC or death [OR: 1.52 (95% CI: 1.19, 1.94);  $p = 0.0008$ ]. Two studies reported increased risk of late onset sepsis with PIA.

**Conclusions:** PIA is associated with increased risk of NEC and death in preterm VLBW infants.

P332

# HIGH C-REACTIVE PROTEIN IN CASES OF SUBCUTANEOUS FAT NECROSIS

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Subcutaneous fat necrosis (SCFN) of the newborn is a form of panniculitis that mostly affects full-term neonates who often have suffered either perinatal asphyxia or hypothermia. Therapeutic hypothermia is now routinely used in newborns to reduce the mortality and morbidity associated with moderate to severe hypoxic ischemic encephalopathy. The true incidence of SCFN in neonates undergoing cooling is unknown. Hypercalcaemia, thrombocytopenia, hypoglycaemia and other complications of SCFN have been well described in the literature. Mild elevation of C-reactive protein, which is consistent with SCFN being an inflammatory process, has been described in only few reports. We report two infants in whom CRP levels were markedly elevated and paralleled the evolving skin lesions.

Both cases were full term infants with hypoxic ischaemic encephalopathy and both were therapeutically cooled. In both cases, diagnosis of subcutaneous fat necrosis was made clinically. CRP was done initially as part

of a septic screen at the onset of skin lesion. In first case, CRP peaked on day 4 to 272 mg/L and still remained very high (110 mg/L) on day 14. While in case 2, CRP peaked on day 3 to 135 mg/L. In both cases, repeated infection screen including viral PCR was negative.

**Conclusion:** Our case report emphasizes that clinicians should recognize that therapeutic cooling may place neonates at increased risk for SCFN. Although one of the differential diagnoses of SCFN is infection; we wish to highlight that the persistently elevated CRP which caused prolonged antibiotic treatment, can reflect the inflammatory component of SCFN.

P333

# UREAPLASMA SPECIES MULTIPLE BANDED ANTIGEN (MBA) VARIATION IS ASSOCIATED WITH THE SEVERITY OF CHORIOAMNIONITIS IN LATE PRETERM PLACENTAS

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**Background:** Intra-amniotic infection accounts for 30% of all preterm births (PTB), with the human *Ureaplasma* species being the most frequently identified microorganism from the placentas of women who deliver preterm. The highest prevalence of PTB occurs late preterm (32–36 weeks) but no studies have investigated the role of infectious aetiologies associated with late preterm birth.

**Method:** Placentas from women with late PTB were dissected aseptically and samples of chorioamnion tissue and membrane swabs were collected. These were tested for *Ureaplasma* spp. and aerobic/anaerobic bacteria by culture and real-time PCR. Western blot was used to assess MBA variation in ureaplasma clinical isolates. The presence of microorganisms was correlated with histological chorioamnionitis.

**Results:** *Ureaplasma* spp. were isolated from 33/466 (7%) of placentas by culture or real-time PCR. The presence of ureaplasmas, but not other microorganisms, was associated with histological chorioamnionitis (21/33 ureaplasma-positive vs. 8/42 other bacteria;  $p = 0.001$ ). Ureaplasma clinical isolates demonstrating no MBA variation were associated with histological chorioamnionitis. By contrast, ureaplasmas displaying MBA variation were isolated from placentas with no significant histological chorioamnionitis ( $p = 0.001$ ).

**Conclusion:** *Ureaplasma* spp. within placentas delivered late preterm (7%) is associated with histological chorioamnionitis ( $p = 0.001$ ). Decreased inflammation within the chorioamnion was observed when the clinical ureaplasma isolates demonstrated variation of their surface-exposed lipoproteins (MBA). This variation may be a mechanism by which ureaplasmas modulate and evade the host immune response. So whilst ureaplasmas colonise the chorioamnion during pregnancy, they are not suspected because of the normal appearance of placentas and amniotic fluid.

P334

# MID-GESTATION INTRA-AMNIOTIC INFECTION WITH UREAPLASMA PARVUM IS RESOLVED WITHIN SPINY MICE (ACOMYS CAHIRINUS) BY TERM DELIVERY: BUT CAUSED CHRONIC INFECTION OF FETAL LUNGS AND PLACENTAE

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**Background:** *Ureaplasma* species in amniotic fluid at the time of second-trimester amniocentesis increases the risk of preterm birth, but most affected pregnancies continue to term (Gerber *et al.* J Infect Dis 2003). We aimed to model intra-amniotic (IA) ureaplasma infection in spiny



mice, a species with a relatively long gestation (39 days) that allows investigation of the disposition and possible clearance of ureaplasmas in the feto-placental compartment.

**Method:** Pregnant spiny mice received IA injections of *U. parvum* serovar 6 (10 µL,  $1 \times 10^4$  colony-forming-units in PBS) or 10B media (10 µL; control) at 20 days (d) of gestation (term = 39 d). At 37 d fetuses (n = 3 ureaplasma, n = 4 control) were surgically delivered and tissues were collected for; bacterial culture, ureaplasma *mba* and *urease* gene expression by PCR, tissue WBC counts and indirect fluorescent antibody (IFA) staining using anti-ureaplasma serovar 6 (rabbit) antiserum. Maternal and fetal plasma IgG was measured by Western blot.

**Results:** Ureaplasmas were not detected by culture or PCR in fetal or maternal tissues but were visualized by IFA within placental and fetal lung tissues, in association with inflammatory changes and elevated WBC counts (p < 0.0001). Anti-ureaplasma IgG was detected in maternal (2/2 tested) and fetal (1/2 tested) plasma but not in controls (0/3).

**Conclusions:** IA injection of ureaplasmas in mid-gestation spiny mice caused persistent fetal lung and placental infection even though ureaplasmas were undetectable using standard culture or PCR techniques. This is consistent with resolution of IA infection, which may occur in human pregnancies that continue to term despite detection of ureaplasmas in mid-gestation.

### P335

#### POSTPARTUM HAEMORRHAGE, TYPE OF LABOUR AND MANAGEMENT OF THE THIRD STAGE OF LABOUR

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**Background:** The rate of postpartum haemorrhage (PPH: estimated blood loss >500 mL) following vaginal birth appears to be rising in Western countries. We explored possible explanations for this in our hospital.

**Method:** Using data derived from the electronic Birthing Outcome System, we reviewed the PPH rate among singleton vaginal births 2011–2012. In 2012, the recommended 3<sup>rd</sup> stage management for augmented or induced labours changed from 10 iu intramuscular oxytocin to Syntometrine.

**Results:** In 2011–2012 (to date), 4033 women had a singleton vaginal birth at Monash. Of these, 2903 (72%) women had had a normal labour, 300 (7%) had had a spontaneous onset of labour and oxytocin augmentation, and 830 (21%) had had a labour induced with oxytocin. Overall, 596 (15%) women had had a PPH. The PPH rate differed significantly by type of labour: normal labour – 359 (12%); augmented labour – 79 (26%); induced labour – 158 (19%) (P = 0.008). Of women who were induced, 92% received oxytocin for 3<sup>rd</sup> stage and 8% received syntometrine in 2011. The rates in 2012 were 22% and 78%, respectively. For women who were augmented the rates for oxytocin and syntometrine use in 2011 and 2012 were 93% and 7%, and 20% and 80%, respectively. Compared to oxytocin, use of syntometrine did not reduce the higher PPH rate for induced or augmented labours (21% vs 21%).

**Conclusions:** The PPH rate for singleton vaginal births is higher for augmented or induced labours compared to normal labour. The use of syntometrine, rather than oxytocin, did not reduce this increased rate.

### P336

#### ASSOCIATIONS BETWEEN ETHNICITY AND SEVERE PERINEAL TRAUMA AFTER VAGINAL BIRTH

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**Background:** It has been suggested that the risk of sustaining severe perineal trauma may vary by maternal ethnicity. We sought to explore this in a single health service.

**Method:** We reviewed the perineal outcome for women of different ethnicities having a vaginal birth in our service 2009–2012. Data were derived from the electronic Birthing Outcome System (BOS), an electronic maternity record completed for all women as part of State government birth reporting.

**Results:** In 2009–2012, among five selected ethnic groups 19,791 women had a vaginal birth: Australian-born (10,922 women), South Asian (4,357), East Asian (1,321), South East Asian (2,800), and West Asian (391). Among these women, the normal vaginal birth rate differed significantly across ethnic groups (Australian: 81%, S Asian: 76%, East Asian: 76%, SE Asian: 82%, W Asian: 87%, P < 0.0001). Perineal outcomes also differed significantly between ethnic groups (eg intact perineum – Australian: 74%, S Asian: 62%, E Asian: 43%, SE Asian: 48%, W Asian: 67%, P < 0.0001; 3<sup>rd</sup>/4<sup>th</sup> degree tear – Australian: 3%, S Asian: 8%, E Asian: 7%, SE Asian: 6%, W Asian: 4%, P < 0.0001; episiotomy – Australian: 17%, S Asian: 29%, E Asian: 29%, SE Asian: 25%, W Asian: 14%, P < 0.0001).

**Conclusions:** The rate of severe perineal trauma after vaginal birth varies with maternal ethnicity. Women of South/SE/East Asian birth have a higher rate of 3<sup>rd</sup>/4<sup>th</sup> degree tears and of episiotomy than Australian-born or West Asian women. These differences were not explained by differences in the need for assisted vaginal birth. Other possible contributors to the differing rates are under analysis.

### P337

#### EFFECT OF REPEAT ANTENATAL BETAMETHASONE ON CHILDHOOD LUNG FUNCTION: A RANDOMISED CONTROLLED TRIAL (ACTORDS)

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**Background:** Repeat doses of antenatal glucocorticoids given to women at risk of preterm birth are more effective than a single course in reducing the risk of neonatal respiratory disease and serious morbidity. However, glucocorticoid-induced maturational effects in the fetal lung are associated with slowing of alveolar septation and there is concern that exposure to repeat doses could adversely affect lung growth and function in the long-term.

**Method:** Women who participated in the ACTORDS trial were randomised to betamethasone or placebo, repeated weekly if they remained at high risk of preterm birth at <32 weeks' gestation, ≥7 d after an initial course of glucocorticoids. Assessment of their children at 6–8 years' corrected age included lung function testing via portable flow spirometry. Spirogram quality was determined by two independent blinded examiners and only data from adequate quality spirograms were included (ATS/ERS 2005 guidelines). Sex, age and height dependent Z-scores for forced vital capacity (FVC), forced expiratory volume in 1 s (FEV1), FEV1/FVC and maximal mid-expiratory flow (FEF25–75%) were compared between groups.



**Results:** Spirometry was attempted in 740 children but adequate spirometry was obtained from only 185 (25%) children. There were no significant differences between children exposed to repeat antenatal betamethasone ( $n = 98$ ) and those exposed to placebo ( $n = 87$ ) for any lung function measures (Table).

**Conclusions:** Exposure to repeat doses of antenatal betamethasone compared with a single course of antenatal glucocorticoids does not adversely affect lung function at early school-age. However, few children in this cohort achieved adequate spirometry and lung function testing should be repeated at an older age.

**Table.** Lung function of children exposed to repeat betamethasone or placebo

| Z-score   | Repeat<br>Betamethasone<br>( $n = 98$ ) | Placebo<br>( $n = 87$ ) | Mean<br>difference<br>(95% CI) | P    | P<br>Adjusted* |
|-----------|-----------------------------------------|-------------------------|--------------------------------|------|----------------|
| FVC       | 0.0 (0.9)                               | 0.0 (0.9)               | -0.1 (-0.3, 0.2)               | 0.67 | 0.55           |
| FEV1      | -0.9 (1.1)                              | -0.7 (1.0)              | -0.2 (-0.5, 0.1)               | 0.25 | 0.23           |
| FEV1/FVC  | -1.5 (0.9)                              | -1.3 (0.9)              | -0.2 (-0.4, 0.1)               | 0.15 | 0.20           |
| FEF25-75% | -1.5 (1.2)                              | -1.3 (1.2)              | -0.3 (-0.6, 0.1)               | 0.13 | 0.14           |

\*Adjusted for gestational age at trial entry, preterm prelabour rupture of membranes, antepartum haemorrhage. All analyses adjusted for non-independence of children from multiple pregnancy.

### P339

#### CHOLESTEROL ABNORMALITIES ARE COMMON IN WOMEN WITH PRIOR GESTATIONAL DIABETES MELLITUS

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**Background:** Gestational Diabetes Mellitus (GDM) is a common medical condition that is associated with substantial increase in risk of maternal and perinatal complications. Perinatal complications include macrosomia, birth injuries and long-term adverse effects of health such as impaired glucose tolerance, obesity and impaired intellect. The primary objective was to investigate the prevalence of persisting diabetes and high cholesterol in postpartum women with prior Gestational Diabetes Mellitus (GDM).

**Method:** All women with a diagnosis of GDM who delivered from April 2010 to 30 June 2012, at Joondalup Health Campus were included in the study. On the postnatal ward women were given an appointment to attend a diabetes follow up clinic at 6–12 weeks postpartum. They were provided with a pathology request for a glucose tolerance test, fasting cholesterol, triglycerides, LDL-C and HDL and cardiac risk ratio. Data were extracted from files and were analysed using descriptive statistics.

**Results:** Of 4970 women delivering, 168 (3.4%) were diagnosed with GDM, of whom 136 attended for postpartum review (85%) at 6–12 weeks. Persisting glucose intolerance and type 2 diabetes were diagnosed in 16% and 6% of the patients. High cholesterol was identified in 54% and 50% had high LDL-C levels. Additionally, 13% of GDM women had high triglyceride levels.

**Conclusions:** Women with GDM have high rates of persisting glucose abnormalities at 6–12 weeks postpartum and half have abnormalities in cholesterol levels. Overall, these findings suggest that women with previous GDM are at increased risk of cardiovascular disease and early postnatal intervention should be offered.

### P340

#### MATERNAL DIETARY INTAKE AND ITS RELATIONSHIP TO BIRTH OUTCOMES IN PREGNANCIES COMPLICATED BY ASTHMA AND CIGARETTE USE

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**Background:** The occurrence of asthma or smoking during pregnancy is associated with poor fetal growth. Few studies have assessed whether diet plays a role in infant outcomes associated with these co-morbidities. Therefore, the aims were to assess dietary intakes in women living in a lower socioeconomic area and to i) compare between asthma and smoking status ii) assess differences in birth outcomes and iii) identify whether maternal dietary intakes affect growth parameters by infant sex.

**Method:** Secondary analysis of women participating in a study on the effect of asthma during pregnancy ( $n = 278$ ).

**Results:** Asthmatic smokers ( $n = 31$ ) had significantly higher intakes of energy, macronutrients and most micronutrients compared to asthmatic non smokers ( $n = 126$ ), control (non asthmatic) smokers ( $n = 38$ ) and control non smokers ( $n = 83$ ). There was no significant difference in infant birth weight, length or head circumference between asthma and smoking status. There was a sexually dimorphic relationship between maternal diet and birth weight, such that higher dietary intakes in asthmatic women positively correlated with female growth; whereas dietary intakes in smokers correlated negatively with female growth. There were no relationships between maternal diet and male birth weight in any of the groups. **Conclusions:** Maternal diet is increased in the presence of cigarette use during pregnancy but does not vary in relation to the presence of maternal asthma alone. Dietary intake appears to play a central role modulating growth of a female fetus in the presence of asthma or cigarette use but has no relationship to the growth of a male fetus.

### P341

#### WOMEN WITH PREGNANCY HYPERGLYCAEMIA: HOW WELL ARE LIFESTYLE INFORMATION BOOKLETS USED?

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**Background:** Providing lifestyle information, often in the form of printed materials, is recommended as the primary therapeutic strategy for women with pregnancy hyperglycaemia. This observational study aimed to assess the uptake of lifestyle information by women with pregnancy hyperglycaemia and to see if this varied by their socio-demographic characteristics.

**Method:** Participants were women with borderline gestational diabetes mellitus (GDM) who received printed diet and exercise information booklets as part of a randomised trial: investigating dietary and lifestyle advice for women with borderline GDM (the IDEAL Trial). Women were advised to use the booklets to record and set monthly goals for their diet and exercise. Outcomes include women's use of the booklets and their socio-demographic characteristics.

**Results:** A total of 69 women and their babies were involved in the study. 49 (71%) women used the booklet at least once with the remaining 20 (29%) women never using the booklet. Compared with non-users, women who used the booklets were more likely to have a medical history of hypertension, a family history of diabetes or hypertension and to have a higher socioeconomic status rating (48.8% advantaged/most advantaged compared with 35% in the non-user group (Table 1).

**Conclusions:** Based on the limited data from this small observational study, over two-thirds of women with borderline GDM used the lifestyle information booklets provided. Women with more resources or aware of their risk through a medical history, or family history, of hypertension or diabetes may be more likely to use printed lifestyle information.

**Table 1.** Women's socio-demographic characteristics by uptake of printed intervention booklets

| Characteristics                            | Booklet used<br>(n = 49) | Booklet not<br>used (n = 20) |
|--------------------------------------------|--------------------------|------------------------------|
| Mother's history of hypertension: N (%)    | 5 (10.2%)                | 1 (5%)                       |
| Family history of diabetes mellitus: N (%) | 24 (49.0%)               | 9 (45.0%)                    |
| Family history of hypertension: N (%)      | 27 (55.1%)               | 8 (40.0%)                    |
| Socioeconomic status: N (%)                |                          |                              |
| – Most disadvantaged                       | 6 (12.4%)                | 3 (15.0%)                    |
| – Disadvantaged                            | 10 (20.4%)               | 7 (35.0%)                    |
| – Average                                  | 14 (28.6%)               | 3 (15.0%)                    |
| – Advantaged                               | 8 (16.3%)                | 5 (25.0%)                    |
| – Most advantaged                          | 11 (22.5%)               | 2 (10.0%)                    |

P342

**THE EXPERIENCES OF TEENAGE MOTHERS IN THE NEONATAL UNIT**Bocock SJ*Taranaki District Health Board, New Plymouth, New Zealand**Email: jane.bocock@tdhb.org.nz*

**Background:** The nurse's knowledge of patients' experiences contributes to gaining a deeper understanding of their health process and enables the nurse to develop a caring, therapeutic relationship. There is very little literature that focuses on the experiences of teenage mothers whose babies are in the neonatal unit. The purpose of the study was to hear and value the stories as told by teenage mothers; inform and strengthen health care practice and identify areas of change if change is necessary.

**Method and Results:** Conversations with four teenage mothers were taped, transcribed and then analysed using van Manen's approach to researching lived experience. From the analysis two main themes emerged – the need to do [mothering] and being in a safe place. A third theme – the emotional roller-coaster – was also identified and ran throughout the stories. The experiences of the teenage mothers were compared and contrasted with the existing literature.

**Conclusions:** A key finding of the research was that teenage mothers, despite their developmental stage, had similar experiences to those of older mothers in the same situation. Additionally, their competence at mothering was greater than might have been expected. Recommendations were made related to 'family centred care' in the neonatal unit to ensure all mothers have the support, information and education they need. Additionally, the research supports the development of strong therapeutic relationships with mothers based on partnership in decision making and care giving which, in turn, sees a transfer of power and control to the mother.

P343

**METFORMIN FOR WOMEN WHO ARE OBESE DURING PREGNANCY VERSUS STANDARD CARE FOR IMPROVING MATERNAL AND INFANT OUTCOMES – A SYSTEMATIC REVIEW**Eames AJ<sup>1</sup>, Grivell RM<sup>1,2</sup>, Dodd JM<sup>1,2</sup>, Deussen A<sup>1,2</sup>*<sup>1</sup>Women's & Children's Hospital, Adelaide, Australia, <sup>2</sup>Robinson Institute, Discipline of Obstetrics & Gynaecology, The University of Adelaide, Australia Email: ajeames@yahoo.com.au*

**Background:** The prevalence of obesity is rising globally and is associated with a higher risk of maternal and infant health complications in pregnancy. There is little evidence surrounding effective interventions to limit gestational weight gain (GWG). Some evidence exists to suggest a role for metformin in treating not only diabetes and PCOS but also obesity. A small number of studies have suggested metformin may be effective in promoting weight loss in non-pregnant obese women. Our aim was to conduct a systematic review of the literature to evaluate the effect of

metformin on insulin sensitivity and limiting GWG in obese women, and its effect on clinical maternal and infant outcomes.

**Method:** Types of Participants: Pregnant women with a BMI  $\geq 30$  kg/m<sup>2</sup>. Types of Studies: Randomized controlled trials which compared the use of metformin alone or in combination with any other treatment modality to limit gestational weight gain to improve maternal and infant health outcomes.

**Search Terms:** We searched the Cochrane Pregnancy and Childbirth Group's Trials Register (February 2012), PUBMED, and the Australian (ACTR) and International (ICTN) Clinical Trials Registry.

**Primary Outcome:** infant birth weight above 4 kg.

**Results:** No published trials were identified. An ongoing UK-based trial comparing metformin with placebo in 400 obese pregnant women was identified.

**Conclusions:** Further evaluation of the role of metformin in limiting gestational weight gain and the effect on maternal and infant clinical outcomes in women who are obese is warranted. An Australian-based randomized trial is about to commence recruitment.

P344

**THE EFFECT OF ANTENATAL DIETARY ADVICE ON MATERNAL BODY COMPOSITION IN WOMEN WHO ARE OVERWEIGHT OR OBESE – FINDINGS FROM THE LIMIT RANDOMISED TRIAL**Kannicappan LM<sup>1</sup>, Deussen AR<sup>1</sup>, Moran LJ<sup>1</sup>, Grivell RM<sup>1,2</sup>, Yelland LN<sup>1</sup>, Dodd JM<sup>1,2</sup>*<sup>1</sup>The University of Adelaide, Robinson Institute, Discipline of Obstetrics & Gynaecology, Women's and Children's Hospital, Adelaide, Australia,**<sup>2</sup>Department of Perinatal Medicine, Women's and Children's Hospital, Adelaide, Australia Email: lavern.kannicappan@adelaide.edu.au*

**Background:** Little is known about the changes in maternal body composition that occur over the course of pregnancy, and whether these measurements can be modified by dietary intervention.

**Methods:** We conducted an ancillary study in the context of the LIMIT randomised trial, evaluating the effect of a dietary and lifestyle intervention for women who were overweight or obese during pregnancy, on maternal and infant health outcomes. Women were eligible for the LIMIT trial with a BMI  $\geq 25$  kg/m<sup>2</sup>, and a live singleton pregnancy. Women were randomised using a central telephone randomisation service to either the dietary and lifestyle intervention group, or to receive current standard antenatal care.

Women participating in the randomised trial had their height and weight measured and body mass index calculated at the time of their first antenatal appointment. At the time of trial entry, a subset of women had their mid-arm circumference, and biceps, triceps and subscapular skin-fold thicknesses measured, and percentage body fat calculated. Gestational weight gain was calculated as the difference in measured weight at 36 weeks' gestation (or nearest to birth). Analyses were conducted using intention to treat principles.

**Results:** A total of 1,582 women were involved in this ancillary study, with 807 women randomised to the dietary and lifestyle intervention group, and 775 women to the standard care group. The two groups were similar at the time of trial entry.

**Conclusions:** Analyses will presented and discussed evaluating differences between treatment groups, changes which occur over the course of pregnancy.

P346

# EARLY VERSUS LATE INTRODUCTION OF HUMAN MILK FORTIFICATION FOR PRETERM INFANTS: A COCHRANE REVIEW

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**Background:** It is standard clinical practice to add human milk fortifier (HMF) to human milk to provide additional nutrients to enhance growth. The ideal time for introduction of HMF is unknown and clinical practice varies widely. The objective of this review is to determine whether early versus late introduction of HMF in preterm infants improves growth and neurodevelopmental outcomes without significant adverse effects.

**Methods:** Standard methods of the Cochrane Neonatal Review Group were used. Early fortification was defined as introduction of HMF at an enteral feed volume of <120 mls/kg/day and late at ≥120 mls/kg/day.

**Results:** Only one trial was found. This trial of 207 preterm infants was primarily designed to study the effect of human milk based HMF compared with bovine based HMF. Infants were randomized to one of three groups: 1) human milk based HMF commenced at an enteral intake of 40 mls/kg/day or 2) 100 mls/kg/d and 3) bovine based HMF commenced at 100 mls/gk/day.<sup>1</sup> Although this trial did not meet the inclusion criteria, given that no other studies have been found to date, it has been included in the review. For the review the groups commencing at 100 mls/kg/day were combined. No difference in weight, gain was found between the groups, nor in the rate of necrotizing enterocolitis.

**Conclusions:** This one study suggests that introduction of HMF at 40 mls/kg/day may be safe. However given that human milk based HMF was used, further trials are required that reflect current clinical practice where bovine based HMF predominates.

## Reference

1. Sullivan S, Schanler RJ, et al. The Journal of Pediatrics 2010;156:562–567.e561.

P347

# LOW DOSE STOSS THERAPY IN NEWBORNS OF PREGNANT WOMEN WITH VITAMIN D DEFICIENCY

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**Introduction:** Newborns of women with Vitamin D deficiency during pregnancy are at risk of both osseous and non osseous complications. It was our aim to determine the effect of low dose STOSS therapy on Vitamin D levels at 6 weeks in newborns of mothers identified with Vitamin D deficiency in the first trimester of pregnancy.

**Methods:** Between September 2008 and January 2009, all term newborns delivered to Vitamin D deficient mothers were given 40,000 U of cholecalciferol following a day 2 Vitamin D level, measured by enzyme immunoassay. Vitamin D was measured again at 6 weeks of age at the time of neonatal clinic review.

**Results:** Ninety five consecutive newborns were identified, 45% male. Following STOSS, Vitamin D [median (range)] levels increased from 62 (nmol/l) (17, 176) on day 2 to 119 nmol/l (27, 366) at 6 weeks. For Vitamin D deficient infants on day 2 of life (n = 33), levels increased from 35 nmol/l (17, 49) to 95 nmol/l (30, 360). Vitamin D increased from 52 nmol/L to 118 nmol/L in exclusively breast fed infants (n = 23) and from 70 nmol/L to 114 nmol/L in formula/combined feeding infants (n = 56). Seven newborns had levels above 250 nmol/L at 6 weeks.

**Conclusion:** Low dose STOSS therapy improves Vitamin D levels in newborns with mild to moderate deficiency but may result in significantly

elevated levels in newborns without Vitamin D deficiency. Post-STOSS levels are not influenced by mode of feeding.

P348

# WHAT ARE THE OTHER HALF DOING? A COMPARISON OF FEED FORTIFICATION PRACTICE BETWEEN NURSING AND MEDICAL STAFF

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**Background:** Human milk fortifiers (HMF) are extensively used in Neonatal Intensive Care Units (NICU). HMF is prescribed by medical staff but nursing staff prepare and administer feeds.

**Method:** Medical and nursing staff from tertiary NICU's in Australia and New Zealand were surveyed independently by online method. Although the surveys queried different aspects of nutrition and feeding practice, direct comparisons regarding fortification practice were possible in five domains. Specifically; feed volume upon commencement of fortification, fortifier strength at commencement, by whom added, where fortifier is added and allowable standing time.

**Results:** 23 out of 28 units provided responses in the domains of interest. Only 1 unit from the 23 had identical answers from both staff, over the 5 domains. Detailed results tabulated below.

**Conclusions:** Medical and Nursing staff amongst Australian and New Zealand NICU's had differing responses to HMF practices for fortification administration. This has may have implications for neonatal care. Communication between these two important care groups may need improvement.

|                                           | Nursing                   | Medical                   | Intra-Unit Agreement |
|-------------------------------------------|---------------------------|---------------------------|----------------------|
| Feeding Volume at which Fortifier started | 120–180 (mls/kg/day)      | 80–170 (mls/kg/day)       | 7/23 (30%)           |
| Strength Fortifier started at             | ½ – Full strength         | ½ – Full strength         | 10/22 (45%)          |
| Who adds fortifier                        | Bedside Nurse – Milk Aids | Bedside Nurse – Milk Aids | 10/19 (52%)          |
| Where fortifier is added                  | Bedside – Formula Room    | Bedside – Formula Room    | 14/19 (74%)          |
| Allowable standing Time                   | 0–24 hours                | 0–24 hours                | 3/22 (13%)           |

P349

# ULTRASOUND IMAGING OF INFANT TONGUE MOVEMENT WITH A NOVEL TEAT COMPARED TO BREASTFEEDING

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**Background:** Concerns that bottle-feeding is detrimental to the establishment of breastfeeding in preterm infants are based on milk flow rates and configuration of the teat. The nipple expands evenly during breastfeeding in contrast to marked peristaltic tongue actions previously recorded on teats. This study compared preterm infants tongue movement during breastfeeding to those with a teat designed to release milk with vacuum and reinforced to limit compression of the teat.

**Method:** Preterm infants (n = 15) of gestational age at delivery of median (range) 28<sup>2</sup> weeks (23<sup>4</sup> to 32<sup>6</sup>) had ultrasound imaging performed during a breastfeed with a nipple shield at a corrected gestational age of 35<sup>2</sup>



weeks (33 to 39<sup>th</sup>) and a bottle-feed at 36<sup>th</sup> weeks (25<sup>th</sup> to 40). Measurements were made of nipple/teat diameter (N/TD) at 2, 5, 10 and 15 mm from the nipple/teat tip in the maximum tongue up and tongue down positions.

**Results:** For both teat and breast, smallest N/TDs were at 2 mm ( $P < 0.001$ ) and largest at 15 mm ( $P < 0.001$ ). Movement between tongue up and tongue down was significant ( $P < 0.001$ ), and was the same at all N/TDs. Movement was 0.7 mm greater for the breast and varied between suck cycles within a feed ( $P < 0.001$ ).

**Conclusions:** The pattern of tongue movement employed during feeding from the novel teat was similar to that of preterm and term breastfeeding. Differences in degree of movement may be due to the use of nipple shields. Inconsistent sucking patterns within a feed are commensurate with the immaturity of the infant.

We acknowledge the support from Medela AG.

### P350

#### THE USE OF PASTEURISED DONOR BREASTMILK

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**Background:** Pasteurised donor breastmilk (PDM) was introduced into Mercy Hospital for Women Neonatal Intensive Care Unit (MHW NICU) in July 2011. Infant eligibility for PDM when mothers' own breastmilk insufficient/unavailable is: born before 32 weeks gestation or birthweight below 1500 grams or recovering from necrotising enterocolitis or abdominal surgery. PDM is continued to the later of 34 weeks post menstrual age (PMA) or 1500 grams.

**Method:** Clinical audit of infants who received PDM in the 12 months to July 2012.

**Results:** 60 infants received PDM. One eligible family refused. (253 were admitted <1500 grams birthweight in the time period). 4 infants were 100% PDM fed, all other infants first feed was mother's own breastmilk. No infant died. One had suspected necrotising enterocolitis. One infant had one episode late onset sepsis. 25 had chronic lung disease with 11 still requiring respiratory support at 36 weeks PMA. At discharge from hospital 21 infants were receiving breastmilk, 27 formula and 12 combination feeds. Simple statistics for gestation, birthweight, day of life started, duration and percentage volume of total feeds are tabulated.

|                             | Median | IQR (25–75) |
|-----------------------------|--------|-------------|
| Birth gestation (weeks)     | 28.6   | 26.4–31.2   |
| Birthweight (grams)         | 1207   | 900–1458    |
| PDM first fed (day of life) | 10     | 4–20        |
| Days of PDM                 | 19     | 10–32       |
| Volume PDM/EBM + PDM (%)    | 49     | 27–83       |

**Conclusion:** Use of PDM in MHW NICU was accepted by families and staff as a supplemental feed when mothers own breastmilk was insufficient/unavailable.

### P351

#### ESTABLISHMENT OF A NURSE-LED DIGITAL IMAGING SCREENING PROGRAM FOR RETINOPATHY OF PREMATURITY

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**Background:** A nurse-led digital retinal imaging program to screen for Retinopathy of Prematurity (ROP) was established in 2012.

**Method:** Two neonatal nurses received training in the use of a RetCam™ by a Paediatric Ophthalmologist in premature infants 'at risk' of ROP. The

training program included educational material, in-service education, and supervised practice. Infants <30 weeks' gestation or <1250 g at birth were screened at pre-specified intervals. At least five images were taken from each eye. Image quality was graded by the ophthalmologist as good, adequate or inadequate for assessing ROP, and the ROP stage, zone and plan for review determined.

**Results:** Following the initial training period of three months, 114 infants (birthweight 1015 ± 270 g, gestation 27.9 ± 2.0 weeks) underwent 294 RetCam™ examinations in 6 months. The median corrected gestation of first examination was 31.6 weeks. Of 588 eyes examined, diagnostic-quality views were obtained in 99.1% of image series. The median number of RetCam™ examinations per baby was 3 (range 2–14).

**Conclusions:** Screening for ROP using digital retinal imaging can be successfully carried out by neonatal nurses following an education program. This model has the potential to become an outreach program in special care nurseries, which may facilitate moving infants to facilities closer to their home.

### P352

#### ANALYSIS OF NEONATAL RETRIEVALS IN NORTH QUEENSLAND OVER THE PERIOD OF 10 YEARS

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**Introduction:** The Townsville neonatal retrieval team transports all the sick newborns from North Queensland to regional Neonatal Unit in Townsville. We analysed the retrievals done by this unit over a 10 year period.

**Method:** Two authors independently verified the Neodata focusing on referring hospital, patient details, and duration of retrievals.

**Results:** During the 10 years, there were 750 retrievals (annual average of 75). There were 438 (58.4%) male babies and 312 (42.6%) female babies. 469 (62.5%) of the newborns were Caucasians, 220 (29.3%) Aboriginal and Torres Strait Islanders (ATSI), 14 (1.8%) Asians and 36 (4.8%) others. The average gestational age was 34.2 week (range 22.6 weeks to 41.6 weeks). 115 (15.4%) babies were <28 weeks, 344 (46%) 28 to 37 weeks and 289 (38.6%) >37 weeks gestation. The mean birth weight (BW) was 2374 g (range 420 g to 4830 g). The number of babies with BW ≤ 1000 g were 94 (12.7%), 1001 g to 1500 g were 110 (14.9%), 1501 g to 2500 g were 188 (25.4%) and 349 (47.2%) were >2500 g. The origin of the babies retrieved were 145 (19.3%) from Cairns, 131 (17.5%) Mackay, 121 (16.1%) Townsville Mater, 103 (13.7%), Mount Isa, 30 (4%) from Thursday Island and the rest from other areas. The mode of retrievals were 474 fixed wing craft (63.9%), 172 (23.2%) road and 96 (12.9%) helicopter. The total duration of the retrieval varied between 0.33 hours and 18.8 hours with the mean of 5.12 hours.

**Conclusion:** This first study of neonatal retrievals in North Queensland revealed a higher rate of ATSI babies, higher retrievals by fixed wing and longer retrieval times.

### P353

#### RISK FACTORS AND TREATMENT OF CATHETER-RELATED THROMBOSIS IN NEWBORN INFANTS: A 12 YEAR REVIEW

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**Background:** Central venous (CVL) and arterial catheters (AC) are used frequently in the intensive care management of newborn infants, but are the leading cause of thrombosis in infants. Improved understanding of line-related risk factors for thrombosis would guide better use of these essential lines.

**Method:** Retrospective cohort review of newborn infants managed at our institution between 2000–12, who had a central venous or arterial (central

or peripheral) catheter. Thrombosis number, site, treatment and potential risk factors were determined.

**Results:** 2709 catheters were inserted in 1580 patients. Fifty-nine catheters (2%) were associated with ultrasound-confirmed thrombus. Eleven (19%) thrombi were associated with AC; 1 radial, 2 brachial, 2 UAC and 6 femoral. 46 (81%) thrombi were associated with CVLs; 15 femoral, 11 iliac, 8 IJV, 4 IVC, 3 subclavian, 3 saphenous and 2 portal vein. Thrombosis was significantly associated with lower birth weight, lower gestational age and older age at line insertion (all  $P \leq 0.05$ ), Table 1. There was no association with duration of line insertion. 17% of infants had a blood-culture positive sepsis, and 33% received red cell transfusion in the week preceding thrombus detection. Unfractionated heparin treatment was commenced in 31 (53%) cases. Low molecular weight heparin was continued at discharge in 19 (32%) of infants with thrombi. No infant was diagnosed with a thrombophilia prior to discharge.

**Conclusions:** Thrombosis remains a significant complication of CVL and ACs. Vigilant clot surveillance should be exercised, especially in infants of lower gestational age and weight. Sepsis and transfusion may be important risk factors.

**Table 1.** Patient characteristics and incidence of catheter-related thrombus

| Patient characteristic,      | No Thrombus<br>Group<br>mean (SD) | Thrombus<br>Group<br>mean (SD) | P<br>value |
|------------------------------|-----------------------------------|--------------------------------|------------|
| Birth weight (kg)            | 2.4 (1.1)                         | 2.1 (1.1)                      | 0.05       |
| Gestational age (weeks)      | 34.8 (5.0)                        | 33.3 (5.2)                     | 0.02       |
| Age at line insertion (days) | 18 (25)                           | 26 (31)                        | 0.01       |
| Duration of line insertion   | 11 (15)                           | 10 (17)                        | 0.45       |

#### P355

##### NEONATAL FALLS

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**Background:** In-Hospital neonatal falls occur, however literature is lacking. Rates of neonatal falls vary from 1.6–4.14/10,000 live births. Our study aims to review our rate of in-hospital neonatal falls.

**Method:** Incidents of neonatal falls for any reason from June 2010 to June 2012 were audited using the hospital incident system to look for potential risk factors and common features.

**Results:** We identified six incidents; five being term infants and a 23-week gestation baby, giving a rate of 16.8/10,000 live births. Five of the incidents were in the post-natal ward, with three in the first day and three occurring on day-two post-delivery. Demographic features included four out of six maternal smoking, three NZ Maori ethnicity, five were breastfeeding in bed without a support person, five occurred in the early morning, and five featured maternal opioids and/or surgical delivery.

**Conclusions:** Of particular concern is the high rate of falls during breastfeeding in bed without a support person present, following a surgical delivery especially in the early morning when mothers are vulnerable due to lack of sleep. These are potentially modifiable risk factors and should be highlighted to staff caring for these infants and mothers.

#### P356

##### POSTPRANDIAL OXYGEN CONSUMPTION IN NEWBORN INFANTS USING INDIRECT CALORIMETRY

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**Background:** Commercially available metabolic monitors have been used to measure oxygen consumption in newborn infants. In this study we

describe the development and validation of a custom built monitor to measure oxygen consumption in spontaneously breathing term neonates, and apply this to the study of metabolic responses to feeding in babies. **Methods:** Oxygen consumption (VO<sub>2</sub>) and respiratory exchange ratio (RER) were measured continuously in term infants between two consecutive feeds using open circuit indirect calorimetry. Measurements were commenced 10 min after feeding and continued until the baby demanded for the next feed.

**Results:** Ten healthy full term infants of average weight for gestation were enrolled in the study. The median (range) of duration of gas analyses (VO<sub>2</sub> and RER measurements) was 128 (90–180) min. The mean (SD) of all VO<sub>2</sub> for the cohort was 7.1 (0.65) mL/kg/min. Discrete ten minute average values of VO<sub>2</sub> showed the highest VO<sub>2</sub> values within the first 60 minutes post feed, and a mean (SD) fall in VO<sub>2</sub> of 15.4% (5.1) from the immediate post-feed period to before the next feed. The mean change in VO<sub>2</sub> per 10 minutes was  $\beta = -0.06$  mL/kg/min (95% CI =  $-0.8, -0.04$ ,  $p < 0.001$ ). The respiratory exchange ratio showed no obvious change during the period between feeds.

**Conclusion:** This study has described a simple and reliable technique of measuring oxygen consumption using custom built equipment. Our findings indicate that a rise in VO<sub>2</sub> is observed within 60 min following feeding and this gradually returns to pre-feed levels.

#### P357

##### CARING FOR THE NEONATAL SKIN – CHALLENGES ENCOUNTERED DURING DEVELOPMENT AND IMPLEMENTATION OF A NEONATAL SKIN CARE GUIDELINE AT A TERTIARY NEONATAL NURSERY

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**Background:** Skin breakdown and infection can lead to significant morbidity or even mortality in newborns, especially in extremely preterm infants. We aimed to develop an evidence based skin care guideline for our neonatal nursery to address routine skin care as well as treatment of skin breakdown; we also wanted to develop quantitative measures for monitoring of skin integrity of our highest risk patients and the treatments used.

**Method:** Literature was reviewed and a draft guideline developed with input from medical and nursing staff. The new guideline was circulated and several presentations given to staff. 'Skin rounds' by the investigator were introduced to educate staff, monitor progress more formally. After a transition period, monitoring of the quantitative measures was begun.

**Results:** While publications about adult wound care are numerous, published evidence on which to base neonatal skin care was limited. Multiple opinions existed at our nursery. The initial introduction of the guideline and related documentation failed. Therefore, a second phase using 'skin rounds' by the investigator was started; with the second phase, progress was monitored and assessed in a more systematic manner. With over two years, the time needed for development and introduction of the guideline exceeded the expected.

**Conclusions:** Many difficulties were encountered during the development and introduction of a skin care guideline: Evidence to base care on was limited, many individual preferences for treatment existed, a large group of staff had to be educated, consistency in documentation was difficult to achieve and measures for quantitative monitoring of care had to be developed.

P359

# VARIATION IN VANCOMYCIN DOSING AND MONITORING ACROSS NEONATAL UNITS OF THE ANZ NEONATAL NETWORK

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**Background:** Following evidence of increasing resistance and treatment failure in adults linked to vancomycin troughs of <10 mg/L, the Therapeutic Guidelines (Antibiotic 2012) recommends a target vancomycin trough level of 12–18 mg/L for neonates. This is in line with The Infectious Diseases Society of America recommendations. However, there is wide variation in recommended doses and target trough levels amongst neonatal dosing references. This study investigated vancomycin dosing practices in Australian and New Zealand NICUs.

**Method:** Telephone survey of all tertiary ANZNN nurseries, including questions about vancomycin dosing and monitoring.

**Results:** 22 of 28 (79%) nurseries responded. Dosing regimens were based on six different rationales; using various combinations of postmenstrual age, chronological age, weight, serum creatinine, and mg per kg dosing. For babies of 24 and 40 weeks gestational age, eight and five different dosing regimens were applicable. Total daily dose ranged from 10 to 30 mg/kg/day and 18 to 50 mg/kg/day respectively. All nurseries used trough levels for monitoring, seven units (32%) used both peak and trough levels. Timing of trough levels varied; prior to the third dose was the most common (9 nurseries), with prior to the second, fourth and sixth and at 48 hours also being used. Nine different ranges of vancomycin trough levels were reported (range 0 to 20 mg/L), with only 13 (45%) of nurseries targeting levels >10 mg/L.

**Conclusions:** There is considerable variation in vancomycin dosing and monitoring practices. More evidence is needed to inform appropriate treatment guidelines for neonates.

P360

# CONTINUOUS VERSUS INTERMITTENT VANCOMYCIN FOR THE TREATMENT OF SUSPECTED NEONATAL SEPSIS

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**Background:** A recently developed continuous vancomycin infusion guideline maintained target concentrations more effectively than intermittent vancomycin infusions in neonates.[1]

**Aims:** To evaluate the guideline in a population of infants with anticipated lower mean corrected gestation and weight.

**Methods:** All infants commenced on vancomycin therapy in the neonatal unit at the Southern General Hospital, Glasgow, between January and July 2012 were included. During the first two months intermittent vancomycin infusions (IVanc) of 10, 15 or 20 mg/Kg (according to creatinine concentration), were given every 8 to 12 hours. Doses were adjusted to maintain troughs of 12–15 mg/L. Subsequently IVanc was replaced with continuous vancomycin infusions (CVanc). A loading dose of 15 mg/Kg was followed by continuous infusion of 20–50 mg/Kg/day, according to corrected gestational age and creatinine concentration. Doses were adjusted to maintain concentrations of 15–25 mg/L.

**Results:** 29 IVanc and 23 CVanc courses were completed. Mean corrected gestational age and weight were lower in the current population than in the previous study (IVanc 33.0 ± 3.9 wks, CVanc 29.9 ± 3.8 Kg versus IVanc 37.6 ± 4.8 wks, CVanc 38.7 ± 2.7 Kg). The proportion of levels within the target range increased from 36% with IVanc to 74% with CVanc (p < 0.001). These results were comparable to the previous study, which reported an increase from 34% to 83% after switching from IVanc to CVanc.

**Conclusions:** Continuous vancomycin infusion maintains target concentrations more effectively than intermittent infusions in neonates across a range of gestational ages and weights.

## Reference

1. Patel AD, Anand D, Lucas C, et al. Intermittent versus continuous infusion of vancomycin in neonates. *Arch Dis Child* 2012;97:e20.

P361

# IMMUNISATION OF PRETERM INFANTS IN THE NEONATAL UNIT

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**Background:** Preterm infants are at increased risk of morbidity from vaccine preventable diseases. Vaccinations are immunogenic in preterm infants. We aimed to assess adherence to national immunisation guidelines for hospitalised preterm infants, and the incidence of immunisation related adverse events.

**Method:** Retrospective audit of preterm infants who were inpatients when eligible for immunisation, with Prevenar, Infanrix hexa, and RotaTaq on day 60 of life. The study was conducted in Newborn Intensive and Special Care (NISC) at the Royal Women's Hospital, Melbourne from Jan 2011 to July 2012. Physiological parameters and level of support during the 48 hours pre and post immunisation were compared.

**Results:** 126 of 131 eligible infants (96%) were immunised in NISC; mean (SD) gestational age (GA) was 26.5 (1.9) w and birthweight 923 (291) g, corrected GA 35.6 (2.3) w and weight 2323 (504) g at immunisation. 93 (74%) were immunised on time, and 33 (26%) late.

There was an increase of at least 5 apnoeic/bradycardic events per day in 13% of infants, an increase in respiratory support in 21%, and an increased rate of fever (>37.5) in 24% in the 48 hours after vaccination compared with the preceding 48 hours.

**Conclusions:** Compliance with the national immunisation guidelines for eligible inpatient preterm infants in NISC was excellent (96%), with 74% immunised at 2 months postnatal age. There was a significant risk of increased apnoea/bradycardic events and respiratory support after vaccination. These may be considered when determining timing of immunisation, including appropriate counseling for parents.

P362

# TWO YEAR NEURODEVELOPMENTAL OUTCOMES OF NEONATES WITH HYPOXIC ISCHEMIC ENCEPHALOPATHY TREATED WITH THERAPEUTIC HYPOTHERMIA USING COOL GEL PACKS – AN EXPERIENCE FROM WESTERN AUSTRALIA

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**Background:** Therapeutic hypothermia is the standard clinical practice for neonates with moderate to severe hypoxic ischaemic encephalopathy (HIE).

**Aim:** To describe the two year neurodevelopmental outcomes of neonates who were routinely cooled for HIE in Western Australia.

**Methods:** Retrospective study. Cases were identified from the neonatal databases. Information was collected from chart review.

**Results:** 65 infants received therapeutic hypothermia, of which 13 had mild, 35 moderate and 17 had severe HIE. The mean rectal temperature during the entire cooling period was 34.1°C (SD 0.68); Target rectal temperatures (33.5 to 34.5) were achieved in 70% instances. Rectal temperatures were above 34.5°C in 18% and below 33.5°C in 12% instances. There were no serious adverse effects attributable to cooling. All 13 infants with



mild HIE survived, of whom developmental outcomes were available on nine; none had severe disability. Among 52 infants with moderate to severe HIE, there were nine deaths (17%) and developmental outcomes were available on 39; the incidence of severe disability was 23%. The risk of death or severe disability was 40% in infants with moderate to severe HIE. Physical growth was adequate at two years of age.

**Conclusions:** Neonates undergoing therapeutic hypothermia with cool gel packs had both good survival rates and long term neurodevelopmental outcomes and met international benchmarks.

### P363

#### DIFFERENCES IN MORTALITY AND MORBIDITY WITH A COMPLETE COURSE OF ANTENATAL STEROIDS COMPARED TO INCOMPLETE/NO ANTENATAL STEROIDS IN EXTREMELY PREMATURE NEONATES

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**Background/aims:** Antenatal steroids have been shown to reduce mortality and short term morbidity in neonates born less than 28 weeks gestation. The aims of the study were to determine the differences in survival, short and long term morbidity for those neonates receiving no antenatal steroids or an incomplete course of steroids in comparison to those receiving a complete course.

**Methods:** A retrospective review of prospectively collected data from 10 NICU'S between January 1998 to December 2004.

**Results:** 2549 neonates were included in the study with 319 (12.5%) not given any antenatal steroids. Hospital mortality was significantly worse without steroids (30% vs 20%;  $p < 0.001$ ). Those with no steroid cover were more likely to have NEC (11% vs 7%;  $p = 0.018$ ) and Grade 3 or 4 IVH (19% vs 12%;  $p = 0.001$ ). Factors predictive of mortality included: lack of antenatal steroids, male gender, lower gestation and hypertensive disease of pregnancy. Long term data was available for 1473 survivors with no difference in neurological outcome between those not receiving steroids and those receiving any steroids.

**Conclusions:** Antenatal steroids confers a survival benefit in neonates born <29 weeks gestation and reduces short term morbidities of NEC and severe IVH, but does not have an impact on long term neurodevelopment.

### P364

#### PLACENTAL MASS IN UNEXPLAINED STILLBIRTH

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**Background:** The rate of stillbirth in Australia is 7.8 per 1000 births. Excluding major congenital abnormalities, the largest single diagnostic group is unexplained stillbirth, accounting for 16% of all stillbirths in normally-formed fetuses. This study sought to identify whether low placental mass might be associated unexplained stillbirth.

**Method:** We examined the database of all singleton fetal deaths submitted to ACT Pathology during the period 2005 to 2010. Data from 140 singleton stillbirths were divided into three groups: 24 cases (17.1%) with fetal growth restriction; 87 cases (62.1%) without growth restriction, and where a specific cause for the fetal death was established; and, 29 cases (20.7%) in the 'unexplained' group. From the 87 'explained' stillbirths, placental weight was not available in 15 cases, leaving 72 for analysis.

**Results:** In 55% of 'unexplained' stillbirth, the placental weight was below the 10<sup>th</sup> centile, compared to 83.3% where fetal growth restriction was identified (OR 0.25, 95% CI 0.05–1.04). For the 'explained' stillbirths where growth restriction was absent, the proportion of placentas below the 10<sup>th</sup> centile was 33.3% (OR 2.46, 95% CI 0.94–6.53).

**Conclusions:** Placental volume below the 10<sup>th</sup> centile is common in unexplained stillbirth. We encourage other groups to examine their own data, and if the trend to small placental volume is found to be common in unexplained stillbirth, it may well suggest that further evaluation of ultrasound estimate of placental volume using 3D/4D techniques might allow development, for the first time, of a screening test to identify otherwise normal pregnancies at risk of intrauterine death.

### P365

#### NEURODEVELOPMENTAL OUTCOMES OF CHILDREN BORN TO MOTHERS WITH INTRAUTERINE CONDITIONS (NOCIC): INTRAUTERINE GROWTH RESTRICTION

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**Background:** Intrauterine growth restriction (IUGR) can have a negative effect on the physical and mental development of newborn babies, with substantial perinatal mortality and morbidity. We aimed to review and determine pregnancy- and birth-related factors that may influence the neurodevelopmental outcome in IUGR babies.

**Method:** Children born >32 weeks with estimated foetal and birth weights <5<sup>th</sup> percentile for gestational age in 2007–2009, who had received care prenatally at a teaching hospital in South-Western Sydney, were identified. Parents were invited to fill out a semi-structured questionnaire to evaluate the child's demographics, and behavioural and developmental progress.

**Results:** Questionnaires were received on 8 children with a mean age of 3.78 years. Abnormal Doppler features were found in 6 children (75%). Two children had poor Apgar scores (25%) and 3 children had perinatal complications (37.5%). Two children had borderline scores (25%) in the Child Behaviour Checklist and 1 child had a borderline fine motor score (12.5%) on the Child Development Chart. Amongst these 3 children, older maternal age, smaller gestational age at birth and lower birth weight were observed, compared to other children without problems. Two of these 3 children were attending speech therapy.

**Conclusions:** Despite the small sample size, our study still advocates further research into identifying factors to improve neurodevelopmental outcomes in IUGR fetuses. When compared with existing evidence in the literature, later gestational age and higher birth weight appear to be associated with better neurodevelopmental outcomes, which may have implications for perinatal care and management of the timing of delivery.

### P366

#### PLACENTAL NEUROSTEROIDOGENESIS AND OUTCOME FOLLOWING PRETERM BIRTH IN GUINEA PIGS

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**Background:** The placenta has a key role in the production of protective neurosteroids such as allopregnanolone. Chronic insults raise the expression of enzymes 5 $\alpha$  reductase (5 $\alpha$ R) type 1 and 2, and allopregnanolone production, potentially a protective mechanism. Our objective was to compare placental and brain neurosteroidogenic capacity, by assessing 5 $\alpha$ R1 and 2 expressions, with neonatal outcome following preterm birth. **Method:** Guinea pigs were delivered by caesarean section at term (GA69;  $n = 12$ ) and preterm (GA62;  $n = 12$ ) and maintained for 24 hours. Preterm

pups that did not survive this period formed a preterm non-survival group ( $n = 12$ ). Placentas and brains were analysed for 5 $\alpha$ R1 and 2 mRNA expression by real-time PCR. Neonatal wellbeing scores were based on respiratory, behavior and posture factors.

**Results:** Placental 5 $\alpha$ R2 expression was significantly higher in preterm neonates than term ( $P < 0.001$ ) and in non-survivors compared to surviving preterm neonates ( $P = 0.04$ ). Placental 5 $\alpha$ R1 expression was greater in non-survivors than those at term ( $P = 0.04$ ). Brain 5 $\alpha$ R1 mRNA expression showed no differences with gestational age or survival. There were no differences in body, brain or placenta weights between the preterm survivors and non-survivors however wellbeing scores were markedly lower in the non-survival group ( $P < 0.01$ ).

**Conclusions:** The finding that 5 $\alpha$ Rs expression was greater in the placenta of preterms support the essential role of the placenta in the provision of neurosteroids prior to term. The observation that 5 $\alpha$ R2 expression was greatest in non-survivors suggests there was pre-existing compromise in these neonates prior to birth. Further investigation is required to determine if rising placental neurosteroid output is an early marker of compromise and potential poor outcome.

P367

### PUBLICLY-FUNDED HOMEBIRTH IN AUSTRALIA: OUTCOMES OVER 5 YEARS

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**Background:** Since 1996, 14 publicly-funded homebirth programs have been established in Australia in all states and territories, except ACT, Queensland and Tasmania. These programs provide low risk women with the option of giving birth at home with midwives who are employed through the public health system. We aimed to collate maternal and neonatal outcomes from these programs.

**Method:** We requested data from 2005–2010 (or from commencement of the program to 2010) from the 13 publicly-funded homebirth programs currently in place at the time of the study. A descriptive analysis was undertaken.

**Results:** Data were provided by nine publicly-funded homebirth programs. Of the 1807 women who intended to give birth at home at the onset of labour, 1506 (83%) did so. A further 315 (17%) transferred to hospital during labour or within one week postnatally. The Perinatal Mortality Rate was 3.3 per 1000 births, and 1.1 per 1000 births when excluding deaths because of expected congenital anomalies. The normal vaginal birth rate was 90%.

**Conclusions:** This study evaluates a substantial proportion of women choosing to have a publicly-funded homebirth in Australia. However, the sample size does not have sufficient power to determine safety. More research is needed on the safety of different birthplaces within Australia.

P368

### A TWO-CENTER RANDOMISED CONTROLLED TRIAL OF OUTPATIENT CERVICAL PRIMING FOR INDUCTION OF LABOUR: A QUALITATIVE EXAMINATION CONFIRMING PSYCHOSOCIAL BENEFIT

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**Background:** This study compares women's experiences with outpatient priming and inpatient priming for induction of labour. The hypothesis was that outpatient priming would improve maternal psychosocial outcomes.

**Method:** A randomised trial was conducted at two tertiary referral hospitals in Adelaide. Women completed a questionnaire at 7-weeks postnatal, which included a space for written comments about the positive and negative aspects of the priming experience. These were independently coded by two assessors not involved in the trial. This study reports on a content analysis of responses from women who received their allocated intervention.

**Results:** 1004 eligible women were approached; 85% consented. About half of women in both groups received their allocated type of care. The questionnaire response rate for both groups was 76%. Approximately 67% of women who experienced priming gave comments (140 outpatient women and 127 inpatient women). Those who received outpatient priming gave more positive comments (mean: 2.14 (SD: 1.98), compared with 1.66 (SD: 1.56); mean diff: 0.47; 95% CI: 0.04 to 0.90) and fewer negative comments (mean 1.55 (SD: 1.82), compared with 2.15 (SD: 2.42); mean diff: -0.60; 95% CI: -1.11 to -0.08). The majority of positive comments (60.2%) were written by women receiving outpatient priming. Four main thematic areas were identified: satisfaction with priming; satisfaction with care overall; readiness for labour; experience with priming, labour and induction.

**Conclusions:** While the differences between the two groups are not large, the results confirm the findings of our quantitative examination indicating that outpatient priming can confer psychosocial benefit.

P369

### A MULTIDISCIPLINARY APPROACH TO MANAGEMENT OF A QUINTUPLET PREGNANCY

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**Background:** Higher order multiple pregnancies present significant challenges to the health services providing care. Here we describe care processes for and outcomes of a spontaneous quintuplet pregnancy in a 47 year old woman with 12 existing children. We highlight the counselling, resource, and planning challenges and describe how a multidisciplinary approach ensured woman centered care and optimal outcomes.

**Method:** From referral at 12 weeks of pregnancy, care was provided through a maternal-fetal medicine service by a consultant obstetrician and a lead senior midwife. From the 20<sup>th</sup> week of the pregnancy regular multidisciplinary meetings involving Maternal Fetal Medicine medical and midwifery staff, neonatologists, neonatal nurses, anaesthetists, theatre nurses and social workers were held to plan for an elective caesarean section at around 32 weeks gestation and to ensure that the needs of both mother and babies were met.

**Results:** Serial fetal surveillance identified severe growth restriction in one fetus at 26 weeks gestation. With the outcomes of the other four fetuses in mind, the mother chose to continue the pregnancy in the knowledge of the likely loss of the growth restricted fetus. The mother was admitted to hospital from 24 weeks due to an inability to function at home. At 30 weeks gestation the demise of the growth restricted fetus was confirmed. At 31 weeks and 2 days a team of 32 health professionals were involved in an elective lower segment caesarean section delivering 4 live infants in good condition and one stillborn baby. The mother and surviving infants made uncomplicated recoveries.

**Conclusions:** While very resource intensive, a multidisciplinary approach to care of women with higher order multiple pregnancy can achieve excellent outcomes.

## P370

## STRATEGIES ON IMPROVING HEALTH CARE SEEKING FOR MATERNAL AND NEWBORN ILLNESSES IN DEVELOPING COUNTRIES: A SYSTEMATIC REVIEW

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**Background:** Maternal and child deaths represent two of the most difficult to achieve targets among the Millennium Development Goals (MDGs). Achieving the MDG 4 and 5 targets of reducing maternal and newborn mortality will require rigorous efforts to scale up evidence-based interventions, especially community-based preventive, promotive and therapeutic strategies as well as availability of commodities and health workers in primary care. In this review we will systematically review the literature to determine the interventions that can improve the care seeking pattern among mother and their newborns in developing countries.

**Method:** We will include intervention studies, prospective cohort studies and qualitative studies that address health care seeking in pregnant women at any gestation age, postpartum women up to 6 weeks after giving birth and neonates less than 28 days of age residing in developing countries. Two review authors will independently assess study quality and extract data. Results will be synthesized by study type.

**Results:** Results will be available by mid 2013. Search strategy identified 20672 papers, of which, 187 has been identified for full text review.

**Conclusion/policy relevance:** A lack of appropriate care seeking for ill mothers and neonates is thought to contribute to high mortality rates; therefore, a major challenge is the appropriate mix of strategies for demand creation as well as provision of services. This review will synthesize the evidence on strategies that can improve health care seeking of mothers and neonates that may lead to improvements in maternal and newborn health will therefore be highly relevant for both policy and practice.

## P371

## CHANGING KNOWLEDGE, USE AND VIEWS OF ANTENATAL MAGNESIUM SULPHATE FOR NEUROPROTECTION OF THE FETUS (2011–12)

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**Background:** Bi-national clinical practice guidelines recommend administration of antenatal magnesium sulphate to women at risk of early preterm (<30 weeks), imminent birth, to reduce the risk of babies dying or having cerebral palsy.<sup>1</sup> The aims of this qualitative study were to explore health professionals' changing awareness, views and use of these guidelines, and to identify strategies for improving implementation.

**Method:** Face-to-face interviews were conducted with randomly selected health professionals (obstetric, neonatal, midwifery staff) at the Women's and Children's Hospital, Adelaide in 2011 and 2012 (24 and 19 respectively). Interviewees were asked a range of structured questions regarding awareness and use of the magnesium sulphate clinical practice guidelines, and perceived factors influencing the uptake of this neuroprotective therapy.

**Results:** The majority of staff in 2011 and 2012 were aware of the guidelines prior to interview (79% and 84% respectively). In 2012, staff were more likely to report prescribing/advising/administering magnesium sulphate 'routinely' or 'sometimes' (2012: 84%; 2011: 46%), and were less likely to report 'never' prescribing/advising/administering (2012: 5%; 2011: 42%). In 2012 interdisciplinary collaboration and communication were identified as key facilitators to implementation. Predicting time of

birth and having sufficient time to administer the treatment remained important perceived barriers.

**Conclusions:** Increasing awareness and self-reported use of antenatal magnesium sulphate for fetal neuroprotection have been observed. The WISH Project will continue to assess whether improved knowledge and positive views are reflected in clinical practice.

## Reference

1. Antenatal magnesium sulphate prior to preterm birth. Adelaide: The University of Adelaide, 2010. Online at: <http://www.adelaide.edu.au/arch/MagnesiumSulphate2010.pdf>

## P372

## A RETROSPECTIVE AUDIT OF COAGULATION SCREENING AND FRESH FROZEN PLASMA TRANSFUSIONS IN PRETERM INFANTS LESS THAN 29 WEEKS

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**Background:** There is paucity of data regarding the normal coagulation parameters for extremely preterm infants and also lack of consensus on the issues of coagulation screening and use of fresh frozen plasma transfusions. This study aimed to derive coagulation data for extremely preterm infants and to study clinical practices regarding coagulation screening and FFP transfusions within a tertiary neonatal unit.

**Methods:** Retrospective data collection including perinatal and neonatal variables affecting coagulation system, coagulation test results, clinical bleeding episodes and use of FFP. Statistical tests applied for difference included two sample t – test for parametric data and Mann whitney test as non parametric test. The p value of 0.05 was considered significant.

**Results:** 81.6% infants in the cohort had a coagulation screen in the first five days of life. The majority of tests were done within 6 hours of age. The threshold for transfusion of FFP appeared to be INR of 1.5–1.6. The APTT values appeared to transiently improve after 6 hours of age. 48% infants who had coagulation screen received FFP transfusion and the median dose of FFP was 15 ml/kg.

**Conclusions:** There are no guidelines for screening coagulopathy and transfusions in extremely preterm infants and there is a need for consensus based guidelines. This audit is a reflection of practice.

## P373

## MANAGEMENT OF CENTRAL VENOUS CATHETER PRACTICES IN TERTIARY NEONATAL UNITS, ACROSS AUSTRALIA AND NEW ZEALAND

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**Background:** The CDC guideline for the prevention of intravascular catheter-related infection described evidence-based practice to reduce nosocomial infection. Our aim was to survey current medical and nursing management of central venous catheters (CVCs) in tertiary NICUs in Australia and New Zealand.

**Methods:** A prospective, cross sectional survey was performed across NICUs of the ANZNN (n = 27) in September 2012. Two types of web-based questionnaires were distributed. CVC insertion surveys were directed to medical directors while CVC 'maintenance' surveys were sent to nurse unit managers.

**Results:** 74% (20/27) medical management and 59% (16/27) on nursing management surveys were completed. Every NICU has a guideline on CVC maintenance and all except one for insertion. In the units using femoral lines, 42.9% have a guideline on insertion and 50% for maintenance. Insertion is restricted to credentialed staff in 60%. Only 31.3% use full maximum sterile barrier for insertion. Skin preparation practices widely



vary. Dressing use and dressing change regimen is standardised; all using a semi-permeable dressing. Duration of cleaning time of the access point varies significantly, however the majority use a chlorhexidine with alcohol solution (68.8%). Line and fluid changes vary from daily to 96 hourly. The majority use sterile gloves and a sterile dressing pack to access the CVC (68.8%). Units favour using stopcocks (62.5%) with a needle-less access point attached (85%). Education is provided on insertion and maintenance in less than half of the units.

**Conclusion:** There is some diversity of current practices and some aspects lack adherence to the evidence-based guidelines.

### P374

#### RISK FACTORS AND MANAGEMENT OF PNEUMOTHORAX IN LATE PRETERM INFANTS IN A SECONDARY UNIT. UNMATCHED CASE CONTROL STUDY

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**Background:** Neonatal pneumothorax (Ptx) is associated with significant respiratory morbidity in late preterm infants requiring uptransfer to tertiary hospitals. Positive pressure ventilation is associated with a higher risk of pneumothorax in preterm infants.

**Method:** We performed a case control study of Ptx in late preterm infants >32 weeks of gestation at birth (n = 28) and their unmatched controls (n = 26) over a time period of 2 years (June 2010–June 2012). Data on various risk factors such as gestational age, birth weight, mode of delivery, resuscitation at delivery, need for respiratory support in the form of CPAP were compared between the two groups. Logistic regression analysis was used to compute statistical significance.

**Results:** Any resuscitation and treatment with CPAP were associated with Ptx both in univariate and multivariable logistic regression analysis. Gestational age, birthweight, caesarian birth were not associated with Ptx.

|                                     | Univariate<br>analysis<br>OR (95% CI),<br>p value | Multivariable<br>regression<br>OR (95% CI),<br>p value |
|-------------------------------------|---------------------------------------------------|--------------------------------------------------------|
| CPAP                                | 4.75 (1.50–15.0),<br>p 0.008                      | 4.54 (1.22–16.8),<br>p 0.024                           |
| Any resuscitation                   | 3.2 (1.03–9.85),<br>p 0.043                       | 2.19 (0.62–7.70),<br>p 0.22                            |
| Caesarian delivery                  | 1.38 (0.47–4.10),<br>p 0.555                      | 1.32 (0.35–4.97),<br>p 0.68                            |
| Birth weight category<br>(>2.5 kgs) | 1.45 (0.81–2.62),<br>p 0.213                      | 2.81 (0.65–12.06),<br>p 0.17                           |

Of the 28 cases, 18 cases were managed at the secondary centre, 10 cases were uptransferred. In total, 20 cases were managed without any intervention, 3 cases required intercostal needling and 5 cases were managed with intercostals catheter insertion.

**Conclusions:** Exposure to positive pressure ventilation is a significant risk factor for Ptx in late preterm infants. Majority of the patients were managed conservatively at the local secondary hospital.

### P375

#### THE DEVELOPMENT OF THE NATIONAL MATERNITY INFORMATION MATRIX

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**Background:** One of the findings of the 2009 review of maternity services, led by the Commonwealth Chief Nurse and Midwifery Officer was a lack of consistent national recording and reporting of maternity-related data. Following publication of this report, the AIHW National Perinatal Epidemiology and Statistics Unit conducted a review of relevant Australian data collections to determine the current depth and breadth of maternal and perinatal health-related data capture in Australia. A total of 45 different data collections were located and reviewed.

**Method:** Using information provided by data custodians, managers and published reports, an index and summary of all available data items pertaining to maternal and perinatal health was developed in 2010. This index was subsequently reviewed and updated in 2011. Information about each data item and the collection it was contained in was recorded in a central database, including descriptive information, definitions, availability and scope of data.

**Results:** The review resulted in the development and publication in 2012 of the Maternity Information Matrix, an online index containing a list and summary of all data items currently collected in Australia pertaining to the health of mothers and babies during pregnancy, birth and the neonatal period as well as maternal and perinatal mortality.

**Conclusions:** The Maternity Information Matrix provides information for researchers, policy makers, data custodians, clinicians and consumers about the range of maternity data currently being collected in Australia and where to locate it.

### P376

#### HYPO- AND HYPERCARBIA IN VENTILATED EXTREMELY PRETERM INFANTS – HOW DO WE RESPOND? AN AUDIT OF CURRENT NICU PRACTICE

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**Background:** Ventilated extremely preterm infants are at risk of hypo- and hypercarbia. Both extremes of arterial carbon dioxide pressure (PaCO<sub>2</sub>) have been associated with poor neurodevelopmental outcome. Close monitoring is therefore required. This audit reviewed treatment responses to occurrence of hypocarbia and hypercarbia.

**Method:** We reviewed arterial blood gas (ABG) results, ventilation charts and clinical notes of 20 extremely preterm babies born in 2010 and ventilated on conventional (CMV) or high frequency oscillatory ventilation (HFOV). Hypocarbia was defined as PaCO<sub>2</sub> ≤ 40 mmHg, hypercarbia as PaCO<sub>2</sub> ≥ 56 mmHg. A treatment response was considered timely when it occurred within one hour, delayed when between 1–3 hours.

**Results:** The cohort's average (SD) gestational age and birth weight was 24.3 weeks (0.6) and 623 g (140 g) respectively. A total of 354 abnormal ABG were analysed: 135 had low PaCO<sub>2</sub> values, 219 had high PaCO<sub>2</sub> values. 244 ABG were sampled from babies on CMV, and 110 on HFOV. Low PaCO<sub>2</sub> values prompted timely ventilatory changes >55% of the time and response rates were significantly better when on HFOV (73% vs. 56%; *p* < 0.05). PaCO<sub>2</sub> ≤ 30 mmHg elicited more rapid responses than abnormal low values closer to the acceptable range (86% vs. 58%; *p* < 0.05). Similarly PaCO<sub>2</sub> > 70 mmHg elicited more rapid responses than high values closer to the acceptable range (85% vs. 54%; *p* < 0.001). Response rates were similar during normal working and after hours.

**Conclusions:** The overall response rate to the extremes of PaCO<sub>2</sub> was satisfactory and better response rates were seen in babies on HFOV.

## P377

# OPTIMISING THE PROVISION OF EXPRESSED BREAST MILK TO PRETERM INFANTS: AN AUDIT OF CLINICAL PRACTICE GUIDELINES FOR WOMEN DELIVERING <32 WEEKS GESTATION

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**Background:** Breast milk reduces the incidence of sepsis, feed intolerance and potential developmental morbidities in preterm infants. Early and frequent expression promotes milk supply. Our current clinical guidelines recommend expression within 6 hours of delivery, and 7 times in the first 24 hours.

**Methods:** A prospective audit of women <32 weeks intending to breast feed following preterm delivery (n = 31). Data are presented as mean (range) or number (%). Chi-squared tests were used to assess frequency data and Pearson's correlations used to assess relationships between continuous data.

**Results:** Time to first expression was <6 hours in 5 women (16%) but none met the recommendation for 7 expressions/24 hours. Time to the first expression was 10.3 (3–28) hours with 3 (0–6) expressions in the first 24 hours and 5 (0–12) in the second. Number of expressions in the first 24 hours was correlated with the number from 24–48 hrs ( $R^2 = 0.45$ ,  $p = 0.01$ ). A greater proportion of women who delivered between 8am–4pm expressed within 12 hours compared to those delivered out with this period ( $\chi^2 = 7.77$ ,  $p = 0.02$ ). No influence on time or frequency of expression was observed for mode of delivery, anaesthetic received or parity.

**Conclusions:** Management of breast milk expression did not follow the current clinical practice standard. While sample size limits interpretation of the current data, factors that influence these processes warrant further investigation prior to implementing alterations in practice to improve provision of expressed breast milk following preterm birth.

## P378

# A STATE-WIDE COLLABORATIVE QUALITY IMPROVEMENT PROGRAMME: DECREASING CENTRAL LINE ASSOCIATED BLOOD STREAM INFECTIONS

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**Background:** Central line associated blood stream infections (CLABSIs) are a well-documented complication of the insertion of central lines in NICUs. The elimination of CLABSIs is a complex problem which requires an understanding of each unit's unique culture. Quality Improvement Collaboratives have become popular internationally and have demonstrated efficacy in reducing CLABSI rates.

**Method:** The group was formed to address the problem of CLABSI in NICUs in NSW. It operates on the principles of evidence based best practice and aims to implement changes in a standardised but unit sensitive manner across NICU's in NSW. We aim to decrease the rates of CLABSI in infants born below 28 weeks gestation. Standardisation of data collected will be used to calculate rates of infection per 1000 central line days in infants <28 weeks restricted to the first 35 days post delivery using Shewart control chart processing methodology. In this way we will close the feedback loop of the QI cycle.

**Results:** This work will describe the process by which the NICUS special interest group was formed and will provide details of critical elements of

the collaboration process including review of infection reduction strategies in each participating NICU, standardization of Peripherally Inserted Central Catheter (PICC) insertion, standardization of data collection and development of infection review processes within each NICU.

**Conclusions:** We have developed a successful model of collaboration between tertiary NICU's in NSW. We aim to decrease CLABSIs using clinical practice improvement principles. We will close the feedback loop with data trending tools which use weighted risk to benchmark state-wide CLABSI rates.

## P379

# REDUCING CENTRAL LINE ASSOCIATED BLOOD STREAM INFECTIONS: AN ACHIEVABLE GOAL

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**Background:** Central line associated blood stream infections (CLABSIs) are a well-documented complication of the insertion of central lines in newborn care units. CLABSIs account for significant short term and long term morbidity and mortality. Including an effect on long term neurodevelopmental outcomes.

**Method:** We conducted a Quality Improvement (QI) project by implementing evidence based guidelines (in the form of a bundle of interventions) in order to reduce the rates of CLABSIs in the NICU. This included implementing stricter definitions of what constitutes a definite infection and a change in the definition of CLABSIs. We studied the results using the CLAB-ICU tally tool to define the rates as number of CLABSIs per 1000 central line days in all babies admitted to the nursery who fitted NICUS admission criteria. We compared two four month periods pre and post introduction of the QI project guidelines.

**Results:** The rate of CLABSIs decreased from 23 per 1000 central line days to 6.9 per 1000 central line days despite the fact that the number of central line days increased markedly.

**Conclusions:** Bundled interventions in the form of quality improvement projects are a valuable and increasingly popular method of introducing many changes at one time. They are a proven tool in sepsis prevention and valuable in raising awareness of important and current issues. The implementation of the project raised the units awareness of the utility of quality improvement exercises to improve the care of our fragile patient population.

## P381

# PCVC INSERTION AND MONITORING: SURVEY OF PRACTICE ACROSS ANZNN

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**Background:** Although Percutaneous Central Venous Catheter (PCVC) has become an integral part of management of premature infants, there are no guidelines on safe practice of PCVC insertion and monitoring.

**Aim:** To determine the practice of PCVC insertion and monitoring amongst neonatologists of Australia and New Zealand.

**Method:** Questionnaires were emailed to neonatologists across 27 tertiary neonatal intensive care units of Australia and New Zealand through survey Monkey website.

**Results:** Responses were obtained from all 27 Units. Sixty seven percent routinely used contrast to delineate PCVC tip. Most (90%) did not monitor thyroid function while the others (10%) monitored occasionally. Seventy seven percent were happy to use intravenous cannula routinely, to insert PCVC and only 14% used guide wire routinely, to negotiate PCVC. Most (88%) were happy to confirm tip on x-ray antero-posterior view alone,

while others (12%) used additional lateral view for lower limb veins. About 46% used ultrasound when x-ray was not helpful in delineating tip. All of them (100%) considered venae cavae to be the ideal position for PCVC tip. Once PCVC is dressed and tip confirmed, majority (81%) would not change dressing routinely, unless soiled. Most (88%) did not do routine check x-ray or ultrasound to monitor for migration. Only 32% were happy to give once daily medication (e.g. caffeine) routinely through PCVC. Most (64%) had never given blood through PCVC while others (36%) had given in desperate situations. About 8% said they would not use infusing pressure range to monitor extravasation.

**Conclusion:** There is a great variation in practice of PCVC insertion and monitoring amongst different units of Australia and New Zealand.

### P382

#### ESTABLISHING A NATIONAL RESEARCH CONSORTIUM FOR STILLBIRTH IN AUSTRALIA

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**Background:** Stillbirth remains a public health problem which can have devastating effects on parents and families, and their care providers. Stillbirth rates have shown little or no improvement in over 20 years and up to one-third are classified as unexplained. A research consortium has been established to address priority research in stillbirth prevention in Australia. The initial study aims to address suboptimal approaches to investigation of stillbirths.

**Method:** An NHMRC funded study forms the platform for this consortium and represents one of the main research initiatives of the Australian and New Zealand Stillbirth Alliance (ANZSA). This prospective study aims to identify the yield from investigations across maternity hospitals following the Perinatal Society of Australia and New Zealand (PSANZ) Clinical Practice Guidelines for Perinatal Mortality (PNM CPG) as part of routine practice. Process data on the establishment of the consortium including recruitment of hospitals, obtaining ethics and governance approval, and development of the database were collected.

**Results:** The consortium consists of 57 collaborators with 30 hospitals recruited. As the study is epidemiological research, applications were submitted to 14 Human Research Ethics Committees, including 2 Aboriginal and Torres Strait committees, and 30 Governance offices, more than would have been required for a clinical trial. A web-based database has been developed to collect the recommended clinical audit data according to the PSANZ PNM CPG and is undergoing formal usability testing.

**Conclusions:** Establishing a national research consortium involves considerable time and resources to comply with regulations of multiple jurisdictions, especially for epidemiological research.

### P383

#### EXPERIENCE WITH SCREENING FOR OBSTRUCTIVE SLEEP APNOEA (OSA) IN PREGNANCY WITH THE RESMED APNEALINK DEVICE – RESULTS OF AN ONGOING PILOT STUDY

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**Background:** Obstructive sleep apnoea (OSA) is characterized by repeated periods of upper airway obstruction with resulting hypoxia and hyper-

capnoea.<sup>1</sup> Potential consequences of chronic OSA include hypertension, metabolic syndrome, and cardiovascular risk. An association between sleep-disordered breathing (SDB), including OSA, and adverse pregnancy outcomes has recently been noted.<sup>2-4</sup> Although the prevalence of OSA in the general female population is between 2 and 5%, there are few data regarding the incidence of OSA in pregnancy. At present, there are no specific guidelines for screening pregnant patients for OSA. Since snoring, daytime fatigue, and non-refreshing sleep are almost universal in late pregnancy, the value of screening questionnaires is questionable. The standard method of diagnosis is spending time in a sleep laboratory (polysomnography), and this is expensive and disruptive. There are also delays in gaining access to sleep laboratories that are incompatible with the short timeframe in pregnancy.

**Methods:** Recently, portable devices designed to screen for OSA have become available on the market. A simple and inexpensive device is the ResMed ApneaLink™ monitor. This device was designed to act as a screening tool to identify patients at risk of OSA. It is a compact device that is easy to use at home, without the need for inpatient sleep study (polysomnography). In addition to flow and snoring, the unit records pulse and pulse oximetry. At the completion of a night's recording, it generates a single-page summary report identifying periods of apnoea and generating a risk profile for OSA. During this ongoing trial, women and their sleep partners are given a standard questionnaire to complete, and are asked to use the ApneaLink™ device for one night during the third trimester. The report is downloaded the following morning and discussed with the women.

**Results:** All participants have found the device easy to use and no technical problems have been encountered. During the initial study period, one woman went on to formal polysomnography and was diagnosed with OSA. She was managed with CPAP for the remainder of the pregnancy, and no adverse outcome was recorded.

**Conclusions:** This ongoing study has revealed that women find the ApneaLink™ home sleep study device acceptable and easy to use. Once this pilot program is complete, we will be seeking partner institutions and funding for a larger formal study.

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### P384

#### RESPIRATORY OUTCOMES FOLLOWING 100 MG/KG VS. 200 MG/KG OF PORACTANT ALPHA: A RETROSPECTIVE REVIEW

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**Background:** The treatment guideline for the management of respiratory distress syndrome at Auckland City Hospital (ACH) newborn unit was amended in July 2010. In keeping with current evidence, the initial dose of Poractant was increased from 100 mg/kg to 200 mg/kg. The outcomes of newborns requiring treatment with surfactant, before and after this change, were reviewed.

**Methods:** Electronic clinical records were reviewed of infants admitted to ACH who received surfactant during the period December 2008 to December 2011. There were two groups: A 100 mg group (Group A) for infants



who received 100 mg/kg of Poractant as an initial dose (Dec 2008 to Jun 2010) and a 200 mg group (Group B), for infants who received 200 mg/kg as an initial dose (Jul 2010 to Dec 2011). Infants with congenital anomalies and those treated with surfactant prior to transfer to ACH were excluded.

**Results:** A total of 256 infants were included in the analysis – 118 in Group A and 138 in Group B. Infants in Group B had a higher median GA (28 vs. 27 weeks;  $p$  0.52) and BW (1065 g vs. 930 g) compared to infants in Group A. Significantly more infants in Group A received more than one dose of surfactant (30.5% vs. 15.9%;  $p$  0.001). Infants in Group B showed a significant reduction in oxygen requirement after the administration of surfactant ( $p$  0.0003).

**Conclusion:** The administration of 200 mg/kg of Poractant led to a significant improvement in oxygenation and a reduction in the need for further doses of surfactant.

### P385

#### ANTENATAL BETAMETHASONE DOES NOT IMPAIR DIAPHRAGM MUSCLE CONTRACTILITY IN PRETERM FETAL SHEEP

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Antenatal maternal glucocorticoids increase fetal surfactant and promote lung maturation. However, glucocorticoids cause skeletal muscle myopathy and impaired diaphragm function in adult subjects. We hypothesise that antenatal glucocorticoid exposure will induce diaphragm muscular atrophy leading to reduced contractile force.

Date-mated Merino ewes received two intra-muscular injections of betamethasone (0.15 mg/kg) or saline (control) at an interval of 24 hours at 2 d, 14 d or 2 + 14 d prior to delivery. Preterm fetal sheep were delivered via caesarean section at 121 d gestation and euthanized. A longitudinal strip of diaphragm muscle was isolated and mounted in an *in vitro* muscle test system. The muscle strip was adjusted to the optimal muscle length (Lo) at which maximum twitch (P<sub>0</sub>) and tetanic (P<sub>0</sub>) forces were recorded. The fatigue resistance (FI) and the susceptibility to stretch-induced damage of the diaphragm muscle were also evaluated.

There were no significant differences observed in body weight, P<sub>0</sub>, P<sub>1</sub>, fatigue, or muscle damage in the betamethasone treated (2 d, 14 d and 2+14 d) fetal animals compared to controls.

|                                     | Control<br>(n = 6) | 2 d<br>(n = 7) | 14 d<br>(n = 8) | 2 + 14 d<br>(n = 8) |
|-------------------------------------|--------------------|----------------|-----------------|---------------------|
| P <sub>0</sub> (N/cm <sup>2</sup> ) | 9.8 ± 0.6          | 11.1 ± 1.0     | 11.4 ± 0.7      | 12.1 ± 0.6          |
| FI                                  | 0.5 ± 0.0          | 0.6 ± 0.0      | 0.5 ± 0.0       | 0.5 ± 0.0           |
| Body weight (Kg)                    | 2.2 ± 0.1          | 2.1 ± 0.1      | 2.2 ± 0.1       | 2.0 ± 0.2           |

Antenatal betamethasone treatment caused no significant difference in preterm fetal diaphragm muscle contractility. Therefore, these data suggest that the beneficial effect of glucocorticoid treatment on lung function is not compromised by alterations in diaphragm contractile function.

**Funding:** NHMRC APP1010665, Viertel SMRF (JJP), WIRF.

### P386

#### CAN CHANGES IN CARDIO-RESPIRATORY PERFORMANCE DURING TIME OFF CONTINUOUS POSITIVE AIRWAY PRESSURE (CPAP) DURING NURSING CARES REFLECT HOW WELL A BABY DOES WHEN THEY CEASE CPAP? AN OBSERVATIONAL PILOT STUDY

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**Background:** A number of babies were observed during time off CPAP for nursing cares to determine how this period is managed, assess their cardio-respiratory status and to determine whether the cardio-respiratory status during this time off is associated with successfully stopping CPAP. **Method:** A sample group was observed and cardio-respiratory status recorded every 30 seconds, during time off CPAP from 1 minute before nasal prongs were removed until 1 minute after prongs were reinserted. **Results:** Most of the 33 patients observed had resolved hyaline membrane disease. The median (range) gestational age at birth was 27<sup>+5</sup> (24<sup>+4</sup> to 30<sup>+0</sup>) weeks. The median (range) post menstrual age at time of test 32<sup>+0</sup> (28<sup>+6</sup> to 38<sup>+6</sup>) weeks. Average time off for cares was 6 minutes. Patients assessed as not ready to cease CPAP did not have a statistically different mean change in heart rate, respiratory rate or oxygen saturations when compared to those assessed as ready to cease CPAP, many had respiratory distress without CPAP. Seventeen babies came off CPAP. Six patients had failed their time off CPAP at 24 hours, with maximum change in heart rate mean (SD) 14.8 (28.2) beats/minute, maximum change in respiratory rate mean (SD) 14 (33) breaths/minute with a maximum change in saturations mean (SD) -6% (12). Five of these had respiratory distress.

**Conclusions:** There is wide variation of cardio-respiratory status during time off CPAP. The observer used respiratory distress and oxygen desaturation as a marker of readiness to stop CPAP. No particular cardio-respiratory features were associated with successfully staying off CPAP.

### P387

#### EFFECT OF GAS FLOW ON HEATED, HUMIDIFIED GASES FOR NEONATAL RESUSCITATION

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**Background:** Dry, cold gas is used for neonatal resuscitation. This cold gas may contribute to low admission temperatures and exacerbate lung injury. We aimed to determine which gas flow rate would provide the fastest provision of heated, humidified gas that could be used for resuscitation.

**Method:** A T-piece resuscitator was connected via a heated patient circuit to a humidifier chamber. An oxygen flowmeter was set to deliver gas at flow rates of 6 L/min, 8 L/min, and 10 L/min. Temperature was recorded at the humidifier chamber (T1), distal temperature probe (T2) and the T-piece (T3) over a 20 minute period at 30 second intervals. Target temperature at T-piece was 36°C, and near 100% humidity assumed when T1-T3 temperature difference was <±1°C.

**Results:** At 8 L/min gas flow, median (interquartile range) time to reach 36°C at the T-piece was 9 minutes (165 seconds) and time to temperature difference (T1-T3 <±1°C) was 4 minutes (30 seconds). A flow rate of 10 L/min reached 36°C at T-piece in 11 minutes (240 secs), and temperature difference (T1-T3 <±1°C) at 5.25 minutes (68 seconds). Time to temperature difference between flow rates was statistically different. 6 L/min flow rate did not achieve target temperatures.

**Conclusions:** There is no significant difference in time to achieve 36°C at the T-piece between any gas flow rates, but near 100% relative humidity is achieved earlier with a flow rate of 8 L/min. We do not recommend flow rates lower than 8 L/min.

P388

# VOLUME OF WATER USED IN HEATED, HUMIDIFIED GASES FOR NEONATAL RESUSCITATION – CLINICALLY MORE EFFICIENT?

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**Background:** Dry, cold gas is used for neonatal resuscitation, and may contribute to lower admission temperature and exacerbation of lung injury. Altering humidifier chamber water volume and gas flow may provide faster provision of heated, humidified gas for resuscitation.

**Method:** A T-piece resuscitator connected via a heated patient circuit to a humidifier chamber (MR290 – autofill chamber) with 108 g or 30 g of water. An oxygen flowmeter was set to deliver gas at 10 L/min. Temperature was recorded at the humidifier chamber (T1), distal temperature probe (T2) and the T-piece (T3) at 30 second intervals for 20 minutes. Target temperature at T-piece was 36°C, and near 100% humidity assumed when T1-T3 temperature difference was  $\leq 1^\circ\text{C}$ .

**Results:** T-piece temperature and near 100% humidity was reached, and remained stable, after 8 minutes using 30 g of water at a flow rate of 10 L/min. Near 100% humidity was assumed at 5.5 minutes using 108 g of water at 10 L/min flow, however T-piece temperature lagged reaching 36°C at 15 minutes.

**Conclusions:** Gas flow at 10 L/min allows provision of heated, humidified gas quickest after set-up, and we recommend this flow rate for neonatal resuscitation. T-piece target temperature is achieved earlier using 30 g of water in the humidifier chamber, while 100% humidity is achieved earlier using the larger volume. Whether warm or humidified gas being of more importance during resuscitation is yet to be determined. To achieve both aims, we recommend using 30 g water in the humidifier chamber and 10 L/min gas flow.

P390

# ATTACHING A TEST LUNG DURING SET-UP OF HEATED, HUMIDIFIED GAS FOR NEONATAL RESUSCITATION – DO WE ACHIEVE TARGET TEMPERATURE EARLIER AT THE PATIENT T-PIECE?

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**Background:** Dry, cold gas is used for neonatal resuscitation, and may contribute to lower admission temperature and exacerbation of lung injury. The addition of a test lung to the T-Piece during set-up may achieve faster provision of heated, humidified gas for resuscitation.

**Method:** A T-piece resuscitator, with a test lung attached, was connected via a heated patient circuit to a humidifier chamber (MR290) filled with 30 g of water. An oxygen flowmeter was set to deliver gas at 10 L/min. Temperature was recorded at the humidifier chamber (T1), distal temperature probe (T2) and the T-piece (T3) at 30 second intervals for 20 minutes. The test lung was then removed. Target temperature at T-piece was 36°C, and near 100% humidity assumed when T1-T3 temperature difference was  $\leq 1^\circ\text{C}$ .

**Results:** Target temperature at the T-piece was not reached with the test lung attached. T-piece temperature reached, and remained stable between 34°C–35°C. T1-T3 temperature difference  $\leq 1^\circ\text{C}$  was not reached due to the lower temperatures seen at the patient T-piece. After test lung removal, temperature at the T-piece climbed to 36°C in  $<10$  seconds.

**Conclusions:** The addition of a test lung to the heated, humidified gas circuit during set-up (10 L/min; MR290; 30 ml water) does not achieve target temperatures earlier, most likely due to gas escape through the PEEP

valve and radiant heat loss from the test lung. Attaching a test lung to the heated, humidified gas circuit is not necessary during set-up.

P391

# LIPOPOLYSACCHARIDE INDUCED WEAKNESS IN PRETERM OVINE FETAL DIAPHRAGM IS ASSOCIATED WITH MITOCHONDRIAL DYSFUNCTION

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**Background:** Diaphragmatic contractility is reduced in preterm lambs after lipopolysaccharide (LPS) exposure *in utero*, 2 d and 7 d before delivery at 121 d gestational age (GA). We hypothesise that impaired diaphragm contractility is mediated by intrauterine inflammation-induced mitochondrial dysfunction and oxidative damage.

**Method:** Pregnant ewes received intra-amniotic saline or 10 mg LPS 2 d or 7 d prior to caesarean section at 121 d GA (term = 150 d). The fetus was euthanased at delivery, immediately prior to tissue sampling. Mitochondrial fractions were prepared from the isolated diaphragm and mitochondrial electron transfer chain activities were evaluated using enzymatic assays. Oxidative stress was investigated by quantifying mitochondrial oxidative protein levels and determining antioxidant gene and protein (superoxide dismutase 2, catalase and glutathione peroxidase 1) expression. The activity of the antioxidant signaling pathway Nrf2 was examined from the cell lysate.

**Results:** A 2 d LPS exposure *in utero* decreased the activity of electron transfer chain complex II and IV (see Table). A 7 d LPS treatment inhibited superoxide dismutase 2 and catalase expression in both gene and protein levels, and Nrf2 pathway activity ( $p < 0.05$ ) compared with control and 2 d LPS groups respectively. Increased oxidative protein accumulation was found in diaphragm mitochondria after a 7 d insult.

**Conclusions:** Intrauterine exposure to LPS induces mitochondrial oxidative stress and electron chain dysfunction in preterm diaphragm, which is further exacerbated by impairment of the antioxidant signaling pathway and decreased antioxidant activity.

**Funding:** NHMRC APP1010665, Ada Bartholomew Medical Research Trust, WIRF.

|             | Control           | LPS 2 d            | LPS 7 d            | p value |
|-------------|-------------------|--------------------|--------------------|---------|
| Complex I   | 0.145 $\pm$ 0.011 | 0.140 $\pm$ 0.010  | 0.128 $\pm$ 0.016  | 0.258   |
| Complex II  | 0.092 $\pm$ 0.004 | 0.063 $\pm$ 0.018* | 0.102 $\pm$ 0.003* | <0.05   |
| Complex III | 0.110 $\pm$ 0.004 | 0.091 $\pm$ 0.018  | 0.085 $\pm$ 0.015  | 0.171   |
| Complex IV  | 0.244 $\pm$ 0.022 | 0.181 $\pm$ 0.026* | 0.163 $\pm$ 0.027* | <0.05   |

\* $p < 0.05$  compared with control and LPS 2 d groups, respectively. Values are mean (SD).

P392

# STAFF PERCEPTIONS OF TWO SYSTEMS FOR DELIVERING NEONATAL CONTINUOUS POSITIVE AIRWAY PRESSURE

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**Background:** Continuous positive airways pressure (CPAP) is a mainstay of non-invasive respiratory support for neonatal patients and different manufacturers offer a range of systems. Until recently, Hudson Prong (HP) CPAP has been the standard system in use at the Royal Hobart Hospital. We evaluated the Fisher and Paykel Midline (MID) system for six months

from April to September 2012. At the end of this time we surveyed staff perceptions of both CPAP delivery systems.

**Method:** Internet-administered survey including questions of perceptions of the advantages and disadvantages of both systems, with opportunities for free text responses. Eligible staff included nursing staff with CPAP training and medical staff working in the nursery.

**Results:** Thirty nine staff responded (30 ventilator-trained nurses, four CPAP-trained nurses and five medical staff); 31% and 39% of eligible and medical staff respectively. 77% of respondents  $\geq 10$  shifts experience with both systems (31%  $\geq 50$  shifts). Overall, 48% of respondents preferred the MID system, and 28% the HP system. Collated responses are shown in the Table. Staff perceived the MID system as being more comfortable for babies, better for head positioning and better for avoiding and managing nasal trauma, and the HP system as easier to apply and faster to commence.

**Conclusions:** Overall, after the evaluation period, clinical staff preferred the MID neonatal CPAP system and reported several advantages over the HP system. However, even after six months' experience, the MID system was perceived as less easy to apply and slower to commence.

| In your opinion, which system ...                            | Number of responses | HP System   |        | No difference | MID System |             |
|--------------------------------------------------------------|---------------------|-------------|--------|---------------|------------|-------------|
|                                                              |                     | much better | better |               | better     | much better |
| ... is better overall?                                       | 39                  | 10%         | 18%    | 25%           | 38%        | 10%         |
| ... do the babies find more comfortable?                     | 37                  | 8%          | 5%     | 30%           | 49%        | 8%          |
| ... is better for maintaining the CPAP pressure?             | 37                  | 11%         | 14%    | 38%           | 38%        | 0%          |
| ... is better for head positioning?                          | 36                  | 6%          | 8%     | 11%           | 67%        | 8%          |
| ... is easier to fit for size?                               | 37                  | 11%         | 8%     | 27%           | 41%        | 14%         |
| ... is easier to apply?                                      | 36                  | 19%         | 22%    | 36%           | 22%        | 0%          |
| ... can be commenced most quickly when CPAP needed urgently? | 37                  | 16%         | 30%    | 30%           | 24%        | 0%          |
| ... is better for avoiding nasal trauma?                     | 37                  | 3%          | 3%     | 16%           | 59%        | 19%         |
| ... is better for managing established nasal trauma?         | 37                  | 0%          | 5%     | 11%           | 70%        | 14%         |

### P393

#### LIMITED AERATION OF THE LUNG AT BIRTH LEADS TO A SIGNIFICANT VENTILATION/PERFUSION MISMATCH

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**Background:** Lung aeration is an established factor in increasing pulmonary blood flow (PBF) at birth, but regional relationships between lung aeration and the increase in PBF are not well understood. We studied the effect of partial lung ventilation on pulmonary blood vessels immediately following birth using simultaneous phase contrast (PC) X-ray imaging and angiography.

**Method:** Newborn rabbits were delivered near-term (~30 d GA; term ~32 d GA) and immediately PC X-ray imaged while an iodinated contrast agent was infused into the jugular vein before and then during unilateral (of the right lung) ventilation, followed by ventilation of both lungs, to visualise the PBF. The quantity and diameter of visible blood vessels as well as integrated intensity line profiles that transected vessels at different locations were measured, providing a relative measure of PBF.

**Results:** During unilateral ventilation of the right lung, the unventilated and liquid-filled left lung increased visible vessel quantity (from  $15 \pm 1$  to  $44 \pm 4$ ), vessel diameter (from  $554 \pm 50 \mu\text{m}$  to  $616 \pm 40 \mu\text{m}$ ) and integrated intensity (from  $6969 \pm 1638 \mu\text{m}\cdot\text{AU}$  to  $12143 \pm 2546 \mu\text{m}\cdot\text{AU}$ ).

The ventilated right lung increased all measured variables to similar values as the unventilated left lung, resulting in no discernible differences between aerated and non-aerated lung regions.

**Conclusions:** Aeration in limited regions of the lungs immediately following birth promotes a global increase in PBF, leading to a highly significant ventilation/perfusion mismatch in the unventilated tissue. These findings were unexpected and imply mechanisms other than regional lung aeration have significant roles in the increase in PBF at birth.

### P394

#### GASTROSCHISIS: A CASE OF UNEXPECTED THIRD TRIMESTER GASTRIC EXTERIORISATION

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**Background:** Serial ultrasound scans form part of routine monitoring of pregnancies with antenatally diagnosed gastroschisis. A recent study on the development of gastroschisis hypothesized that exteriorisation of abdominal contents can continue until birth.<sup>1</sup> We report a case where externalization of the stomach occurred late in the third trimester.

**Method:** Case report.

**Results:** An abdominal wall defect was suspected on antenatal scan at 11 weeks gestation, and subsequently confirmed as gastroschisis, with partially externalised small and large bowel but an intra-abdominal stomach. Serial scans up to 36 weeks identified growth parameters reducing from the 20<sup>th</sup> to below the 2<sup>nd</sup> centile, with reducing amniotic liquor volume. At final scan before birth (37 weeks), the stomach could not be identified within the abdominal cavity. Caesarean section was performed 2 days later, prompted by concerns over fetal wellbeing and cardiotocographic abnormalities. At birth, the entire large bowel, small bowel and stomach were noted to be exteriorised. Single stage surgical reduction could not be performed due to respiratory compromise and a two-stage procedure was performed with a silo. Complete closure was performed on day six and enteral feeds were gradually established. The baby was discharged home by six weeks of age.

**Conclusions:** This case confirms that in gastroschisis, abdominal contents can continue to exteriorise until birth and hence the findings at birth may differ from those on antenatal scans. This is important when counselling parents and planning surgical intervention.

#### Reference

1. Brugger PC, Prayer D. Development of gastroschisis as seen by magnetic resonance imaging. *Ultrasound Obstet Gynecol* 2011;37(4): 463–70.

### P395

#### TRUE CONGENITAL PANCREATIC CYST: CHALLENGES OF DIAGNOSIS AND MANAGEMENT

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**Background:** True pancreatic cyst is a rare congenital abnormality. We present a case where the diagnosis was made intra-operatively, and describe the surgical management.

**Method:** Case report and literature review.

**Results:** In an otherwise normal pregnancy, 19 week ultrasound identified a  $14 \times 19 \text{ mm}$  midline cystic structure. The lesion persisted throughout pregnancy and following term delivery, a soft small fluctuant supra-umbilical mass was palpable. Postnatal abdominal X-ray, ultrasound, MRI



and upper GI contrast studies confirmed a  $37 \times 40 \times 40$  mm non-communicating cystic structure posterior to the stomach at the level of the gastric antrum. Suck feeds were established easily. Gastric duplication was strongly suspected and laparotomy performed on day ten. At surgery, the cyst was unexpectedly identified as pancreatic in origin. To reduce the risk to the pancreas, common bile duct and retroperitoneal structures, near total resection with judicious use of cautery to the residual cyst wall was performed. Histopathology confirmed congenital true pancreatic cyst. Recovery was uncomplicated and without recurrence five months to date. Twenty five other cases of true congenital pancreatic cyst have been reported. Diagnostic approaches include abdominal X-ray, Ultrasound, CT, MRI, Upper gastrointestinal contrast studies and management has included laparotomy, roux en Y internal drainage, endoscopic or radiological drainage. Near total resection with cautery to the residual cyst wall has not previously been described. Other differentials include gastric duplication, choledochal cyst, mesenteric cysts, meconium pseudocysts and cysts of renal origin.

**Conclusions:** Differential diagnosis of congenital upper abdominal cysts includes true pancreatic cyst and other congenital abnormalities. Current medical imaging may not provide a definitive diagnosis.

P396

# CONGENITAL HAEMOPHAGOCYTIC LYMPHOHISTIOCYTOSIS IN A PRETERM BABY: A CASE REPORT

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**Background:** Congenital Haemophagocytic Lymphohistiocytosis is a rare autosomal recessive disorder of unregulated activation of T cells and macrophages. Only 7 cases in preterms have previously been reported in the literature.

**Case:** The patient was born to non-consanguineous Nigerian parents, who presented at 34 weeks with a 24 hour history of decreased movements and a pathological CTG. A live male infant was delivered via LSCS with birth weight of 3040 grams. The initial Hb was 71 g/L at birth (with a high nucleated red cell count) with marked hepatosplenomegaly. Thrombocytopenia and an unconjugated hyperbilirubinemia developed within the first 12 hours of life. The patient received a single red blood cell transfusion, two exchange transfusions and multiple platelet transfusions. In the following days the conjugated bilirubin increased significantly, and both AST and ferritin were raised.

There was significant deterioration clinically on day 4 with development of a metabolic acidosis, hypotension, hypoglycaemia and seizures. MRI brain showed evidence of hypoxic ischaemia damage, and in consultation with the parents intensive care support was withdrawn and the patient died soon after.

Histology from autopsy showed haemophagocytosis by histiocytes and together with clinical and laboratory results confirmed the diagnosis.

**Conclusions:** This case illustrates the key clinical and histopathological findings of Congenital Haemophagocytic Lymphohistiocytosis. The clinical features resemble that of congenital infection, and thus obstetricians and neonatologists should be aware of this differential diagnosis.

P397

# BLEOMYCIN TOXICITY IN AN INFANT TREATED WITH SCLEROTHERAPY FOR VASCULAR MALFORMATION OF THE NECK

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**Background:** We present the case of a male infant diagnosed with an anterior cystic neck mass at 21 weeks gestation. This had grown, propor-

tional to the fetus, until delivery at 34 weeks gestation. The tracheo-oesophageal displacement index was 18 mm, indicating a critically compromised airway. It was postnatally identified as a low flow venous vascular malformation which on MRI had affected the posterior pharynx, tongue, anterior neck and mediastinum.

**Method:** The baby was delivered using an EXIT procedure following preterm labour. He remained dependent on an endotracheal tube for airway patency and tracheostomy was performed, after debulking of the mass, at 5 months of age. A sclerotherapy regime was initiated at three months of age, with a combination of agents: 100% ethanol (3.5–13 ml per session), histoacryl: lipiodol mixture (4.5–9 ml per session) and bleomycin (3–10 IU per session). These were performed under general anaesthesia using fluoroscopic guidance.

**Results:** On several occasions following sclerotherapy, the infant developed a transient oxygen requirement and inflammatory response. After the eighth administration of sclerosants, this progressed to intractable respiratory failure. Pneumonitis secondary to bleomycin was suspected. He required ventilation in 100% oxygen, HFV, and iNO and was transferred on ECMO to an associated tertiary hospital for ongoing life support. He passed away ten days later due to complications of right-sided cardiac failure.

**Conclusions:** Vascular malformations affecting the airway are difficult to manage, with high risks of morbidity and mortality. More study is required to examine the safety and efficacy of sclerosant agents in younger patients.

P398

# CASE REPORT OF A FAMILY WITH A NOVEL 14Q21.1–21.2 DELETION

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**Background:** We present a novel 14q micro-deletion identified in a family with multiple congenital abnormalities.

**Method:** Case report.

**Results:** A mother with bilateral talipes, bicornuate uterus, facial hypotonia and two previous preterm deliveries (29 and 34 weeks) presented with a fetus whose antenatal findings included bilateral talipes, microcephaly and normal liquor volumes. Following spontaneous breech labour at 27 weeks, a live 922 g female infant was birthed by caesarean section.

Postnatal clinical findings included bilateral fixed talipes, micrognathia, high arched palate, broad nasal bridge, sacral dimple and facial hypotonia. Issues in the neonatal period included difficulties with feeding and managing oral secretions, severe recurrent episodes of apnoea and bradycardia and gastro-oesophageal reflux. Management involved glycopyrolate, omeprazole and nasojejunal feeds. Clinical issues post-term have included ongoing severe feeding issues and mild motor developmental delay, initially managed with home nasogastric feeding before eventual fundoplication and gastrostomy at six-months corrected gestation.

Molecular karyotyping revealed a novel interstitial deletion of approximately 3.85 megabases on the long arm of chromosome 14 at cytogenetic band q21.1–q21.2. Review of the family identified facial hypotonia and bilateral talipes in a female sibling and the mother. Genetics screening of the parents identified the deletion to be maternal in origin. Screening of the siblings is not being performed at this time however other clinical effects of this micro-deletion may become apparent in the future.

**Conclusion:** This novel 14q21.1–21.2 micro-deletion has variable phenotypic expression, which includes dysmorphic features, feeding difficulties, facial hypotonia, bilateral talipes and developmental delay.

P399

# TRANSIENT NEONATAL MYASTHENIA GRAVIS – A CASE STUDY

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**Background:** A term baby boy was born in a district hospital by LSCS for failure to progress, the pregnancy was complicated with maternal Myasthenia Gravis (MG), managed with fortnightly intragram. The infant was born weighing 3.79 kilograms with apgars 8 & 9, and was noted to have a weak cry at birth associated with bulbar weakness, increased oral secretions leading to desaturations requiring regular suctioning. The infant was transferred to Grace Centre of Newborn Care for further investigations and management of unclear underlying diagnosis.

On admission the infant remained hypotonic with bilateral ptosis, a weak cry, a weak suck, and minimal spontaneous movements. Frequent bradycardia with oxygen desaturations required suction every one to two hours. On examination no other dysmorphic features noted. A Neostigmine trial was planned due to maternal MG. The AChR antibody level was – 8 nmol/l and subcutaneous Neostigmine 0.04 mg/kg was commenced every three hours which resulted in transient improvement lasting 30 minutes–one hour. On day six of life Neostigmine was changed to oral pyridostigmine and infant continued to show gradual improvement in muscle strength. Atropine was commenced for ongoing issues with secretions.

**Method:** A retrospective review of case notes and care plans enabled the detailed history and management to be gathered in to a presentation which will highlight the challenges for nurses who are faced with a difficult diagnosis.

**Conclusions:** Clinical diagnosis of Transient Neonatal Myasthenia Gravis is made on symptoms and trial with an anticholinesterase agent (Neostigmine, Pyridostigmine). Transient Neonatal MG can be life-threatening and early supportive care is the mainstay of treatment.

P400

# MATERNAL CARDIAC FUNCTION IN WOMEN DEEMED AT HIGH RISK OF PRE-ECLAMPSIA WITH SUBSEQUENT NORMAL PREGNANCY OUTCOME

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**Background:** Women can be screened for early onset pre-eclampsia (ePET) at 12 weeks and treated with Aspirin to prevent the onset of ePET (Number Needed to Treat = 40). Women destined to develop ePET have a hyperdynamic circulation prior to being symptomatic. Maternal cardiac function may be a valuable tool for improving the positive predictive value of screening. We define normal parameters of cardiac function in women at high risk of ePET at first trimester screening.

**Method:** Women at high risk of ePET were offered serial echocardiographic assessment. Maternal cardiac output (CO) and total peripheral resistance (TPR) were calculated. Women who developed pre-eclampsia or delivered <34 weeks were excluded. Cross-sectional data from assessments at 14 and 18 weeks and longitudinal data from 14 to 30 weeks are reported.

**Results:** 32 normotensive women were included. They had a mean CO of 4.6 L/min and 5 L/min at 14 and 20 weeks respectively. The mean CO increased 15% to 5.3 L/min between 14 and 24 weeks then reduced through the third trimester to a mean CO of 5 L/min. TPR was 1564, 1436 and 1473 dynes/sec/cm<sup>5</sup> at 14, 20 and 30 weeks gestation. These findings are higher than previously reported.

**Conclusions:** Maternal cardiac output normally increases through the second trimester of pregnancy, peaking at 24 weeks gestation at 5.3 L/min. These data are consistent with previous reports in an unselected population. The TPR results follow the normal trend of decreasing from the first to second trimester with nadir through 20–30 weeks. These data allow comparison with pregnancies affected by ePET to determine whether they can be differentiated by a hyperdynamic circulation.

P401

# FEMORAL FACIAL SYNDROME (FFS)

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The Femoral Facial Syndrome (FFS) is a very rare health condition of uncertain inheritance comprising of femoral hypoplasia and facial dysmorphism which may include, upslanting eyes, short nose with a broad tip, long philtrum, thin upper lip, micrognathia, and cleft palate. We report an additional observation detected by ultrasound at 28 weeks and diagnosed at birth. The important criteria for diagnosis of FFS include significant femoral hypoplasia and at least two of the facial features. With these criteria, we arrived with the diagnosis of FFS by having a bilateral short and bowed femurs, micrognathia and cleft palate. In addition to the malformations usually described in this syndrome, there were left club-foot, single umbilical artery and persistent left superior vena cava. Few cases are reported worldwide. Most reported cases have been sporadic, and no definite genetic link has been established at this time which is consistent with the normal Chromosomal Analysis of our patient. The etiology is unknown but maternal diabetes has frequently been reported, just like our case diagnosed with Overt Diabetes. The pathogenesis is multifactorial and it has been suggested that a disruption of normal carbohydrate homeostasis at a critical point of embryogenesis may be important in the development of FFS. Genetic counseling may be of benefit for patients and their families. It is important to introduce prevention strategies before conception. Congenital Anomaly Scan is an important part of work-ups especially among diabetic mothers. This would help us reduce the incidence of misdiagnosis antenatally.

P402

# NEONATAL MYOCARDIAL INFARCTION AND SPECKLE TRACKING ECHOCARDIOGRAPHY

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**Background:** Neonatal myocardial infarction (MI) is an uncommonly diagnosed condition in the newborn. It is associated with asphyxia, structurally abnormal heart, thromboembolism and abnormalities of the coronary arteries. MI can complicate up to 30% of babies with asphyxia. The diagnosis is supported by traditional echocardiographic measurements, raised cardiac enzymes and electrocardiogram (ECG) abnormalities. The clinical reliability of cardiac enzymes as a marker for neonatal MI is limited, especially during therapeutic hypothermia. ECG abnormalities are not always present, and not specific for MI. Traditional echocardiographic measurements can detect significant MI, but are not sensitive enough to detect more subtle changes of MI. Novel echocardiographic techniques using acoustic markers from frame to frame (speckle tracking echocardiography) have shown promising results in quantifying adult MI.

**Methods:** This is a retrospective case-series of 5 babies with neonatal myocardial infarction. We describe speckle-tracking echocardiography as an additional method of diagnosing and quantifying MI.

**Results:** Five neonates with a diagnosis of MI are described. Two of the babies had been cooled for asphyxia and the others were investigated for clinically suspected congenital heart disease. Cardiac enzymes were raised in only 2 of the 5 infants. ECG abnormalities were found 2 infants. Speckle

tracking echocardiography showed ventricular wall motion abnormalities in all infants and was useful in monitoring progression of the condition. **Conclusions:** Speckle tracking may be a valuable tool for diagnosing and quantifying neonatal MI.

## P403

# STUDY OF CLINICAL AND ECHOCARDIOGRAPHIC PARAMETERS THAT PREDICT LIGATION OF PATENT DUCTUS ARTERIOSUS IN PRETERM NEWBORNS

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**Background:** The need and timing of surgical ligation of Patent Ductus Arteriosus (PDA) in preterm newborns is an ongoing clinical dilemma. **Method:** Retrospective review of case-notes and electronic records of preterm newborns (GA < 30 weeks) admitted to NICU at Flinders Medical Centre (January 2009–April 2012).

**Data collected:** gestation, birth weight, serial echocardiographic parameters, ibuprofen and steroid treatment, ventilation parameters, length of stay and duration of home oxygen. Statistical analysis was performed using T test.

**Results:** Out of 161 infants born <30 weeks, 92 infants were identified to have PDA. Ten infants requiring surgical ligation of PDA had large PDA > 2.5 mm (Group A). Twenty other infants were identified to have PDA > 2.5 mm but responded to conservative management (Group B). Both groups were of similar gestation [mean (SD) 25.9 (2) wk Vs 26.7 (2) wk] and BW [898 (240) gm Vs 973 (257) gm], had similar maximum diameter of PDA [mean (SD) 3.2 (0.6) mm Vs 3.0 (0.4) mm,  $p = 0.17$ ] with similar LA:Ao ratio [1.72:1 Vs 1.6:1,  $p = 0.31$ ].

Infants in Group A had more severe lung disease [postnatal steroid 8.2 Vs 3 doses, HFOV duration 7.1 d Vs 2.6 d, IMV duration 21.1 d Vs 14.9 d, CPAP duration 60.2 d Vs 34.1 d, need of home oxygen in 40% Vs 25% infants]. Length of stay was similar in both groups [discharge at 41.4 wk Vs 41.7 wk]. Two of the non-ligated babies died following withdrawal of treatment after grade 4 IVH.

**Conclusions:** Difficulty in weaning respiratory support was a better predictor for need of surgical ligation than any other parameter studied. PDA diameters were similar in both groups on serial echocardiograms.

## P404

# BENEFITS OF PROSTAGLANDIN E<sub>1</sub> USE DURING TRANSFER OF THE HYPOXAEMIC NEWBORN WITH SUSPECTED DUCT-DEPENDENT HEART DISEASE OR PERSISTENT PULMONARY HYPERTENSION

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**Background:** In the hypoxic newborn infant distinguishing duct-dependent congenital heart disease (DDCHD) and persistent pulmonary hypertension (PPHN) may be challenging. Prostaglandin E<sub>1</sub> (PGE<sub>1</sub>) is indicated in DDCHD to maintain ductal patency, but in non-DDCHD exposes infants to potential side-effects including apnoea, which may complicate transfer and clinical course. This study evaluated the consequences of PGE<sub>1</sub> use during transport of infants with suspected DDCHD or PPHN.

**Methods:** Retrospective study of infants ≤10 days old with suspected DDCHD/PPHN transferred from non-tertiary centres to the Royal Children's Hospital, Melbourne between 2007 and 2011. Combined database analysis was performed, including assessment of PGE<sub>1</sub> use and outcomes.

**Results:** Of 142 eligible infants, 81 had DDCHD, 51 had PPHN and 10 had neither. PGE<sub>1</sub> was commenced in 50% of infants, 63% of DDCHD group,

and 9 (18%) of PPHN group. Median (range) PGE<sub>1</sub> dose was 20 (5–100) nanograms/kg/min. There were no differences in demographics between PGE<sub>1</sub> and no PGE<sub>1</sub> groups. PGE<sub>1</sub> use was not associated with higher rates of ventilation or inotrope use (Table 1).

PGE<sub>1</sub> use was associated with significantly shortened length of stay in PPHN, {3 (1–14) vs. 14 (13–27) days,  $P = 0.004$ }, but not DDCHD patients {19 (21–37) vs. 13 (13–32) days,  $p = 0.12$ }.

There was no significant difference in mortality between PGE<sub>1</sub> and no-PGE<sub>1</sub> groups, in either PPHN or DDCHD subgroups.

**Conclusions:** PGE<sub>1</sub> was associated with faster recovery in PPHN, and was not associated with adverse events during transfer. These findings support early use of PGE<sub>1</sub> in the hypoxic infant regardless of whether DDCHD or PPHN is suspected.

**Table 1.** Demographic and transport characteristics, and cardio-respiratory therapies during retrieval of infants with suspected DDCHD and PPHN

|                                     | All infants<br>(n = 142) | PGE <sub>1</sub> group<br>(n = 71) | Non-PGE <sub>1</sub><br>(n = 71) | P*     |
|-------------------------------------|--------------------------|------------------------------------|----------------------------------|--------|
| <b>Demographic data</b>             |                          |                                    |                                  |        |
| Gestation, median (IQR)             | 39 (38–40)               | 39 (38–40)                         | 39 (37–40)                       | 0.12   |
| Birth weight (g), median (IQR)      | 3.3 (2.9–3.8)            | 3.5 (2.9–3.7)                      | 3.3 (2.9–3.8)                    | 0.50   |
| <b>Transport characteristics</b>    |                          |                                    |                                  |        |
| Distance (km), median (IQR)         | 63 (21–150)              | 63 (21–140)                        | 63 (26–152)                      | 0.72   |
| Land, n (%)                         | 101 (71)                 | 48 (68)                            | 53 (75)                          | 0.46   |
| Air, n (%)                          | 41 (29)                  | 23 (32)                            | 18 (25)                          | 0.46   |
| Duration, hours, mean (SD)          | 5.4 (3.1)                | 5.9 (3.9)                          | 4.9 (1.9)                        | 0.05   |
| <b>Cardio-respiratory therapies</b> |                          |                                    |                                  |        |
| Self-ventilating, n (%)             | 62 (44)                  | 33 (46)                            | 29 (41)                          | 1.0    |
| CPAP, n (%)                         | 11 (8)                   | 4 (6)                              | 7 (11)                           | <0.001 |
| IMV, n (%)                          | 68 (48)                  | 34 (48)                            | 34 (48)                          | 0.85   |
| Fluid Bolus, n (%)                  | 107 (75)                 | 64 (90)                            | 43 (61)                          | <0.01  |
| Inotropes, n (%)                    | 42 (30)                  | 22 (31)                            | 20 (28)                          | 0.85   |

\*P value: comparison of PGE<sub>1</sub> and non-PGE<sub>1</sub> groups.

## P405

# DIAGNOSTIC ACCURACY IN THE RETRIEVAL OF INFANTS WITH SUSPECTED DUCT-DEPENDENT CONGENITAL HEART DISEASE

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**Background:** Duct dependent congenital heart disease (DDCHD) may be difficult to distinguish from other diagnoses, notably persistent pulmonary hypertension of the newborn (PPHN) in the hypoxic infant. Affected infants born in regional hospitals require transfer to a tertiary paediatric centre for echocardiographic diagnosis. Identification of predictive factors to distinguish DDCHD from non-DDCHD would assist in appropriate early management during transport.

**Methods:** Retrospective cohort review of infants with suspected DDCHD or PPHN transferred by the Newborn Emergency Transport Service, Victoria (NETS) between 2007 and 2011. Clinical, physiological and x-ray data were compared between DDCHD and non-DDCHD (including PPHN) groups.

**Results:** Of 142 eligible infants, 81 had DDCHD and 61 had non-DDCHD, of whom 51 had PPHN. Diagnostic accuracy of DDCHD by transport team was 77% (positive predictive value 75%), Table 1.



At univariate analysis, presence of heart murmur, abnormal pulses, upper and lower limb BP difference >10 mmHg, cardiomegaly, initial SpO<sub>2</sub> of <92% and pre-post ductal SpO<sub>2</sub> difference >10% were significantly associated with DDCHD. Labile SpO<sub>2</sub> (difference between maximum and minimum ≥10%), abnormal lung parenchyma, mean blood pressure <40 mmHg, pH < 7.25, lactate >5, PaO<sub>2</sub> < 50 mmHg and FiO<sub>2</sub> > 0.5 were significantly associated with non-DDCHD.

On multivariate analysis, only labile SpO<sub>2</sub> and mean BP < 40 mmHg were significantly associated with non-DDCHD.

**Conclusion:** Clinical diagnosis of DDCHD is challenging. No single clinical factor reliably distinguished DDCHD from non-DDCHD. Labile SpO<sub>2</sub> and hypotension (mean BP < 40 mmHg) were associated with non-DDCHD.

**Table 1.** Provisional diagnosis by transport team and definitive echocardiographic diagnosis in infants with suspected duct-dependent congenital heart disease

| Provisional diagnosis by transport team | Echocardiographic Diagnosis |                   |
|-----------------------------------------|-----------------------------|-------------------|
|                                         | DDCHD (n = 81)              | No DDCHD (n = 61) |
| Suspected DDCHD (n = 98)                | 73                          | 25                |
| No DDCHD (n = 44)                       | 8                           | 36                |

#### P406

##### DIAGNOSIS AND MANAGEMENT OF INFANTS WITH TACHYARRHYTHMIA RETRIEVED BY THE NEWBORN EMERGENCY TRANSPORT SERVICE (NETS) VICTORIA

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**Background:** Newborn infants with tachyarrhythmia born outside a tertiary centre require retrieval by the Newborn Emergency Transport Service (NETS). We reviewed the management prior to, and during retrieval, and clinical outcomes.

**Method:** Retrospective cohort study of newborn infants with tachyarrhythmia retrieved by NETS to the Royal Children's Hospital (RCH), Melbourne, between January 2007 and January 2011. Clinical, treatment and outcome data obtained from case note review.

**Results:** 46 eligible infants were identified, data were available for 32. Supraventricular tachycardia (72%) and atrial ectopic tachycardia (11%) were the most common arrhythmias. NETS diagnosis was correct in 86% of cases. Prior to NETS arrival 23 infants received facial ice/vagal manoeuvres, 13 received adenosine and one received electrical cardioversion. Arrhythmia recurrence during transfer and treatment data are also provided in Table 1; 7 infants received at least one dose of adenosine following facial ice/vagal manoeuvres. Six infants had persistent arrhythmia; 2 were ventilated during transfer. Data was available for 28 infants following admission to RCH, of whom 23 (82%) had further arrhythmias, 8 (29%) received anti-arrhythmic medication in addition to adenosine, and 4 (14%) required electrical cardioversion. All infants survived to discharge at a mean of 8 days if further episodes or 5 days if no subsequent arrhythmias. Twenty-four were discharged on anti-arrhythmia medication (83% beta blockers), and 3 were discharged on two anti-arrhythmics.

**Conclusions:** NETS transfer on average 11 infants per year with tachyarrhythmia. Most neonates remained stable during retrieval but had subsequent tachyarrhythmias during admission and majority were discharged on medication.

**Table 1.** Management of infants with tachyarrhythmia prior to and during NETS retrieval Numbers in parentheses refer to %

|                                            | Episodes and treatments of arrhythmia in infants retrieved by NETS |           |            |                            |
|--------------------------------------------|--------------------------------------------------------------------|-----------|------------|----------------------------|
|                                            | No episodes                                                        | 1 episode | >1 episode | Persistent tachyarrhythmia |
| <b>Management prior to NETS retrieval</b>  |                                                                    |           |            |                            |
| No intervention                            | –                                                                  | 8 (25)    | 17 (53)    | 7 (22)                     |
| Facial ice/vagal manoeuvres                | –                                                                  | 2         | 6          | 1                          |
| Adenosine                                  | –                                                                  | 6         | 11         | 6                          |
| Electrical cardioversion                   | –                                                                  | 2         | 8          | 3                          |
|                                            | –                                                                  | 1         | 0          | 0                          |
| <b>Management by NETS during retrieval</b> |                                                                    |           |            |                            |
| No intervention                            | 16 (50)                                                            | 4 (12)    | 6 (19)     | 6 (19)                     |
| Facial ice/vagal manoeuvres                | 16                                                                 | 3         | 2          | 4                          |
| Adenosine                                  | 0                                                                  | 1         | 4          | 2                          |
| Ventilated for transfer                    | 0                                                                  | 1         | 4          | 2                          |
|                                            | 0                                                                  | 0         | 0          | 2                          |

#### P407

##### INTERSTATE CARDIAC TRANSFERS FROM SOUTH AUSTRALIA. THE MEDSTAR EMERGENCY MEDICAL RETRIEVAL SERVICE EXPERIENCE SINCE INCEPTION IN 2009

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**Background:** MedSTAR was founded in 2009 as the stand alone retrieval service for neonates, children and adults in South Australia. Neonatal and paediatric cardiac surgery in South Australia is limited to PDA ligation only. All other neonates and children requiring cardiac surgery are transferred to the Royal Children's Hospital, Melbourne by the MedSTAR Kids service.

**Method:** This descriptive, retrospective chart review evaluated the frequency and safety of neonatal and paediatric transfers for cardiac surgery. **Results:** In the 3 years since the inception of MedSTAR a total of 1800 (approx.) paediatric and neonatal retrievals have occurred, 150 (approx.) of which were transfers to the Royal Children's Hospital, Melbourne for cardiac surgery. Of these, (approx) 20% patients were ventilated, (approx) 20% were receiving a prostaglandin E2 infusion. There were no deaths and no adverse incidents reported. The average transport time was 6 hours (approx).

**Conclusions:** Transfers for cardiac surgery comprise approximately 5% of the overall workload for the MedSTAR Kids service. We are able to perform such transfers safely and with no reported morbidity. This data suggests a local cardiothoracic service is not required in South Australia however utilisation of retrieval services for these cases is time consuming and limits the availability of teams for other critical patients in the state.

#### P408

##### ALLEN'S TEST, IS IT RELIABLE IN THE NEWBORN BABIES?

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**Introduction:** Radial artery cannulation, a common procedure in the NICU, can be associated with complications which include ischemia. The Modified Allen's test (MAT), assessing the collateral circulation in the hand, is recommended before the radial artery cannulation. The incidence of normal MAT in adults is 73% and 11% in neonates.

**Aim:** The aim of our 6 month prospective study is to determine whether there are any difference in the incidence of normal MAT based on the babies' gender, birth weight and gestation. We also studied the inter observer variation.

**Methods:** All babies admitted to the unit were eligible for the study. Two examiners independently performed the MAT on the first day of admission. A normal Allen's is when reperfusion time is under 10 seconds.

**Results:** A total of 151 babies were recruited: mean gestation 36.7 weeks (range 23 to 41.3 weeks); 37% of babies <37 weeks gestation, 63% term; 55.6% male, 44.4% female; 37% were <2500 g and 67% were >2500 g. The incidence of normal MAT was 26.5% for examiner A, and 19.2% for examiner B (p value 0.7). The incidence of normal MAT was: 47.4% for term and 21.4% for preterm babies (p = 0.001), 21.4% in low birth weight group babies compared to 47.4% for >2500 g babies (p = 0.001). There was no difference between male and female, right and left hands.

**Conclusion:** The incidence of normal MAT in newborn babies is lower than adult population and is lower in babies who are preterm and low birth weight.

P409

# CHEST SHIELDING FOR PREVENTION OF A HAEMODYNAMICALLY SYMPTOMATIC PATENT DUCTUS ARTERIOSUS IN PRETERM INFANTS RECEIVING PHOTOTHERAPY

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**Background:** Phototherapy is the most widely used therapy for neonatal unconjugated hyperbilirubinaemia especially in preterm infants. It has been proposed that shielding the chest from phototherapy light may prevent a haemodynamically significant patent ductus arteriosus (PDA). The chest shield may be made of any photo-opaque material designed to prevent phototherapy light from reaching the ductus. It is hypothesised that the skin of preterm infants permits transmission of light during phototherapy causing photo-relaxation of the musculature of the DA. Animal studies have shown that chest shielding blocks transmission of light and prevents photo-relaxation of the DA.

**Method:** The standard search strategy of the Neonatal Review Group was used.

**Results:** Two randomised controlled trials (Rosenfeld 1986; Travadi 2006) met the inclusion criteria. Both studies have methodological concerns. No study reported symptomatic PDA. Rosenfeld (1986) reported a significant reduction in haemodynamically significant PDA (74 infants, RR 0.23, 95% CI 0.05, 1.01). Meta-analysis of the 2 studies found a significant reduction in PDA (128 infants, RR 0.66, 95% CI 0.44, 0.98). Rosenfeld (1986) reported a significant reduction on treatment with indomethacin (74 infants, RR 0.12, 95% CI 0.02, 0.88).

**Conclusions:** There is insufficient evidence to recommend use of a chest shield during phototherapy for prevention of PDA in preterm infants. Further trials may be warranted, particularly in settings where infants are not receiving prophylactic or early echocardiographic targeted cyclooxygenase inhibitors for PDA.

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P410

# RATES AND PREDICTORS OF EARLY INTERVENTION FOR A COHORT OF EXTREMELY PRETERM OR EXTREMELY LOW BIRTH WEIGHT INFANTS FROM THE ROYAL WOMEN'S HOSPITAL, MELBOURNE

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**Background:** Early intervention (EI) in the community includes a range of multidisciplinary services. Extremely preterm (EP) and extremely low birthweight (ELBW) infants are often referred to EI. This study aimed to describe the rates of EI for a cohort of EP or ELBW infants and examine factors that may influence the receipt of EI.

**Method:** We performed a retrospective audit of data collected at 2 years from 80 children. Predictors for receiving EI were neonatal risk characteristics (birthweight, gestation, male sex, small for gestational age, chronic lung disease, intraventricular haemorrhage and sepsis), socioeconomic status (SES) and developmental delay. Predictors were assessed using logistic regression.

**Results:** The overall rate of receipt of EI during the first two years was 68% (n = 54). Rates of EI for children were 80%, 75% and 70% for moderate to severe cognitive, language and motor delay respectively. Rates of EI for children with moderate to severe delay in 3 domains of development were 80%, and 1 domain 67%. Birthweight was the only neonatal predictor significantly related to the receipt of EI (OR 0.79; 95% CI 0.63-1.00, p = 0.045). Children with moderate to severe delay were more likely to be receiving EI than those with no delay (OR 4.20; 95% CI 1.41-12.52; p = 0.01). All children with CP were receiving EI at two years regardless of delay.

**Conclusions:** Developmental delay and birthweight predicted receipt of EI. There was an unmet need for EI for EP and/or ELBW infants and a proportion of children with all levels of delay did not receive EI.

P412

# EARLY CEREBRAL FRACTIONAL TISSUE OXYGEN EXTRACTION IN VERY PRETERM NEWBORNS: THE IMPORTANCE OF TIME

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**Background:** Inadequate oxygen delivery increases the risk of restricted oxygen consumption and early poor outcome in the very preterm newborn (VPN). Oxygen extraction, the ratio of oxygen consumption to delivery, best quantifies this balance. Cerebral fractional tissue oxygen extraction (cFTOE) discriminates the risk of death or brain injury in VPN at 14 hours of life.

**Aim:** To extend evaluation of cFTOE in VPN early post delivery.

**Method:** In newborns ≤30 weeks, cFTOE was calculated from contemporaneous measurement of tissue oxygenation index (TOI) (Hamamatsu NIRO200) and cutaneous oximetry (Masimo) at 6, 12 and 24 hours. Brain injury was determined by sonography on Day 1, 3 and 7. Data is presented as median (range) or n (%).

**Results:** Twenty five babies were studied [gestation 28.1 wks (25.2-29.8), birth weight 1056 g (470-1690). 9 (36%) male]. All cranial sonographs were normal. TOI increased from 6 to 12 hours of age (p = 0.08) resulting in a significant fall in cFTOE (p = 0.03). There was no change in either measure between 12 and 24 hrs. There was no relationship between gestation and TOI and no difference between cFTOE calculated using either cutaneous oximetry or co-oximetry (ABL 700 Radiometer).

**Conclusion:** High cFTOE falls over the first hours of life in VPN. This likely relates to improved cerebral blood flow. The current data suggests that any cFTOE threshold needs reference to time of measurement in order to be useful to the bedside clinician. Right radial cutaneous oximetry

may be a useful surrogate for invasive arterial co-oximetry in this population.

P413

#### DEVELOPMENTAL AND BEHAVIOURAL OUTCOMES OF UNCOMPLICATED MONOCHORIONIC DIAMNIOTIC TWINS BORN IN THE THIRD TRIMESTER

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**Background:** Recently, decreasing rates of perinatal mortality for singletons have led to an increased focus on the outcomes of high-risk multiple pregnancies. This study presents a cohort of 34 uncomplicated MCDA twins, delivered in the third trimester; a population not addressed previously in the literature.

**Method:** 34 children were included in this study. Developmental and behavioural problems were screened for using 3 questionnaires, the Child Behaviour Checklist, The Child Development Chart and a parent survey. All questionnaires were delivered and returned by mail. Developmental and behavioural problems were defined as abnormal Child Behaviour Checklist and/or Child Development Chart scores. Previously established developmental/behavioural problems reported by parents were also considered.

**Results:** 5 children (14.7%) were identified as having developmental or behavioural problems by the screening tests. A further 7 (20.6%) were identified by parents as having previously existing problems, although these did not correspond with abnormal scores in our testing. Lower than average motor scores were observed in the population as a whole, although these were not at abnormal levels. In a comparison of all children affected or not affected by developmental/behavioural problems, a correlation was observed between better developmental outcomes and higher birth weight and gestational age.

**Conclusions:** The results of this study indicate better developmental and behavioural outcomes for uncomplicated monochorionic twins with increasing gestational age and birth weight. Analysis of these results was limited by small cohort size; however, further responses have now been collected and the intention is to include these cases in the study also.

P414

#### PERICONCEPTIONAL UNDERNUTRITION IN SHEEP LEADS TO DECREASED VOLUNTARY PHYSICAL ACTIVITY IN ADULT OFFSPRING

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**Background:** Maternal undernutrition increases offspring obesity and metabolic disease risk. We hypothesized that periconceptional undernutrition in sheep from 60 days prior to conception through day 30 of gestation (UN) would decrease voluntary physical activity in adult offspring compared with offspring from well nourished controls (CON).

**Method:** Voluntary activity was measured in offspring at 18 months of age using GPS units worn by each animal for approximately 48 h in the paddock. The GPS units recorded coordinate locations continuously throughout the 48 h period. Data were analyzed using multiple regression analysis, with explanatory variables including sex, nutrition group, birth weight, average time between GPS measurements (ATBGM), and percentage of time during the measurement period spent in daylight. ATBGM was included because there was a strong negative correlation between ATBGM and metres walked per h.

**Results:** Mean ( $\pm$  SE) distance walked was greater for CON than UN animals, and greater for females than males ( $110.2 \pm 6.5$  m/h for CON females,  $110.7 \pm 6.3$  m/h for CON males,  $105.1 \pm 5.3$  m/h for UN females and  $95.5 \pm 5.8$  m/h for UN males,  $p = 0.02$  for nutrition group effect and for sex effect). There was no sex by nutrition group interaction, no effect of birth weight, and no effect of twin status.

**Conclusions:** These data indicate that undernutrition around the time of conception leads to a significant decrease in voluntary physical activity in adult offspring. This could contribute to an increased risk of obesity and metabolic disease in offspring born to undernourished mothers.

P415

#### ETHANOL EXPOSURE DURING LATE OVINE GESTATION ALTERS IRON HOMEOSTASIS IN THE FETAL LIVER

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**Background:** Chronic, high-dose ethanol exposure during fetal life can affect liver development and can increase susceptibility to infection after birth. Our aim was to determine the effects of a moderate level of ethanol exposure in late gestation on the morphology, iron status and inflammatory status of the ovine fetal liver.

**Method:** Pregnant ewes were chronically catheterized at 91 days' gestation (DG; term ~147DG) for daily i.v. infusion of ethanol (0.75 g/kg maternal body weight,  $n = 8$ ) or saline ( $n = 7$ ) over 1 h from 95–133DG. Plasma was collected to determine concentrations of ethanol and iron. At necropsy (134DG), fetal livers were collected for histological and molecular analysis; placentae were collected for molecular analysis.

**Results:** Ethanol concentration of exposed ewes and fetuses was 0.12 and 0.11 g/dL respectively at 1 h after infusion onset. Fetal liver weight, general liver morphology, hepatic cell proliferation and apoptosis, perivascular collagen deposition, and interleukin (IL)-1 $\beta$ , IL-6 and IL-8 mRNA levels were not different between groups. Ethanol exposure significantly decreased hepatic content of ferric iron and gene expression of the iron-regulating hormone *hepcidin*, and tumor necrosis factor (TNF)- $\alpha$ ; placental *ferroportin* mRNA levels were increased (all  $p < 0.05$ ). Plasma iron concentration was not different between control and ethanol-exposed groups; however, control fetuses had significantly higher iron concentrations than their mothers, whereas iron concentration was similar between ethanol-exposed ewes and fetuses.

**Conclusions:** Daily ethanol exposure during the third-trimester-equivalent in sheep does not alter fetal liver morphology; however, decreased fetal liver ferric iron content and altered *hepcidin* and *ferroportin* gene expression indicate iron homeostasis is altered.

P416

#### APPLICATION OF THE INFANT/TODDLER SENSORY PROFILE QUESTIONNAIRE IN POST-SURGERY AND HEALTHY INFANTS

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**Background:** The aim of this study was to investigate the sensory processing of infants who have undergone major surgery in the neonatal period using the Infant Toddler Sensory Profile at 12 months of age. In particular, to compare the sensory profiles of infants who underwent major cardiac and non-cardiac surgery, with a group of healthy infants and ITSP norms. **Method:** A total of 118 infants, 41 who had undergone cardiac surgery, 47 non-cardiac surgery, and 30 healthy controls were drawn from a larger population-based (statewide) longitudinal study of development following



infant surgery. Families were given the Infant Toddler Sensory Profile Questionnaire at their child's one year follow up clinic appointment. The ITSP is a parent-questionnaire that yields aged-normed scores on five 'Quadrants' (dimensions) of sensory functioning and five Sensory Processing Systems. Categories 'More than' or 'Less than' peers fall in the range for 'Consult and Follow-up'.

**Results:** A one-sample binomial test of proportions was used to compare group frequencies with ITSP norms. Results showed that significantly more control group children had scores in the range of 'Consult and Follow-up' ( $p$ 's < 0.05).

**Conclusions:** These findings are concerning in an age-normed test and questions the feasibility of applying the ITSP to this population. It may be that small numbers contributed to this result. We recommend that the ITSP be locally normed with a larger sample of Australian infants. We suggest caution when using tests that were normed on populations that differ from the intended research or clinical population.

#### P417

##### PERCEIVED NEEDS OF FAMILY MEMBERS OF CRITICALLY ILL CHILDREN ADMITTED TO AN INDIAN PAEDIATRIC INTENSIVE CARE UNIT

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**Background:** While the admission of a child to intensive care is very stressful for the child and parents, there is a paucity of literature regarding the needs of these families. This study aimed to ascertain the needs of the family and their needs as perceived by nurses working in a Paediatric intensive care unit (PICU) in an Indian tertiary setting.

**Method:** A semi structured interview using a modified and validated critical care family needs inventory scale (Molter, 1983) was conducted with thirty family members within 48–72 hours following admission of their child to PICU. The same tool was simultaneously given to thirty nurses working in PICU. Family members completed the scale again if their child stayed in PICU for more than eight days to identify whether their needs differed over time.

**Results:** Both parents and nurses perceived emotional needs as most important, however differences were identified between parent and nurse dyads in the ranking of needs, with a statistically significant difference found for cognitive needs ( $P < 0.05$ ). The study also found that informational needs of families decreased over time ( $P < 0.01$ ) whereas emotional, social and physical needs increased.

**Conclusions:** Parents of children admitted to intensive care have many needs, which were perceived both by parents and nurses. However, nurses were not aware of which needs were significant for each individual family. Based on the results of this study guidelines were developed for nurses. A similar study with parents in an Australian surgical neonatal intensive care unit in Sydney will commence in 2013.

#### P418

##### THE DEVELOPMENT AND IMPLEMENTATION OF AN ACCREDITED NEONATAL NURSING SPECIALITY TRANSITION PROGRAM

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**Background:** Neonatal intensive care settings globally are struggling to maintain a skilled nursing workforce. Within Australia a State government plan to ensure the health workforce is recruited and trained to meet the

needs of the health consumers, has led to the development of speciality critical care nursing transition programs. Providing focused education over a time limited period, successful program completion is measured by clinical competence within the speciality setting.

**Method:** In 2012 at a Tertiary Children's hospital the Neonatal Intensive Care Unit (NICU) proposed the development of a speciality transition program would assist in meeting speciality specific education and clinical competence needs. Over a three month period a group of unit based key stakeholders developed a neonatal nursing specialty transition program utilising an outcomes based curriculum that consisted of formal and informal learning strategies. The program was endorsed by the organisations curriculum governance committee in July 2012 and accredited by two universities. The programs accreditation entitles participants at the completion of the program to seek recognition of prior learning for a subject towards an NICU Graduate Certificate.

**Results:** The program was implemented in November 2012, consisting of face to face education, self-directed learning activities, competency assessments and assessable components. The program implementation and transition outcomes are measured utilising self-confidence, clinical competence and knowledge assessment tools.

**Conclusion:** Whilst challenges are inherent in the development and implementation of speciality transition programs they are envisaged to have a positive long term impact on recruitment, retention, career pathways, and increased patient safety due to increasingly qualified and supported nursing staff.

#### P419

##### DEVELOPING A TRAINING PROGRAMME WITHIN A RETRIEVAL SERVICE: IS THERE A NEED AND WHAT MODALITIES CAN BE USED TO DELIVER TRAINING TO STAFF?

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**Background:** NETS Victoria performs over 1000 retrievals per year. In order to perform safely and efficiently, it is vital that transport teams have sufficient knowledge, skills and attitudes relating to neonatal retrieval. There was a perceived need within NETS Victoria for further medical educational development. We aimed to identify these needs.

**Method:** A literature review was performed to evaluate the evidence-base behind proposed educational methods including the introduction of innovative techniques such as e-learning and simulation into the service and a questionnaire designed. Multidisciplinary feedback about learning needs was sought using the key principles underlying educational development via this questionnaire.

**Results:** 6/9 (66%) of fellows and 11/32 (34%) of nurses have returned questionnaires to date. Only 4/17 (24%) respondents felt their needs were met by the current programme. There were no domains in which staff felt that they had no experience. 10/17 (60%) regularly accessed current inter-hospital education. 13/17 (76%) were in favour of further developing education provision. Procedural training was a priority for targeted education. Staff rated teaching about equipment, procedure training and case reviews as most useful. Journal club, computer assisted learning and developing a formal transport course were deemed least useful.

**Conclusions:** There is a need to develop further educational opportunities within NETS Victoria. In order to meet the learning needs of a multidisciplinary team, challenges include optimising orientation and clinical supervision of transport personnel, maximising regular non-transport activity by introducing e-learning portfolios and the provision of an education programme, all within the constraints of a busy transport service.

## P420

# **VENTILATION PARAMETERS DURING RESUSCITATION: COMPARISON OF TWO DIFFERENT DEVICES IN A MANNEQUIN MODEL WITH AND WITHOUT DISTRACTION**

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**Background:** Positive pressure ventilation is a common intervention in neonatal resuscitation. Distraction, type of device and experience may influence performance. We aimed to compare clinician's ability to ventilate a mannequin using a self inflating bag (SIB) against a T-Piece (TP) with and without distraction.

**Method:** 50 medical and nursing staff were tested using both devices in 2 standardised case scenarios. The devices tested were 1) Ambu® SIB (Mark IV Baby resuscitator, 450 mL capacity) with a PEEP valve and manometer attached and 2) Neopuff® TP. Participants targeted PIP 30 cmH<sub>2</sub>O, PEEP 5 cmH<sub>2</sub>O, inflation rate (IR) 60 inflations/minute. A respiratory function monitor measured PIP, PEEP, IR and Expired tidal volume (TVe). Performance during 3 minutes of distraction was compared to 30 s of no distraction (baseline).

**Results:** 10.287 inflations were analysed. Respiratory parameters are shown in the table. During distraction, PIP was within a range of 25–35 cmH<sub>2</sub>O in 91.6% of SIB inflations compared to 99.9% of TP inflations ( $p < 0.001$ ), but TVe range was not significantly different ( $p = 0.45$ ). Level of experience made no difference in devices' performance ( $p = 0.51$ ). 90% of operators preferred TP.

**Table.** Respiratory parameters measured in both case scenarios

| Parameters<br>Mean (SD) | Distraction<br>n = 8909 inflations |                 | No distraction<br>n = 1378 Inflations |                |
|-------------------------|------------------------------------|-----------------|---------------------------------------|----------------|
|                         | TP<br>n = 4382                     | SIB<br>n = 4527 | TP<br>n = 673                         | SIB<br>n = 705 |
| PIP                     | 29.3 (0.7)                         | 28.9 (3.5)      | 29.3 (0.6)                            | 29.0 (2.3)     |
| PEEP                    | 4.2 (0.6)                          | 5.5 (0.9)       | 4.2 (0.6)                             | 5.5 (0.9)      |
| IR                      | 53 (13)                            | 56.2 (13.5)     | 53.6 (10.3)                           | 56.6 (11.7)    |
| TVe                     | 10.3 (1.9)                         | 9.6 (1.5)       | 10.2 (1.8)                            | 9.7 (0.9)      |

**Conclusions:** Although many operators preferred a TP device, performance when using a SIB with pressure manometer was comparable to the TP. Variation in PIP during distraction had no impact on TVe. This may be relevant in resuscitation settings where TP is not available.

## P421

# **INTESTINAL MALROTATION IN AN EXTREME PREMATURE INFANT: A CASE REPORT**

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**Background:** Intestinal malrotation is a congenital condition that can have disastrous, potentially fatal sequelae with incidence of 1 in 500 live births. Malrotation is more commonly seen in first year of life. But this condition is very rare in extremely premature infants with very few reported incidents.

**Case report:** Baby was born at 24 + 3 weeks and was admitted to NICU for management of prematurity. His general condition was reasonably stable. Enteral feeds started on day 4 and gradually upgraded while he was on parenteral nutrition. He developed distended abdomen with bile stained aspirates which was managed as stage 1 NEC. He had ongoing feed intolerance up to day 34 with worsening distension. X-Ray showed pneumoperitoneum and he underwent laparotomy.

Intraoperatively volvulus with gangrene and perforation was found. 25 cms of bowel was resected and Ladd's procedure done, with colon and small bowel brought out as stomas. Trophic feeds were started on day 40 and reanastomosis done on day 68.

**Discussion:** Malrotation occurs when the normal process of embryonic gut rotation is arrested or altered. The abnormal mesenteric attachment of affected bowel is prone to volvulus. Peritoneal or Ladd's bands are congenital bands of tissue that pass from the ileocaecal area to the right para-colic gutter, across the duodenum, and can cause extrinsic compression of the duodenal lumen. Although NEC is more common, one of the differentials may be malrotation though rare but of disastrous consequence. If found early it can be treated with Ladd's procedure and outcomes may be good.

## P422

# **COMPARISON OF PERCUTANEOUS ENDOSCOPIC AND LAPAROSCOPIC GASTROSTOMY INSERTION IN CHILDREN**

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**Aim:** To compare the outcomes and complications between Percutaneous Endoscopic Gastrostomy (PEG) and Laparoscopic Gastrostomy (LAP) in children.

**Methods:** Retrospective review of 346 patients from July 1998 to December 2010 who had their gastrostomies inserted at Sydney Children's Hospital, Randwick.

**Results:** Out of these 369 patients, 260 patients underwent LAP, 86 PEG and 23 open gastrostomy procedures. The PEG and LAP groups were comparable from the perspectives of age, gender, indication, year of insertion, complication rate. The overall complication rate for PEG was 54.7% and 44.6% for LAP ( $p = \text{NS}$ ). The overall complication rate for open procedures was 73.9% ( $p = 0.01$ , when this was compared to LAPs and PEGs). The early complication rate for PEGs was 10.5%, and 2.7% for LAPs ( $p = 0.006$ ). There were no early complications in the open procedure. The late complication rate was 41.9% for PEGs and 43.1% for LAPs, ( $p = \text{NS}$ ), and 78.3% for open gastrostomy, ( $p = 0.001$  when compared to PEGs and LAPs). In the PEG group, 2 patients developed gastro-colonic fistula and 1 developed peritonitis as a complication.

**Conclusion:** The findings of this study supported current literature in that PEGs had a higher overall complication rate than LAPs, although the difference was not statistically significant. Interestingly the PEGs had a significantly higher early complication rate compared to LAPs and the only serious complications occurred in the PEG group. However, both PEGs and LAPs were significantly superior to the traditional open gastrostomy procedure in terms of complication rates. More data from future prospective studies are needed to corroborate our findings.

## P423

# **POST NEC INTESTINAL STRICTURES: RISK FACTORS AND CLINICAL PROFILE**

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**Background:** Irrespective of the mode of management of acute episode, a substantial number of infants with Necrotising Enterocolitis (NEC) develop intestinal strictures. Strictures manifest at variable period after NEC and cause significant morbidity and mortality. We sought was to describe the natural history, outcomes and risk factors for development of intestinal strictures following NEC.

**Method:** A retrospective cohort study of infants with NEC over 10 year period.

**Results:** Out of 61 infants with NEC, 15 (24.5%) had Bell's stage I, 22 (36%) had stage II and 24 (39.5%) had stage III disease. 28 (46%) infants required surgical intervention and 33 (54%) were conservatively managed. Total 18 (29.5%) infants, 10 (35.5%) from surgical group and 8 (24%) from conservative group developed strictures. Only presence of high white cell

count and non-specific dilatation of intestines on abdominal X-ray during acute NEC differed significantly in the stricture and no-stricture groups. Strictures were diagnosed at median interval of 34 days after NEC and showed clear preference (67%) for colon. At diagnosis of strictures, 2 infants (11%) had intestinal perforation and 2 (11%) had sepsis. 8 infants failed to thrive and 9 developed cholestasis. Median length of hospitalisation for infants in stricture group was 33 days. No mortality attributable to strictures occurred in this series.

**Conclusions:** Prediction of development of intestinal stricture following NEC is difficult. Strictures can present with life threatening complication such as intestinal perforation or sepsis. A substantial number of infants with strictures fail to thrive, majority have any morbidity and need prolonged hospitalisation.

## P424

# PN-ASSOCIATED CHOLESTASIS (PNAC) IN INFANTS FROM A LEVEL III PERINATAL/SURGICAL NICU – A DESCRIPTIVE STUDY

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**Background:** Parenteral nutrition-associated cholestasis (PNAC) is common in infants requiring PN for more than 4 weeks. Factors such as prematurity, low birth weight, lipid formulations and sepsis contributes to the increased risk. The aim of our study was to investigate the clinical characteristics and outcomes of infants in our NICU who has received prolonged PN.

**Method:** We conducted a retrospective review of infants from a 5 year epoch (Jan 2007 to Dec 2011) who had received  $\geq 25$  days of continuous PN. Clinical and laboratory data were abstracted from NICU, pharmacy databases. Pathology results were tabulated weekly (weeks 1–4) from PN commencement.

**Results:** 143 infants who received prolonged PN were identified: 54.5% (78/143) male with mean ( $\pm$  SD) gestational 28.3 ( $\pm$  3.8) weeks, birth weight 1179 ( $\pm$  748) gram. 23 infants had Bell Stage 2/3 NEC; with 6 pneumoperitoneum diagnosed. 28 infants received Ursodeoxycholic treatment. Mean length of stay was 118 ( $\pm$  83) days. As shown in Table 1, only 52% (74/143) infants had conjugated bilirubin and 20% (29/143) had gamma-GT measurements performed on week 4 post-PN commencement.

**Table 1.** Levels of conjugated bilirubin and GGT after commencement of PN

|                                     | Week 1             | Week 2             | Week 3             | Week 4             |
|-------------------------------------|--------------------|--------------------|--------------------|--------------------|
| No of samples                       | 121                | 91                 | 70                 | 74                 |
| Conjugated bilirubin ( $\mu$ mol/L) | 12.6 ( $\pm$ 13.5) | 16.4 ( $\pm$ 26.1) | 24.1 ( $\pm$ 33.7) | 29.7 ( $\pm$ 32.4) |
| No of sample                        | 17                 | 15                 | 27                 | 29                 |
| Gamma-GT (IU/L)                     | 65.7 ( $\pm$ 47.4) | 52.5 ( $\pm$ 36.2) | 72.8 ( $\pm$ 40.2) | 76.3 ( $\pm$ 49.0) |

**Conclusions:** The level of conjugated bilirubin increases with the duration of PN in infants treated in our unit. However, our study also identified quality issues in terms of the monitoring of liver function.

## P425

# RENAL FUNCTION IN INDIGENOUS AND NON-INDIGENOUS PRETERM INFANTS

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**Background:** Preterm birth affects ~8% of all births in Australia with an increased incidence in the Indigenous population (~14%). Preterm infants are born at a time when nephrogenesis, the formation of nephrons in the kidney, is still ongoing. Renal function in preterm neonates may be adversely affected by renal immaturity and/or postnatal injury. However, to date, little is known about the effect of preterm birth and associated pre- and post-natal factors on renal function.

**Method:** Preterm Indigenous (I) and non-Indigenous (nI) infants were grouped according to gestational age at birth:  $\leq 28$  weeks (extremely preterm; n(I) = 6, n(nI) = 10), 29–31 weeks (very preterm; n(I) = 5, n(nI) = 11) and 32–36 weeks (moderately preterm; n(I) = 10, n(nI) = 10). Twenty-four hour pooled urine samples were obtained at days 3–7, 14, 21 and 28 of life via a cotton pad fitted to the nappies. From these samples, creatinine clearance (CrCl; estimate of glomerular filtration rate) and fractional excretion of sodium (Fe<sub>Na</sub>; estimate of tubular function) were calculated.

**Results:** During the first week of life, CrCl increased with gestational age at birth, while Fe<sub>Na</sub> decreased with increasing gestational and postnatal age in both groups. Interestingly, unlike CrCl, Fe<sub>Na</sub> during the first month of life was greater in extremely preterm Indigenous infants than non-Indigenous infants, indicating a difference in tubular function.

**Conclusions:** This study demonstrates that renal function in preterm neonates is significantly dependent on gestational and postnatal age in both Indigenous and non-Indigenous preterm infants, and that the kidneys of extremely preterm Indigenous Australians may be more vulnerable.

## P426

# THE POLYNESIAN PERINATAL PARADOX

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**Background:** Historical accounts of Polynesian populations by the early European explorers suggested that these populations enjoyed exceptionally good health. Anthropologists described individuals in these communities as some of the finest specimens in the human race. The aim of this review is to explore the hypothesis that Polynesians enjoy better perinatal outcomes even though they are economically disadvantaged.

**Methods:** Literature search for observational studies on perinatal outcomes among Polynesian populations. Medline, EMBASE and Google Scholar were searched electronically, using the keywords 'Polynesian', 'Maori', 'Pacific Islanders', 'Perinatal outcome', 'perinatal mortality' as keywords. All observational studies were included.

**Results:** Only six studies of acceptable quality were retrieved. Pacific women living in New Zealand had significantly lower preterm delivery rates and SGA when compared to Caucasians. Maori women in New Zealand had significantly lower rates of cesarean sections than non-Maoris. In one Australian study, Pacific women had a 30% greater risk of perinatal mortality than Australians in NSW, but no increased risk of late stillbirth in New Zealand. In Hawaii, the incidence of Down's syndrome for women over the age of 35 was reported to be significantly lower among Pacific Islanders than in Europeans. In a separate study in Hawaii, rates of LBW were significantly lower among Pacific women than the US national averages.

**Conclusions:** Some perinatal outcomes in Polynesian populations are better than in Caucasians living in the same area even though these women are socially and economically disadvantaged. There may be multiple explanations for this paradox, including social, environmental and genetic factors.



P427

# DOES PI3K-C2A HAVE A ROLE IN MATERNAL ENDOTHELIAL DYSFUNCTION IN PREECLAMPSIA?

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**Background:** Many of the features of preeclampsia are considered to arise, at least in part, from widespread maternal endothelial dysfunction. Recently, the class II  $\alpha$ -isoform of phosphatidylinositol 3-kinase (PI3K-C2 $\alpha$ ) was shown to be essential for normal endothelial cell function, including maintenance of vascular integrity and barrier function (1). We aimed to assess whether PI3K-C2 $\alpha$  expression was altered in preeclampsia.

**Methods:** At caesarean section, blood vessels were collected from healthy pregnant women (n = 6) and from women with preeclampsia (n = 5) and mRNA isolated. Human umbilical vein endothelial cells (HUVECs) were isolated and cultured *in vitro* for 6 hours in the presence of serum from either healthy pregnant women or preeclamptic women (n = 4 each group). mRNA was then isolated. PI3K-C2 $\alpha$  mRNA expression was measured in all samples by qPCR.

**Results:** There were no differences in PI3K-C2 $\alpha$  mRNA expression levels in blood vessels from preeclamptic and healthy pregnant women. Similarly, there was no change in PI3K-C2 $\alpha$  mRNA expression in HUVECs treated with serum from preeclamptic women compared with those treated with serum from healthy pregnant women.

**Conclusion:** It appears unlikely that altered endothelial cell PI3K-C2 $\alpha$  function contributes to the widespread endothelial dysfunction observed in women with preeclampsia.

## Reference

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P428

# CALCIUM CHANNEL BLOCKERS FOR INHIBITING PRETERM LABOUR: A COCHRANE SYSTEMATIC REVIEW

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**Background:** Preterm birth is a major contributor to perinatal mortality and morbidity. Calcium channel blockers (CCB) may be a better tocolytic with a lower adverse event profile than betamimetics agents.

**Method:** We report an update of a Cochrane review on the role of CCB as a tocolytic agent. Standard methods of the Cochrane Collaboration were used including fixed and random effects meta-analysis. Results are presented a Risk Ratio (RR) and Number Needed to Benefit (NNTB) with 95% Confidence Interval (CI).

**Results:** This update includes 29 trials with a total of 2840 women (an additional 17 trials, 1808 women) with an overall moderate risk of bias due to the lack of placebo control in all trials. Comparing CCB (mainly nifedipine) with any other tocolytic agent (mainly betamimetics) a significant reduction was seen in one primary outcome measure of birth within 48 hours of initiation of treatment (RR 0.79 [0.67, 0.93]; NNTB 23 95% CI 14 to 71). The use of CCB also reduced rates of preterm birth less than 34 weeks (RR 0.82 95% CI 0.70 to 0.97), respiratory distress syndrome, necrotising enterocolitis, and intraventricular haemorrhage, and

maternal adverse events (discontinuation of therapy: RR 0.22 95% CI 0.12 to 0.41).

**Conclusions:** CCB for women in preterm labour have benefits over other tocolytics in terms of postponement of birth (allowing time for administration of antenatal corticosteroids and transfer to higher level care) and some neonatal outcomes. However, no difference was shown in perinatal mortality and longer term outcomes could not be assessed.

P429

# THE COMPARISON OF NEWBORN CARE REQUIRED FOR BABIES WITH LOW APGAR SCORES BORN IN BUSINESS HOURS WITH THOSE AFTER-HOURS IN AN OUTER METROPOLITAN PAEDIATRIC UNIT

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**Background:** Studies from tertiary obstetric units show no diurnal variation in neonatal morbidity. However, the link between the time of delivery and the neonatal care required for babies born in Australian Peripheral Centres after hours has not been specifically looked at.

**Methods:** A retrospective cohort study of 23 neonates with Apgar score <7 at 5 minutes, at Campbelltown Hospital, January-July 2012. The data from chart reviews was categorised into business-hours (0800–1800) and after-hours (1800–0800), and outcomes were examined.

**Results:** The average gestation age was 39 and 37 weeks, for after-hours and business hours births respectively (T-test, p < 0.04).

The majority of babies (70%, n = 16) born after-hours, 50% (n = 8) being in theatres (Fisher's Exact Test, 2 sided p = 0.40). Cord gas was available for 35% (n = 8) of all babies.

Of the modes of deliveries after-hours, Emergency Caesareans (50%, n = 8) accounted for highest proportion followed by vacuum deliveries (31%, n = 5). Vaginal deliveries (58%) and Emergency Caesareans (28%) were commonest within business hours (Pearson Chi-square, 2-sided p = 0.08). During after-hours, most babies were admitted in Special Care Nursery (65%, n = 11), followed by NETS (Neonatal and Paediatrics Emergency Retrieval Transport System) retrievals (17%, n = 3). Nursery (50%, n = 3) and post-natal (33%, n = 3) ward admissions were commonest within business hours (Pearson Chi-square, 2-sided p = 0.08).

**Conclusions:** There is a suggestion babies born after hours required higher rates of intervention and retrieval. This raises questions into adequate resourcing, and careful timing of induction of labour and delivery modes in business hours.

P430

# WHAT IS THE OPERATIVE DELIVERY RATE OF THE OCCIPUT TRANSVERSE POSITION IN SECOND STAGE OF LABOUR?

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**Background:** Occiput Transverse (OT) position has a reported prevalence of 19%–37% at the onset of labour and 3%–16% at birth. There is scant evidence in the literature on the operative delivery rate of OT position in the second stage of labour.

**Aim:** To assess the operative delivery rate of the occipito-transverse position in the second stage of labour compared with occipito-anterior (OA).

**Methods:** Data were combined from two prospective studies of ultrasound determined fetal head position early in the second stage of labour. Data regarding demographics, mode of delivery, labour, postnatal period and neonatal outcomes were also collected. Logistic regression analysis was performed to assess the independent contribution of OT position to operative delivery (caesarean section, vacuum or forceps).

**Results:** 49% (39/80) women in the OT group and 31% (92/296) in the OA group had an operative delivery. Caesarean section was performed in

11% (9/80) and 5% (14/303) respectively. The OT group had a longer second stage (mean 2 hr 17 min versus 1 hr 49 mins). OT position early in the second stage of labour was associated with operative delivery in the logistic regression analysis (OR 2.03, 95% CI 1.1–3.7,  $p = 0.02$ ). Other factors which independently predicted operative delivery were nulliparity, abnormal second stage cardiotocograph, and epidural analgesia.

**Conclusion:** OT position early in the second stage of labour is strongly associated with operative delivery. There is potential to explore interventions such as manual rotation for reducing operative delivery for OT position.

## P431

## EXAMINING FETAL MONITORING USE AND INDICATIONS IN A LOW RISK MATERNITY UNIT

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**Background:** Casey Hospital opened as one of three campuses within Southern Health in 2005 providing maternity services to low risk women in the south eastern suburbs of Melbourne. Birthing numbers have grown from 500 to almost 1500 births per annum. In this time there has been an increase in the use of continuous fetal monitoring in labour and rates of caesarean section. This posed the question; has the use of continuous fetal monitoring at Casey Hospital contributed to an increased caesarean section rate without improved neonatal outcomes, and is this consistent with best practice.

**Method:** Births numbers per calendar year between 2005 and 2010 were audited comparing caesarean section rate; apgar scores less than seven at five minutes and numbers of women having continuous fetal monitoring in labour. This prompted a further audit of all births recorded from January 2012 to June 2012 examining whether indications for continuous monitoring were consistent with current RANZGOG guidelines.

**Results:** The incidence of continuous fetal monitoring increased up to 21 percent and emergency caesarean section up to 14 percent. Apgar scores less than seven at five minutes remained relatively consistent with a small upward trend.

**Conclusions:** Increased use of continuous fetal monitoring for low risk women at Casey appears to have no beneficial effect on neonatal outcomes but has prompted an investigation examining whether indications for monitoring were consistent with best practice. The results of the second audit were incomplete at the time of abstract submission but will be available at the conference.

## P432

## CORRELATION OF PRELABOR RUPTURE OF MEMBRANES AT TERM AND CESAREAN DELIVERY FOR CEPHALOPELVIC DISPROPORTION

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**Background:** Prelabor rupture of membranes (PROM) may represent an undiagnosed problem in the labor process, which may include cephalopelvic disproportion (CPD). Delay in the diagnosis of CPD in patients with PROM may lead to hazardous complications of both mother and child. This study aims to determine the correlation of PROM and the incidence of caesarean section for CPD.

**Method:** The study sample consisted of 116 patients who had PROM. Data were compared with 913 women admitted with intact membranes. Factors including maternal age, parity, age of gestation (AOG), and birth weight were analysed using univariate and multivariate analysis.

**Results:** 1029 low risk, term patients were included in this study. 913 (88.72%) had intact membranes, 116 (11.27%) had PROM on admission. Univariate analysis showed association between PROM and caesarean delivery for CPD ( $p = 0.024$ ), with 1.5 times greater (CI: 95%, 1.070–2.217)

risk of caesarean delivery among those with PROM compared to those with intact membranes. Variables significantly associated with increased incidence of caesarean delivery for CPD were birth weight  $\geq 4000$  grams (CI 95% 1.003–35.908), AOG  $< 41$  weeks (95% CI, 1.108–2.340), and maternal age  $< 35$  years (95% CI, 1.112–2.375). Strongest predictors of caesarean delivery in PROM patients are nulliparity and AOG  $< 41$  weeks.

**Conclusions:** This study showed a positive correlation between PROM and increased caesarean delivery for CPD (RE: 1.54, 95% CI). Among patients with PROM, presence of fetal and maternal factors, including birth weight  $\geq 4000$  grams, AOG  $< 41$  weeks, and nulliparity, could account for the increased risk of caesarean delivery.

## P433

## EVALUATION OF PRINTED NUTRITION EDUCATION MATERIAL IN OVERWEIGHT AND OBESE WOMEN DURING PREGNANCY – FINDINGS FROM THE LIMIT RANDOMISED TRIAL

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**Background:** The LIMIT Study was a large multicentred randomised controlled trial evaluating a lifestyle intervention to limit gestational weight gain in overweight and obese women. Women participating in the intervention arm of the LIMIT Study received a comprehensive lifestyle intervention supervised by a dietitian and written printed education materials to assist them make healthy diet and exercise choices. The aim of this current study was to determine the uptake of printed education materials.

**Method:** Participants in the intervention group were asked to evaluate the written diet and exercise educational materials four months after the birth of their baby by completing a written questionnaire. Participants were asked structured questions related to the usefulness and acceptability of the questionnaires and responded using a 5 point Likert scale. Responses were tabulated.

**Results:** A total of 517 of women completed questionnaires about the printed education materials. Women who completed the questionnaire found the books useful in helping them to make healthier food choices and making time to exercise.

Of these women 61.3% strongly agreed/agreed that the diet information book helped them make healthier food choices in pregnancy. However 49.3% said they never/rarely used the book. 48.7% strongly agreed/agreed that the exercise book helped them make time to exercise. However, 61.5% never/rarely referred to the book.

**Conclusions:** Women who responded to the questionnaire viewed the contents of the printed education materials favorably though did not continue to use the resources after their babies were born.

## P434

## DIETARY ADVICE IN PREGNANCY FOR PREVENTING GESTATIONAL DIABETES MELLITUS – A COCHRANE REVIEW

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**Background:** Gestational diabetes mellitus (GDM) is associated with significant adverse outcomes for women and their babies. This Cochrane review evaluates the role of dietary advice in pregnancy in preventing GDM and improving maternal and infant health outcomes.

**Method:** We used standard Cochrane methodology (last search February 2012).

**Results:** Eight trials, with a moderate risk of bias, (involving over 1600 women) were included in this review. Any dietary advice compared with no advice was associated with a trend towards reduction in GDM (RR 0.40; 95% CI 0.16 to 1.01; 3 trials, 898 women). In one trial, women receiving probiotics and dietary fat-focused advice had a lower incidence of GDM than women receiving standard dietary advice alone (RR 0.38; 95% CI 0.20 to 0.74; 149 women). Women receiving any dietary advice compared with no advice had significantly lower pregnancy weight gain (MD -4.89 kg; 95% CI -9.19 to -0.58; 3 trials, 933 women). Babies born to mothers who consumed a low-GI compared with a high-GI diet were significantly less likely to be large-for-gestational age (RR 0.09; 95% CI 0.01 to 0.69; 1 trial, 62 women), had a lower ponderal index and birthweight centile. No significant differences were seen for perinatal mortality, macrosomia, or mode of birth across any comparisons.

**Conclusions:** Some evidence supports the use of dietary advice for the prevention of GDM and associated adverse outcomes. Larger, well-designed trials focused on the optimal dietary intervention are required to make recommendations for clinical practice. Some are underway.

#### P435

##### BREASTFEEDING AND MOTHER'S MENTAL ILLNESS

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**Background:** To investigate the relationship between breastfeeding and postpartum mental illness.

**Method:** This is an historical cohort study based on linked population data using the New South Wales (NSW) Midwives Data Collection (MDC) and Admitted Patients Data Collection (APDC). All mothers with live births between 2007 and 2008 in NSW, Australia were followed up for one year for hospital admissions.

**Results:** Any breastfeeding rate in NSW between 2007 and 2008 was 87.11% and full breastfeeding rate was 80.63%. The first hospital admission for diagnoses of unipolar depression, adjustment disorders or mental illness due to substance use were earlier (37, 31 and 23 days respectively) in non-breastfeeding mothers compared with full breastfeeding mothers. Mothers who did not breastfeed were more likely to be admitted to hospital in the first year postpartum for schizophrenia (adjusted relative risk (ARR) = 2.07, 95% confidence interval (CI): 1.34–3.23); bipolar affective disorders (ARR = 1.98, 95% CI: 1.09–3.58); mental illness due to substance use (ARR = 1.79, 95% CI: 1.28–2.50) compared with full breastfeeding. On the other hand, mothers who did not breastfeed were less likely to be admitted to hospital in the first year postpartum for anxiety disorders (ARR = 0.64, 95% CI: 0.46–0.89) compared with full breastfeeding.

**Conclusion:** Breastfeeding decreases the risk of mother's hospital admissions for schizophrenia, bipolar affective disorders and mental illness due to substance use during the first year postpartum.

#### P436

##### DIFFERENTIAL EFFECTS OF CIGARETTE SMOKE ON BIRTH WEIGHT BY MATERNAL BODY MASS INDEX

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**Background:** The adverse effect of cigarette smoking on birth weight is well known, but the interaction with maternal obesity is poorly understood. The objective of this study was to assess the effects of cigarette smoking on birth weight in different categories of maternal body mass index (BMI).

**Methods:** An obstetric database of 13473 live singleton pregnancies delivered at Nepean Hospital between 1998 and 2003 was studied. Maternal body mass index categories were defined as BMI < 25, BMI > 30 and BMI 25–30. Pregnancy characteristics included in the analysis were self-reported number of cigarettes smoked, age, parity, weight, height, and BMI. Multivariate regression analysis was performed for each category of maternal body weight, to identify the significant predictors of birth weight and the effect of smoking.

**Results:** The mean maternal age was 27.2. Mean birth weight was 3377 g (SD 588 g). The mean BMI was 27.1 (SD 6.1). There was no significant correlation between BMI and number of cigarettes smoked. Regression analysis showed that for women with BMI < 25, the effect of smoking on birth weight was -13.4 g per cigarette, for BMI 25–30 the effect was -12.9 g per cigarette and for BMI > 30 the effect was -10.9 g per cigarette.

**Conclusions:** The adverse effects of cigarette smoking on birth weight may be modulated by maternal body index, with babies of obese women being somewhat less susceptible to the effect of tobacco smoke than lean women.

#### P437

##### THE EFFECT OF ANTENATAL DIETARY AND LIFESTYLE ADVICE ON MATERNAL PSYCHOLOGICAL HEALTH IN WOMEN WHO ARE OVERWEIGHT OR OBESE – FINDINGS FROM THE LIMIT RANDOMISED TRIAL

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**Background:** Overweight and obesity are recognised risk factors for depression, anxiety and other psychological conditions. While there are well documented risks associated with obesity during pregnancy and childbirth, the effect of increasing maternal BMI on depressive and anxiety related symptoms and quality of life in the perinatal period has been poorly documented.

**Methods:** We conducted a randomised controlled trial evaluating the effect of a dietary and lifestyle intervention for women who were overweight or obese during pregnancy on maternal and infant health outcomes. Women were eligible for the LIMIT trial with a BMI ≥ 25 kg/m<sup>2</sup>, and a live singleton pregnancy.

Women completed questionnaires assessing quality of life (SF-36 Health Survey Questionnaire), depression (Edinburgh Postnatal Depression Scale), and anxiety (the Short Form Spielberger State Trait Inventory), at trial entry, at 28 and 36 weeks' gestation, and 4 months post-partum. Potential differences between treatment groups and over time were explored using mixed effects models.

**Results:** Almost 2,000 women contributed data to this analysis. While there were no significant differences in measures of quality of life, depression and anxiety between the two treatment groups, the risk of depression and anxiety improved over the course of pregnancy and the post-partum period. Components of the SF-36 deteriorated over the course of pregnancy, particularly those related to physical functioning, but improved in the post-partum period.

**Conclusions:** Providing an antenatal dietary and lifestyle intervention to pregnant women who were overweight or obese was not associated with an increased risk of adverse psychological outcomes.



P438

**PREDICTING POST-NATAL DEPRESSION IN MOTHERS WHO DELIVER VERY PRETERM**Hagan R<sup>1</sup>, Hagan RA<sup>2</sup>, Campbell C<sup>1,3</sup>, & French N<sup>1,3</sup><sup>1</sup>UWA Centre for Neonatal Research & Education, School of Paediatrics, <sup>2</sup>Department of Psychological Medicine, King Edward Memorial Hospital Western Australia, <sup>3</sup>Neonatal Follow-up Program, King Edward Memorial Hospital Western Australia Email: C/- Catherine.Campbell@health.wa.gov.au**Background:** To explore factors associated with depression in mothers of very preterm (VP) infants during the first year.**Method:** Cohort study of mothers enrolled in a randomised trial on prevention of postnatal depression or its pilot study. 225 mothers assessed using the Schedule for Affective Disorders and Schizophrenia (SADS) and DSM-IV criteria for depression, stress disorders and maternal mental health history were examined at 2 weeks, 2, 6 and 12 months postnatal. **Results:** 79 mothers (35%) met criteria for a diagnosis of depression during their child's first year. 42 were diagnosed while their child was still in hospital. 75 mothers (33%) had a history of depression and 33 mothers (14%) had an antenatal episode of depression during this pregnancy. Depression was significantly more common in these mothers: 42% with positive history only; 73% with antenatal episode only; 83% with both; versus 21% with neither. Logistic regression showed antenatal depression, past history of depression and maternal pre-eclampsia as the only factors significantly associated with an episode of depression. Socio-demographic factors, desire for children, planned pregnancy, assisted pregnancy and a negative labour and delivery experience were not associated with an elevated risk. Fifty of the 79 mothers (63%) who developed depression had either a history of depression or an antenatal episode and could have been identified on admission to the unit.**Conclusions:** Depression is common in mothers following very preterm delivery. Most of these mothers can be identified by their clinical history.

P439

**WALKING TO LIMIT GESTATIONAL WEIGHT GAIN AND KEEP FIT DURING PREGNANCY – FINDINGS FROM THE WALK RANDOMISED TRIAL**Sui Z<sup>1</sup>, Yelland LN<sup>1</sup>, Turnbull D<sup>3</sup>, Dodd JM<sup>1,2</sup><sup>1</sup>The University of Adelaide, Robinson Institute, Discipline of Obstetrics & Gynaecology, Women's and Children's Hospital, Adelaide, Australia, <sup>2</sup>Department of Perinatal Medicine, Women's and Children's Hospital, Adelaide, Australia, <sup>3</sup>The University of Adelaide, School of Psychology, Adelaide, Australia Email:zhixian.sui@adelaide.edu.au**Background:** Overweight and obesity in pregnancy is associated with a number of adverse maternal and infant health outcomes. The aim of this randomised trial was to evaluate the effect of a supervised walking group to improve physical activity for pregnant women who are overweight or obese.**Method:** We conducted a nested randomised trial evaluating the effect of a supervised walking group for women who were overweight or obese during pregnancy, on maternal physical activity and other health outcomes. Women were eligible for the LIMIT trial with a BMI  $\geq 25$  kg/m<sup>2</sup>, and a live singleton pregnancy. Women who were randomised to the intervention group of the LIMIT trial were further randomised to participate in a supervised walking group in addition to receiving the standard written intervention materials (the WALK group), or to receive the written information only (the control group). The walking group was supervised by a trained researcher, and designed to cover a distance of 2.2 km. The sessions were held in a variety of both indoor and outdoor settings across the Adelaide metropolitan area.

Baseline demographic details, physical activity (self reported by questionnaire and converted to Metabolic Equivalent of Task (MET units) at baseline and 36 weeks' gestation), and clinical maternal and infant outcomes were collected.

**Results and conclusion:** Outcomes from the WALK randomised trial will be presented and the implications for further research and clinical practice discussed.

P440

**ENABLERS AND BARRIERS TO MAKING HEALTHY CHANGE DURING PREGNANCY IN OVERWEIGHT AND OBESE WOMEN**Sui Z<sup>1</sup>, Turnbull DA<sup>2</sup>, Dodd JM<sup>1</sup><sup>1</sup>Discipline of Obstetrics & Gynaecology, University of Adelaide, Adelaide, Australia, <sup>2</sup>School of Psychology, University of Adelaide, Australia Email: zhixian.sui@adelaide.edu.au**Background:** High Body Mass Index (BMI) during pregnancy is associated with high risk of adverse health outcomes. Making healthy change during pregnancy is beneficial, especially for overweight and obese women. The aim of this study was to explore the barriers and enablers identified by overweight and obese women that preclude and facilitate making healthy change during pregnancy.**Methods:** 26 women who were overweight or obese during pregnancy were interviewed at 28 weeks gestational age. Interviews were transcribed verbatim and analysed using a grounded theory approach.**Results:** Overweight and obese women cited the same barriers to making healthy change during pregnancy: physiological (eg, pregnancy complications and tiredness); emotional (preference and mood); cognitive (eg, lack knowledge); interpersonal (eg, lack support); and environmental (eg, availability of healthy choice). Similar enablers were identified by overweight and obese women: physiological (eg, relief pregnancy symptoms); emotional (eg, approval and commitment); cognitive (eg, knowledge and belief); interpersonal (eg, family support); and environmental (eg, help from community). More barriers than enablers were given, indicating that these women were more sensitive to barriers than the enablers for making healthy change. Factors viewed by some women as barriers were viewed as enablers by others.**Conclusions:** When designing interventions to promote eat healthily and being active during pregnancy, individually focused interventions must be implemented in conjunction with environmental-level interventions to facilitate behaviour change.

P441

**CONTINUOUS VERSUS BOLUS INTRAGASTRIC TUBE FEEDING FOR PRETERM AND LOW BIRTH WEIGHT INFANTS WITH GASTRO-OESOPHAGEAL REFLUX DISEASE**Foster JP<sup>1</sup>, Richards R<sup>2</sup>, Psaila K<sup>3</sup><sup>1</sup>University of Western Sydney, Sydney, Australia, <sup>2</sup>Liverpool Hospital, Liverpool, Australia, <sup>3</sup>University of Western Sydney, Sydney, Australia Email: j.foster@uws.edu.au**Background:** Gastro-oesophageal reflux (GOR) is the term used to describe the reflux of stomach contents into the oesophagus. When retrograde movement of gastrointestinal contents results in sequelae such as injury to the oesophagus or supra-oesophageal structures, it is considered pathologic and the infant is said to have gastro-oesophageal reflux disease (GORD). GORD is particularly common in preterm and low birth weight infants. Chronic lung disease of infancy (CLD) is a chronic condition that usually evolves after premature delivery and respiratory distress syndrome. Small, fragile infants with CLD are prone to GORD, which may complicate enteral feeding and worsen an already compromised respiratory system by causing asymptomatic aspiration or triggering bronchospasm. Our aim in this systematic review was to determine if continuous or intermittent bolus intragastric tube feeding reduces the number of episodes and duration of GOR in preterm and low birth weight infants.**Method:** The standard search strategy of the Neonatal Review Group as described in the Cochrane Library was used.**Result:** We found no trials that met the inclusion criteria for this review.**Conclusions:** There is insufficient evidence to recommend either strategy. Well-designed and adequately powered randomised controlled trials are needed.

## P442

## THE DEVELOPMENT AND APPLICATION OF A NUTRITION CALCULATOR: 'THE NUTRICALC'

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**Background:** On average 9.3 parenteral nutrition (PN) prescriptions are written each day in our NICU. Standardised PN formulations reduce prescribing error, however a recent review of our practice suggested that recommended nutritional intakes were not being met in many infants. We developed a nutritional intake calculator app ('The Nutricalc') to assess if compliance with PN guidelines could be improved.

**Methods:** 100 prescriptions were prospectively reviewed with the Nutricalc and compliance with the European Society for Paediatric Gastroenterology, Hepatology and Nutrition PN guidelines was assessed.

**Results:** 77% of PN prescriptions did not achieve target energy intake and 35% failed to meet target protein intake. Mean (SD) energy and protein deficit was 25.6% (19.6%) and 34.6% (24.6%) of recommended daily intakes respectively. Using the Nutricalc, potential alterations to improve nutritional intake were identified in 27.3% of energy deficient and 48.6% of protein deficient prescriptions, providing an additional 15.6 kcal/kg/day of energy and 0.7 g/kg/day of protein on average.

48% and 11.4% of energy and protein deficient prescriptions respectively, were in infants receiving maximal PN according to our unit protocol. Lack of central venous access, fluid restriction or additional medication prevented further optimisation in the remaining cases.

**Conclusions:** Energy and protein deficits are common with standard PN formulations. The use of an integrated bedside nutritional intake calculator, may help clinicians identify those infants who are not receiving recommended nutritional intake and optimise PN prescribing.

## P443

## HUMAN MILK XANTHINE OXIDASE AND NEONATAL SALIVARY NUCLEOTIDE PRECURSORS GENERATE HYDROGEN PEROXIDE; A NOVEL PATHWAY WITH POTENTIAL ROLE IN REGULATING ORAL MICROFLORA

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**Background:** Xanthine oxidase (XO) is a molybdeno-flavoprotein occurring with high activity in the milk fat globule membrane and is involved in the final stage of degradation of purine nucleotides. It catalyzes the sequential oxidation of hypoxanthine to xanthine and uric acid, accompanied by production of hydrogen peroxide and superoxide anion. Human saliva has been extensively described for its composition of proteins, electrolytes, cortisol, melatonin and some metabolites such as amino acids, but little is known about nucleotide metabolites.

**Method:** Saliva was collected with swabs from 60 babies at full-term 1–4 days, 6-weeks, 6-months and 12-months. Unstimulated fasting (morning) saliva samples were collected directly from 77 adults. Breast milk was collected from 24 new mothers. Saliva was extracted from swabs and ultra-filtered. Nucleotide metabolites were analyzed by RP-HPLC with UV-photodiode array and ESI-MS/MS mass spectrometry. XO activity was measured as peroxide production from hypoxanthine. Bacterial inhibition over time was assessed using CFU/mL or OD.

**Results:** Median concentrations (μmol/L) of salivary nucleobases and nucleosides for neonates/6-weeks/6-months/12-months/adult respectively were: uracil 5.3/0.8/1.4/0.7/0.8, hypoxanthine 27/7.0/1.1/0.8/2.0, xanthine 19/7.0/2.0/2.0/2.0, adenosine 12/7.0/0.9/0.8/0.1, inosine 11/5.0/0.3/0.4/0.2, guanosine 7.0/6.0/0.5/0.4/0.1, uridine 12/0.8/0.3/0.9/0.4. Deoxynucleosides and dihydropyrimidines concentrations were essentially negligible. XO activity (Vmax:mean ± SD) in breast milk was 8.9 ± 6.2 μmol/min/L and endogenous peroxide was 27 ± 12 μmol/L; mixing breast milk with neonate saliva generated ~40 μmol/L peroxide, which inhibited *Staphylococcus aureus*.

**Conclusions:** Salivary metabolites, particularly xanthine/hypoxanthine, are high in neonates, transitioning to low adult levels between 6-weeks to 6-months (P < 0.001, Mann-Whitney). Peroxide occurs in breast milk and is boosted during suckling as an antibacterial system.

## P444

## SMOFLIPID USE IN LONG TERM PARENTERAL NUTRITION

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**Background:** Short-term use of SMOFlipid has beneficial effects on liver function and biochemical markers of inflammation and oxidation. It is not known if these beneficial effects persist with long-term parenteral nutrition (PN) administration. We aimed to investigate the effect of SMOFlipid on liver function in surgical infants requiring long-term PN.

**Methods:** Retrospective review of liver function tests for infants receiving SMOFlipid for >7 days between January 2011 and June 2012. Data were analysed according to type of lipid and duration of PN received. Cholestasis was defined as (conjugated bilirubin (CB) >50 micromol/litre).

**Results:** 63 infants received SMOFlipid. 1 infant was excluded due to significant liver dysfunction following a hypoxic insult. Mean (SD) gestational age was 34.4 (4.9) weeks, birth weight 2266 (1010) grams and duration of PN 40 (29) days. 22 infants received clinoleic for a mean (SD) of 18 (13) days then transitioned to SMOFlipid. 40 infants received SMOFlipid only.

CB was significantly lower in those infants who received SMOFlipid only for 4–8 weeks compared with those who received clinoleic & SMOFlipid (Table 1).

In both lipid groups CB and ALT increased with duration of PN, with the exception of ALT in infants who received 0–4 weeks of SMOFlipid. ALT was significantly higher in infants who received 0–4 weeks and >8 weeks of SMOFlipid compared to those who received clinoleic & SMOFlipid for the same duration.

**Conclusions:** Early SMOFlipid use should be considered for those infants likely to require PN for greater than 4 weeks, however no advantage was found with short or long term use.

Table 1. CB and ALT for PN duration and lipid group. Mean (SD)

| PN Duration Group | Clinoleic & SMOFlipid |                 |               |                 | SMOFlipid only |                 |               |                 |
|-------------------|-----------------------|-----------------|---------------|-----------------|----------------|-----------------|---------------|-----------------|
|                   | n =                   | Cholestasis n = | CB μmol/litre | ALT units/litre | n =            | Cholestasis n = | CB μmol/litre | ALT units/litre |
| 0–4 week group    | 5                     | 2               | 13.1 (15.8)   | 29.5 (6.8)      | 24             | 2               | 18.1 (19.9)   | 89.5 (79.1)**   |
| 4–8 week group    | 10                    | 4               | 8.5 (7.3)     | 44.1 (28.9)     | 8              | 0               | 4.2 (4.1)*    | 33.7 (11.3)     |
| 8 week group      | 7                     | 2               | 21.7 (20.9)   | 43.5 (29.8)     | 8              | 3               | 23.8 (19.4)   | 56.2 (32.7)†    |

\*p = 0.0087.

\*\*p = &lt;0.0001.

†p = 0.0002.

P445

## CAN BIOMARKERS BE USED TO PREDICT THE OUTCOMES OF NECROTISING ENTEROCOLITIS?

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**Background:** Necrotising enterocolitis (NEC) is the most serious gastrointestinal disease of prematurity with up to 50% of cases requiring surgery and 30% dying. Morbidity can be significant amongst survivors with increased rates of neurodevelopmental impairment. The aetiology is multifactorial and no specific treatment exists. The objective of this study was to identify biomarkers that may predict adverse NEC outcomes.

**Method:** This was a retrospective cohort study on eligible infants cared for in the neonatal unit at RWH between January 2001–December 2011. Inclusion criteria were: birth gestation <32 completed weeks and birth weight <1500 grams. Records of infants diagnosed with NEC ≥ Stage 2 were reviewed for data on demographic factors, disease profile, biomarkers of disease severity, and the outcomes of surgery and death.

**Results:** 2,462 infants met the inclusion criteria. A total of 96 had been diagnosed with NEC ≥ Stage 2; 43% of cases required surgery and 44% died. Lactate and CRP at diagnosis and 72 hours later were the most promising predictive biomarkers (See Table 1). Specifically, lactate at diagnosis was moderately predictive of death and at 72 hours after diagnosis was very predictive of death.

**Conclusions:** The use of specific biomarkers to predict adverse outcome at an earlier stage in the course of NEC has the potential to provide an opportunity for earlier intervention.

**Table 1.** Area under the curve analysis for specific biomarkers and outcomes

| Biomarker                     | Surgery<br>AUC [95% CI] | Death<br>AUC [95% CI] |
|-------------------------------|-------------------------|-----------------------|
| CRP at diagnosis              | 0.52 [0.40–0.64]        | 0.56 [0.45–0.68]      |
| CRP 72 hrs post-diagnosis     | 0.54 [0.41–0.67]        | 0.59 [0.46–0.71]      |
| Lactate at diagnosis          | 0.62 [0.46–0.78]        | 0.75 [0.61–0.88]      |
| Lactate 72 hrs post-diagnosis | 0.54 [0.34–0.73]        | 0.93 [0.84–0.99]      |

AUC = area under the curve, CI = confidence interval, CRP = C-reactive protein, hrs = hours.

P446

## DOES DESIGN MAKE A DIFFERENCE? COMPARISON OF NOISE, LIGHT AND TEMPERATURE IN OPEN PLAN AND TWO COT ROOM DESIGN

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**Background:** Current research recommends small room design (SRD) in Neonatal Intensive Care Units (NICU). SRD reduces noise and allows the individualisation of light and temperature to meet each neonate's requirements. In August 2012 Canberra NICU relocated from an open plan (OPD) to a two cot room (TCR) design. An environmental study was undertaken to compare noise, light (lux) and temperature between OPD and TCR.

**Method:** As per *Australian Standards* five *Edge* dosimeters were configured to log data using noise sensitivity scales: A-weighting to measure an average sound level over 8 hours (LAeq8h) and C-weighting to measure high noise levels (LCpeakdB). A *HOB0* logger measured light and temperature. Recordings were logged every minute. Meters were positioned in proximity to individual cots/isolettes to assess the environment. Meters were relocated, recharged and downloaded daily. Records were logged in

intensive care (IC), high dependency (HD) and special care (SC) cot spaces in both designs.

**Results:** In OPD noise recordings were logged in 34 different cot spaces, light and temperature readings in 7 spaces, averaging between 960–1560 readings per space. Results are shown in Table 1. Preliminary analysis of OPD data has shown noise was highest in HD, light in IC and temperature in IC. Analysis of TCR data and the comparison between the two designs is currently in progress with results to be made available for PSANZ 2013.

**Conclusion:** Analysis of noise, temperature and light in OPD shows that neonates were subjected to large variations and that noise was significant.

**Table 1.** Open Plan NICU Results

| Open Plan NICU  | LCpeakdB |       | LAeq8h |      | Lux  |       | Temperature |      |
|-----------------|----------|-------|--------|------|------|-------|-------------|------|
|                 | Min      | Max   | Min    | Max  | Min  | Max   | Min         | Max  |
| Intensive Care  | 107.0    | 136.7 | 60.4   | 63.6 | 00.0 | 484.4 | 16.4        | 25.1 |
| High Dependency | 110.3    | 137.9 | 60.9   | 65.4 | 10.8 | 258.3 | 23.1        | 24.4 |
| Special Care    | 115.6    | 126.8 | 62.9   | 64.5 | 10.8 | 129.2 | 23.1        | 25.8 |

P447

## IMPROVING SKINCARE IN THE NICU-CANBERRA NEONATAL SKIN RISK ASSESSMENT SCALE (CNSRAS)

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**Background:** During their admission to a Neonatal Intensive Care Unit (NICU) 10–20% of neonates may acquire a skin injury. In 2010 the Canberra NICU conducted a prevalence and incidence study (PAIS) to assess neonatal skin injuries. Study results confirmed the high number of skin injuries in neonates and the importance of identifying and documenting risk factors, causes, types and management of skin injuries. The PAIS research team acknowledged the need for a risk assessment tool to improve neonatal management and outcomes.

**Method:** The group reviewed currently available tools and found them to be inappropriate for neonates. A new tool, based on the result of the literature review and results of the PAIS for the assessment of skin injuries in neonates was developed.

**Results:** The Canberra Neonatal Skin Risk Assessment Scale (CNSRAS) lists eight risk categories: gestational age, sensory perception, activity/mobility, moisture, respiratory support, visual examination, blood collection and nutrition. These categories are graded from 1–4 with 4 being the highest risk of injury. The total score indicates the neonate's level of risk to skin injuries grading them from low to extreme risk. The scale then guides the nurse caring for the neonate with a table of suggested actions, reassessment guidelines and an individualised care plan. Results of the 12 month evaluation of CNSRAS will be presented.

**Conclusion:** CNSRAS will provide a scale that will facilitate early risk identification and prevention of skin injuries essential in providing holistic nursing care in a neonatal unit.

P448

## QUINTUPLET DELIVERY: A NEONATAL UNIT'S EXPERIENCE

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**Background:** Higher order deliveries present organizational and clinical challenges to a nursery. Quintuplets were delivered at Monash Medical Centre in 2012.

**Method:** In preparation for the delivery, additional medical and nursery staff were rostered on call, from 25 weeks gestation. The newborn transport service was alerted in case all five babies could not be accommodated in a single nursery. An increased number of resuscitators were also



provided to the hospital. To minimize the risk of errors, colour-coded hats and linen were prepared for each infant, with matching labels for staff involved in the delivery. Elective caesarean section was performed at 34 weeks gestation following demise of one fetus. Prior to attending the birth, staff teams were allocated to individual babies. Cot allocations, in theatre and the nursery, were pre-allocated. The surviving four babies were admitted in pairs to separate rooms in the NICU to manage staff workloads on admission, and limit identification errors.

**Results:** Co-ordination from delivery to admission was a smooth process. Staff reported a high level of satisfaction with this outcome. All babies had an uneventful clinical course and were discharged home before the expected date of confinement.

**Conclusions:** Extensive planning can facilitate efficiency and safety during the delivery and admission of higher order multiples.

P450

#### OSTEOPENIA OF PREMATURITY: A SURVEY OF PRACTICES IN NEW ZEALAND

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**Background:** Osteopenia of Prematurity (OP) is common in preterm babies. There are several biochemical markers and radiological investigations used for screening. These may not be sensitive enough or impractical. Currently there is no standardised screening or management guideline available. (1)

**Method:** To evaluate current screening and management methods for OP in neonatal units across New Zealand, anonymous online questionnaires were sent to Paediatricians and Neonatologists in level 2 and 3 neonatal units across New Zealand. Data were collected using 'SurveyMonkey' between February and May 2012.

**Results:** Seventy two (57%) of the 127 clinicians responded to the survey of which 66 (92%) confirmed screening for OP. Thirty two (48%) screened infants <32 weeks gestation. Testing included serum alkaline phosphatase (ALP) (n = 66, 100%); serum calcium (n = 48, 72%) and phosphate (n = 50, 75%). The frequency of screening was weekly (n = 19, 28%) or fortnightly (n = 29, 44%). Treatment was based on level of ALP (n = 47, 71%), and included phosphate (n = 22, 33%); combination of phosphate and vitamin D (n = 13, 20%); vitamin D (n = 11, 17%); and triple therapy with calcium, phosphate and vitamin D (n = 7, 10%). Decision to stop treatment was mostly based on the improvement of ALP levels (n = 42, 64%).

**Conclusions:** While most clinicians in New Zealand considered screening for OP, there is a wide variation in practice. Increasing awareness necessitates the development of an evidence-based best practice guideline for its screening and management.

#### Reference

1. Harrison CM, Johnson K, McKechnie E. Osteopenia of prematurity: a national survey and review of practice. *Acta Paediatrica* 2008;97: 407–413.

P451

#### BIRTHS FROM ASSISTED CONCEPTION AT A REGIONAL REPRODUCTIVE CENTRE – IMPACT ON THE TERTIARY NEONATAL UNIT AT NORTH BRISTOL NHS TRUST, BRISTOL, UK

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**Background:** The assisted reproductive technologies have been widely in use in the treatment of sub-fertility over the last 30 years. Currently 1.7% of all births in the UK occur after assisted conception.

**Objectives:** We aimed to look at

- the birth rate due to assisted conception at North Bristol NHS Trust
- the neonatal unit admission rate of babies born following assisted conception

- the impact of these births on the neonatal unit workload at Southmead Hospital

**Method:** We identified women who received assisted conception treatment at Bristol Centre for Reproductive Medicine and delivered their babies during April 2009 to December 2009 at Southmead hospital. We have collected the data about the babies' admission to the neonatal unit and calculated the intensive care (ITU), high dependency (HDU) and special care activity generated by them.

**Results:** 2% of the births at Southmead hospital were following assisted conception. The admission rate of the babies due to assisted conception was 44%, which is much higher than the admission rate of babies born after natural conception i.e. 7.7%. They constitute 10.8% of total admissions to our neonatal unit. The ITU, HDU and special care activity due to babies of assisted conception constitute 2.2%, 2.6% and 4.9% of total ITU, HDU and special care activity in our unit respectively.

**Conclusions:** Babies born after assisted conception at North Bristol NHS trust have very high admission rate to the neonatal unit. However, the majority of them received special care for very short period.

P452

#### NEONATAL SEPSIS IN NICU – A REGIONAL POPULATION STUDY OF TRENDS

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**Background:** This study aimed to determine the trends in early- (EOS) and late-onset (nosocomial) sepsis (LOS), the spectrum of organisms responsible, and outcomes in infants <32 weeks gestational age (GA) with neonatal sepsis.

**Method:** Retrospective cohort study of the New South Wales (NSW) and Australian Capital Territory (ACT) Neonatal Intensive Care Units' Data Collection (NICUS) of all gestations admitted between 1995 and 2009.

**Results:** Included were 31 794 neonates of all gestational ages. 4394 episodes of neonatal sepsis occurred in 3571 infants over 15 years. There was a marked reduction in rates of both EOS and LOS per 1000 patient enrolments over this period (from 19.07 to 12.67 for EOS, and 150.1 to 102.4 for LOS) ( $P < 0.001$ ). Coagulase negative Staphylococcus species (CoNS) remained the major organism responsible for both EOS and LOS, cultured in 22.9% of all EOS cases and 61.8% of all LOS. EOS in infants <32 wks GA was associated with higher rates of mortality than those without (34.4% vs. 10.9%) ( $P < 0.001$ ). EOS was an independent risk factor for mortality ( $P < 0.001$ , OR 2.79, 95% CI 2.0–3.9).

**Conclusions:** Rates of neonatal sepsis fell significantly over the study period, but there was no change in the major causative organisms of either EOS or LOS. Despite its low incidence, EOS remains a major cause of morbidity and mortality in infants born at <32 wks GA.

P453

#### CRP VERSUS I/T RATIO IN SUSPECTED NEONATAL SEPSIS

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**Background:** The management of neonatal sepsis is difficult because clinical presentations are non-specific, and laboratory testing lacks sensitivity and specificity. This retrospective study evaluates the use and effectiveness of C-reactive protein (CRP) and immature: total neutrophil ratio (I/T-ratio) as markers of the status of neonatal infection during antibiotic therapy.

**Methods:** Retrospective patient data (n = 136) was collected on neonates of gestational age 33–42 weeks admitted to The Northern Hospital special care nursery (SCN) for suspected neonatal sepsis treated with at least 2

days of IV antibiotics from September 2010 to August 2011. Data included: maternal risk factors; neonatal risk factors; clinical signs of neonatal infection and outcomes; CRP (measured via the turbidimetric method) and I/T-ratios (raised CRP > 8.0 mg/L; I/T-ratio >0.2). The cohort was divided into clinically well and unwell subgroups. For paired data, CRP and I/T-ratio were compared statistically (well n = 35, unwell n = 80). Regression analysis was performed on CRP and I/T-ratio versus antibiotic duration (well n = 42, unwell n = 94).

**Results:** More than 40% of all paired CRP and I/T-ratio data were inconsistent with regard to infection status. In the unwell group the paired data was significantly discordant ( $p < 0.05$ ). Substantial correlation was only observed for CRP ( $R^2 = 0.95$ ) versus antibiotic duration in the unwell group. Following antibiotics, all neonates had good outcomes.

**Conclusion:** I/T-ratio is not useful as a marker of the status of clinical infection; neither is CRP in the clinically well neonate. In the unwell neonate, CRP provides guidance on duration of antibiotic therapy. Current antibiotic management provides satisfactory outcomes.

#### P454

### SKIN ANTISEPSIS SURVEY IN AUSTRALIA-NEW ZEALAND NEONATAL NURSERIES

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**Background:** Extremely preterm sick infants, are vulnerable to blood-stream infection due to poor skin barrier, and invasive catheters in situ. Chlorhexidine (CHG) based antiseptic for skin preparation before central line insertion and dressing changes has been recommended for use in infants over 2 months of age. There is a lack of uniformity regarding usage of skin antiseptics in newborn infant. There is no recent Australia – New Zealand (ANZ) guidelines regarding the use of CHG in newborn infants. Our intention was to survey and identify the extent of usage of CHG and other skin antiseptic solutions and its adverse effects in various neonatal nurseries across ANZ.

**Method:** A web based survey questionnaire was sent to the level 4–6 NICU across ANZ.

**Results:** 29 (91%) of 32 units responded, The type and concentration of CHG used in the NICU varied a considerably. Commonest adverse effect of CHG was skin erythema and of Iodine was hypothyroidism.

|                            | CHG<br>(aqueous) | CHG<br>(alcohol) | Iodine   |
|----------------------------|------------------|------------------|----------|
| Term Infants (n = 28)      | 12 (43%)         | 16 (56%)         | 0        |
| Preterm infants (n = 29)   | 21 (72%)         | 5 (17%)          | 3 (11%)  |
| Strength of CHG in preterm | <0.1%            | 0.1–0.5%         | >0.5%    |
| PIVC (n = 23)              | 5 (22%)          | 8 (35%)          | 10 (43%) |
| Central lines (n = 23)     | 6 (26%)          | 7 (30%)          | 10 (44%) |

**Conclusions:** Although CHG is clearly effective as skin antiseptic solution, a further research into its optimal strength and associated skin adverse effects needs to be established. A more consistent national policy and database of adverse events of skin antiseptic preparations would be beneficial.

#### P455

### DOES EXOGENOUS PROBIOTIC CROSS-COLONISATION OCCUR BETWEEN PRETERM INFANTS?

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**Background:** Cross-colonisation, occurring when a microorganism is transferred from one individual to another, can lead to nosocomial outbreaks of infections. The aim of this study was to assess the rate of cross-colonisation of probiotic organisms in a neonatal unit investigating the effect of a combination of exogenous probiotics (*Streptococcus thermophilus*, *Bifidobacteria infantis* and *Bifidobacteria lactis*) on late onset sepsis in preterm infants (the ProPrems trial).

**Method:** This was a surveillance substudy of The ProPrems trial undertaken in the Royal Women's Hospital neonatal unit. Faecal samples were obtained from eligible infants at Time point 1 (during the ProPrems Trial) and Time point 2 (6 months after ProPrems completion). Bacterial DNA was extracted and tested using specific real-time PCR's for detection of the 3 probiotic strains. Colonisation was defined as detection of DNA from at least 2 of the 3 probiotic organisms.

**Results:** Of the 43 faecal swabs tested at Time point 1, 8 were colonised with the exogenous probiotics. Six were from ProPrems trial participants and 2 were not, indicating possible cross-colonisation as a result of being cared for in a NISC administering exogenous probiotics in some patients. Of the 44 infants' swabs tested at Time point 2, only 1 was colonised.

**Conclusions:** This study demonstrated a low incidence of probiotic cross-colonisation amongst newborn infants in a neonatal unit trialing probiotic supplementation in preterm infants.

#### P456

### WOMEN'S KNOWLEDGE OF FETAL MOVEMENTS: A SURVEY OF PREGNANT WOMEN

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**Background:** Maternal perception of decreased fetal movements (DFM) is associated with increased risk of adverse outcome. Studies exploring women's knowledge and information provided about DFM are limited.

**Method:** Women attending antenatal care at 36 weeks gestation at a large tertiary maternity hospital were surveyed to determine: 1) maternal knowledge about fetal movements (FM) and DFM; 2) sources of information; and 3) preferences for receiving information. We examined associations between maternal knowledge of FM and DFM and demographic characteristics.

**Results:** 526 women were recruited, 62% of those eligible. 67% of women reported receiving information about FM during pregnancy. Almost two thirds thought it was normal for FM to decrease towards the end of pregnancy. 67% disagreed that being aware of FM would help identify if the baby was unwell and 83% of women would 'normalise' DFM and not take further action. If concerned about DFM, 65% would wait longer than 24 hours before seeking medical advice.

Nearly 80% of women received information from a midwife. 55% of participants would prefer to get their information from midwives clinics, followed by 26% preferring obstetricians' clinics.

Univariate analysis showed that social factors such as ethnicity, language, and low levels of education were associated with less knowledge. Results of multi-variate analysis will also be presented.

**Conclusions:** Women's knowledge of FM and DFM in late pregnancy appears suboptimal and women at increased risk appear to be less well informed. Attention is urgently needed to improve the quality of information provided to women particularly for disadvantaged women.

#### P457

##### WHY SO HIGH?: UNEXPLAINED STILLBIRTHS IN QUEENSLAND, 1995–2004

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**Background:** In Queensland, each stillbirth is registered in the Perinatal Data Collection and assigned a clinical cause of death by an expert clinical panel according to the Perinatal Society of Australia and New Zealand Perinatal Death Classification system (PSANZ-PDC). This is the first in-depth analysis of these data, examining trends over time in clinical causes of stillbirth among singleton births.

**Method:** Data on all singleton births for the entire geographic area of Queensland were assembled for the period 1995 to 2004. Population based trends in PSANZ-PDC categories of stillbirth and levels of investigation were assessed in two time periods (1995–1999, 2000–2004). Subsequently, data to 2010 will be presented.

**Results:** The stillbirth rate among singleton births was stable around 6.2/1000 total births with a reduction in autopsy rates from 40.6% to 35.0% ( $p = 0.002$ ). Excluding terminations of pregnancy, the leading category of stillbirth was unexplained antepartum fetal death which remained stable at 2.14/1000 singleton births (34.3% of all singleton stillbirths). No changes in the other categories were evident. Examining subcategories of the unexplained group, autopsy rates decreased from 53.0% to 44.0% ( $p = 0.004$ ). Within this group in 2000–2004, placental examinations were not performed or unknown if performed in 20.9%. Evidence of uteroplacental insufficiency was reported in 15.4% of cases.

**Conclusions:** Unexplained stillbirths continue to account for a significant proportion of singleton stillbirths in Queensland. Issues with data quality suggest that a significant proportion of these deaths may not be unexplained. Increased effort to improve investigation and classification of stillbirth is urgently required.

#### P458

##### A POPULATION STUDY OF DEMOGRAPHIC CHANGES AND OUTCOMES OF VERY PREMATURE MULTIPLE BIRTHS INFANTS ADMITTED TO NICU IN AUSTRALIA AND NEW ZEALAND

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**Background:** Multiple pregnancies are associated with an increased risk of adverse outcomes. We aim to determine the changes in the demographics, particularly of assisted conception (AC), and outcomes amongst very premature NICU multiple birth infants (MBI) over time.

**Method:** Perinatal characteristics and neonatal data (1995–2009) were obtained from the ANZNN data set. The 48,177 infants <32 weeks studied

included 34,131 (70.8%) singletons, 12,340 (25.6%) twins, 1,571 (3.3%) triplets and 135 (0.3%) higher order multiples.

**Results:** MBI amongst ANZNN cohort increased significantly from 26.0% in 1995 to 30.9% in 2009 while AC has increased from 6.5% to 11.7% ( $p < 0.001$ ). AC (4.2% of singletons, 21.2% twins, 52.6% triplets and 71.1% higher order multiples) was associated with higher maternal age. MBI were of higher gestation than singletons (mean 28.6 v 28.4 weeks,  $p < 0.001$ ). Mortality and major morbidities significantly reduced over the three 5-year epochs for both singletons and multiples. MBI of 21–27 weeks gestation, adjusted for confounders, had higher a mortality risk than singletons (Adjusted Hazard Risk 1.22, 95% CI 1.13 to 1.32) while those of 28–31 weeks gestation multiples had a lower mortality (AHR 0.70, 0.61 to 0.81). Adjusted for confounders including maternal age, AC did not contribute an independent influence on outcomes and elective CS was associated with a lower risk 0.75 (0.61–0.92).

**Conclusions:** There were increased multiple births and AC amongst NICU admissions of very premature infants. The mortality risk discrepancy between gestation groups is unexplained and may be related to population bias of NICU admissions or other socio-economic factors.

#### P459

##### PLACENTAL MESENCHYMAL DYSPLASIA: A CASE WITH HELLP SYNDROME, DIC AND PLACENTAL ABRUPTION.

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**Background:** Placental mesenchymal dysplasia (PMD) is a rare placental condition characterized by placentomegaly and can resemble a molar pregnancy due to 'grape-like vesicles' observed macroscopically and on ultrasound. It has been described in association with Beckwith-Wiedemann syndrome, intrauterine growth restriction and intrauterine fetal demise.

**Method:** We describe PMD in a 36 year old gravid-4 para-2 transferred to our care at 34 weeks' gestation with HELLP syndrome and DIC. She underwent an emergency caesarean section, at which time a placental abruption was discovered, however the female neonate was delivered in good condition. The placenta appeared grossly abnormal and was sent for histopathological examination and cytogenetic testing.

**Results:** The trimmed placenta weighed 570 g, greater than the 90th centile for gestation. Fetal surface vessels were markedly distended by blood clot. Half of the maternal surface was a normal colour, the other half was pale with a depressed surface showing extensive fibrin and several cysts up to 10 mm in diameter. Microscopically there were areas of old thrombus with secondary cystic degenerative changes and some vessels had pale staining walls with a degenerative mucoid appearance. Some villi showed stromal cystic changes, however features of molar disease were not observed. Cytogenetics showed karyotype 46 XX and no chromosomal abnormality was detected. Postpartum investigations revealed no thrombophilia.

**Conclusion:** PMD is a rare condition which has not previously been described in the context of HELLP and DIC. In this case, early delivery led to a favourable outcome for mother and baby.



## P460

# MATERNAL WEIGHT GAIN PATTERN AND BODY COMPOSITION ALTER PLACENTAL PPARGC1A METHYLATION IN OVERWEIGHT AND OBESE PREGNANT WOMEN

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**Background:** High maternal pre-pregnancy BMI and gestational weight gain (GWG) have been linked with increased fetal growth and adiposity and weight at birth. Increasing evidence suggests that maternal factors associated with obesity can modify placental function, indirectly adding to the drive to fetal overgrowth. Peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PPARGC1A) is a major transcriptional factor co-activator in the regulation of lipid transport and metabolism. DNA methylation of the PPARGC1A gene in cord varies with maternal BMI, suggesting the latter might act via this mechanism to affect gene expression and action in other gestational tissues, including the placenta, to contribute to abnormal lipid supply and growth.

**Method:** We measured DNA methylation at five sites (%) in the genes of PPARGC1A and IGF2 in term placentas of a subset nested within a prospective cohort, the Control Arm of the LIMIT randomised trial (n = 65). The relationship of DNA methylation to maternal measures from trial entry to 36 weeks gestation was examined by ANOVA and linear regression, with P < 0.05 taken as significant.

**Results:** DNA methylation of either gene did not vary with maternal BMI or BMI class. Methylation at two PPARGC1A sites did vary with weekly weight gain (p = 0.012) and weekly weight gain class (<0.3 kg/wk, 0.3–0.6 kg/wk &gt;0.6 kg/wk) (p = 0.021). Maternal bicep skin fold change was also associated with methylation at two sites (p = 0.029) and average methylation of PPARGC1A (p = 0.043).

**Conclusions:** In overweight and obese pregnant women, the pattern of change in weight and skin fold thickness rather than BMI per se alters and increases DNA methylation of a major transcriptional factor co-activator in the regulation of lipid transport and metabolism in the placenta. If placental gene PPARGC1A expression is altered as a consequence, this may represent one mechanism whereby gestational weight gain affects fetal growth and size at birth.

## P461

# CHILDHOOD DEVELOPMENTAL OUTCOMES FOLLOWING LASER TREATMENT FOR TWIN-TO-TWIN TRANSFUSION SYNDROME (TTTS)

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**Background:** Laser therapy is a well recognised treatment for significant TTTS. We investigated the early childhood neurodevelopmental outcome of children post Laser treatment for TTTS in our centre.

**Methods:** Children of women, referred from NSW or interstate centres and treated between March 2006 and June 2008, were assessed with WIPPSI-III and a general health questionnaire. Major neurodevelopmental impairment (NDI) was reported as IQ < 70 or cerebral palsy (CP). Borderline cognitive impairment was defined by IQ 70–79.

**Results:** Amongst the 37 pregnancies treated, 62 infants were discharged home and the overall foetal survival rate was 84%. A total of 52 children (84% follow up rate) from 31 pregnancies were assessed. Average age at assessment was 47 (range 30–69) months. The majority of pregnancies were Quintero Stage III (74%). Average gestation at birth was 32 weeks.

There was a trend for worse outcome with higher Quintero stage. The majority, 39 (75%) children were found to be neurodevelopmentally normal; and 9 (17%) had borderline cognitive development; and 4 (8%) had a major NDI. Of these, 2 preterm siblings had congenital hypothyroidism with low IQ, a preterm child with periventricular leukomalacia had non-ambulatory diplegia (3%) with IQ within normal range, and another with cognitive impairment.

**Conclusions:** There was a modest level of neurocognitive impairment post laser therapy for TTTS, mainly borderline cognitive development, lesser so major NDI. There was a low incidence of cerebral palsy. Routine developmental and neurological follow-up of these children is recommended.

## P462

# BIRTH ASPHYXIA REDUCES OVARIAN FOLLICULAR RESERVE IN THE SPINY MOUSE

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**Background:** Using a model of birth asphyxia in the spiny mouse, we have previously shown that a maternal diet supplemented with creatine improves survival and postnatal growth in offspring and prevents newborn brain, diaphragm and kidney injury otherwise caused by birth asphyxia. This study assessed the consequences for ovarian follicular reserve and the protective potential of creatine.

**Method:** Pregnant spiny mice were fed a control (Co) or 5% creatine-supplemented diet (Cr) from day 20 of gestation (term ~39 days). On day 38, pups were delivered by c-section, or birth asphyxia was induced by placing the excised uterus containing all fetuses in a saline bath for 7.5–8 mins. Surviving neonates were cross-fostered to a nursing dam for 33 days. At post-mortem, ovaries were collected into bouins fixative for stereology or sterile saline for follicle isolation and culture to determine in vitro maturation (IVM) potential.

**Results:** Ovarian volume and follicular reserve (FR) were significantly depleted in offspring exposed to birth asphyxia (FR: Co c-section 3178 ± 288; Co asphyxia 2177 ± 237; p < 0.05). Further, the size of primordial follicles was significantly reduced after birth asphyxia. Creatine supplementation somewhat rescued the ovary, with this group not differing from the c-section or birth asphyxia groups (FR: 2469 ± 334). Preliminary IVM results do not differ between groups.

**Conclusions:** We believe this to be the first study to examine the effects of birth asphyxia on ovarian follicular reserve and we show that the ovary is vulnerable to hypoxic injury. Whether this depleted follicular reserve affects reproductive potential remains to be determined.

## P463

# THE SYNTHETIC GLUCOCORTICOID DEXAMETHASONE ALTERS BRANCHING MORPHOGENESIS IN THE SPINY MOUSE PLACENTA

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**Background:** High levels of maternal glucocorticoids during pregnancy can alter the developmental trajectory of some fetal organs. These perturbations are often more profound for the male fetus. Exposure to high concentrations of glucocorticoids at mid pregnancy in the spiny mouse resulted in sexually dimorphic alterations of placental structure. The current study aimed to investigate the expression of genes important in the development of the placental vasculature.

**Method:** Pregnant spiny mice were killed at gestational ages between days (d) 15 and 37 of gestation (term d39); or treated with dexamethasone (DEX) or saline for 60 h at 0.5 of gestation. The effects of fetal sex, gestational age, and DEX on placental labyrinth branching were analyzed by qPCR.

**Results:** Developmentally, the male placenta had higher *TGFB1* expression, than the female, and earlier peaks in expression of *VEGFA* [d19 (male) vs. d37 (female)] and *KDR* [d19 (male) vs. d20 (female)]. Immediately following DEX administration (d23) *BMP4*, *TGFB1* and *WNT4* were significantly increased in the labyrinth of both sexes. The expression of *KDR* and *GREM1* were increased in DEX-exposed male placentas as this time, whereas *KDR* was decreased in females and *GREM1* was unchanged. Two-weeks post treatment (d37) the effect was similar for both sexes with *BMP4*, *TGFB1*, *WNT4*, *GREM1*, and *VEGFA* expression decreased, and *KDR* expression increased compared to controls.

**Conclusions:** These results show that fundamental differences exist between a male and female placenta; and constitutive and glucocorticoid-induced changes in expression of genes involved in branching morphogenesis may be influenced by fetal/placental sex.

#### P464

##### MELATONIN AS A POTENTIAL NEUROPROTECTIVE THERAPY IN INTRAUTERINE GROWTH RESTRICTION

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**Introduction:** Oxidative stress is a key mechanism involved in pathologies associated with intrauterine growth restriction (IUGR), including altered brain development. Animal studies have established the efficacy of maternally administered melatonin as a fetal neuroprotective agent in IUGR.

**Methods:** Pregnant women with an estimated fetal weight lower than the 10<sup>th</sup> percentile for fetal gestational age and abnormal fetoplacental Doppler values were included in an observational study. Weekly maternal blood samples were obtained. At birth, umbilical cord blood, amniotic fluid and placenta were collected. 8-Isoprostane and malondialdehyde were measured to assess oxidative stress. Data were analysed using an unpaired t-test and presented as mean  $\pm$  standard deviation.

**Results:** To date, results are from a total of 13 control and 9 IUGR pregnancies. Malondialdehyde concentrations are elevated in umbilical vein and umbilical artery plasma in IUGR (n = 2) compared to controls (n = 12) (16.3  $\pm$  2.0 vs 5.4  $\pm$  12.0 nmol/mL, p = 0.0041; 3.4  $\pm$  1.0 vs 7.3  $\pm$  3.7 nmol/mL). Malondialdehyde concentrations in the amniotic fluid falls below the sensitivity of the assay in control samples (n = 3), but a mean concentration of 1.6 nmol/ml is observed in IUGR fluid (n = 3). Whilst widely variable, 8-Isoprostane trends towards increased concentrations in maternal plasma in IUGR pregnancies (Controls (n = 13) 16.5  $\pm$  7.6, IUGR (n = 6) 30.3  $\pm$  25.7 nmol/ml).

**Conclusions:** Our results to date demonstrate that human IUGR is associated with increased oxidative stress in maternal and fetal fluids. As demonstrated in animal studies, antioxidant melatonin treatment to the mother in IUGR pregnancy may improve fetal and neonatal wellbeing and reduce neuropathology. A human pilot study testing this hypothesis is underway.

#### P466

##### EYE CONTACT OR ICONTACT: HOW DO PARENTS PREFER TO RECEIVE INFORMATION IN NEONATAL INTENSIVE AND SPECIAL CARE (NISC)?

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**Background:** Good neonatal care should be family centred. Providing families with accurate information about their baby is an essential component of working in partnership with parents. The aim of our study was to find out how parents prefer to receive information about their baby in NISC.

**Method:** Parents were surveyed the week before their baby was discharged or transferred from NISC. Parents were invited to complete the questionnaire anonymously. Non English speaking parents were offered an interpreter. The survey consisted of 6 parts: 'Accessing Information', 'Understanding how we care for your baby', 'Understanding new information', 'Asking Questions', 'How well do we communicate with you?' and 'Getting to know you and your family'. Questions were a combination of Yes/No, multiple choice and open-ended questions. Responses were entered into a Survey -Monkey database.

**Results:** 205 surveys were distributed to parents with 102 returned (48%); of these 57% were females and 43% male. 76% of parents identified a one-to-one meeting with a health professional as their most preferred way of receiving information; less preferred were group sessions, the internet and Apps. Social networking was the least preferred way of receiving information. More than 90% thought that written information would help them remember what happened during a meeting with doctors.

**Conclusions:** In the current era of social networking and internet availability, families value receiving information about their baby directly from health professionals, together with written information for reference.

#### P467

##### MAGNESIUM SULPHATE FOR WOMEN AT TERM FOR NEUROPROTECTION OF THE FETUS - A COCHRANE REVIEW

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**Background:** Magnesium sulphate, administered to women at risk of early preterm birth, reduces the risk of preterm babies dying or having cerebral palsy. As it is unknown whether similar benefits are seen for the term fetus, this Cochrane review aimed to assess the effects of magnesium sulphate given to women at term for fetal neuroprotection.

**Method:** We used standard Cochrane methodology (search: September 2011). Randomised controlled trials comparing antenatal magnesium sulphate given to women at term with placebo, no treatment, or a different fetal neuroprotective agent were eligible. The primary infant outcomes were death (fetal, neonatal or later death), cerebral palsy, and the combined outcome (death or cerebral palsy).

**Results:** One trial (involving 135 women with mild pre-eclampsia), at low risk of bias, was included. A further six studies are awaiting further assessment (data for women administered magnesium sulphate at  $\geq 37$  weeks have been requested). The included trial did not report on the review's primary outcomes. No significant differences were seen between magnesium sulphate and placebo for Apgar score <7 at 5 minutes, gestational age at birth, postpartum haemorrhage or caesarean section. Magnesium sulphate was associated with increased maternal warmth and flushing (RR 3.81; 95% CI 2.22, 6.53), however no difference in cessation of therapy due to adverse effects was observed.

**Conclusions:** There is currently insufficient evidence to assess the effects of magnesium sulphate for neuroprotection of the term fetus. Larger,

well-designed trials are required, reporting long-term follow-up of infants to allow assessment of neurosensory disabilities, including cerebral palsy.

P468

# A QUALITATIVE STUDY OF WOMEN'S VIEWS ON DIAGNOSIS AND MANAGEMENT FOR BORDERLINE GESTATIONAL DIABETES

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**Background:** Lifestyle intervention has been found to be beneficial for women with gestational hyperglycaemia. This qualitative study aimed to explore women's experiences after being diagnosed with borderline gestational diabetes mellitus (bGDM), their attitudes about treatment for bGDM, and factors important in achieving any lifestyle changes.

**Method:** We conducted semi-structured, face-to-face interviews with women involved in the IDEAL trial that is investigating dietary and lifestyle advice for women with bGDM. Interviews were transcribed verbatim and data were analysed thematically.

**Results:** A total of 22 women were interviewed. After a diagnosis of bGDM, 14 (64%) women reported not being concerned or worried. Most women (n = 21) thought bGDM treatment was very important or important and one woman was unsure. Most women (18: 82%) planned to improve their diet and/or exercise to manage their bGDM and four women (18%) planned no changes. The enablers and barriers for achieving the intended lifestyle changes were highly individualised. The most frequently mentioned enabler was being more motivated to make lifestyle changes to improve the health of their baby and/or themselves (14: 78%) and the most frequent barrier was tiredness and/or being physically unwell (11: 61%).

**Conclusions:** A diagnosis of bGDM caused some concern to one third of women interviewed. The majority of women believed managing their bGDM was important and they planned to improve their lifestyle. Although women nominated many different factors that might influence their lifestyle choices, their own and their baby's future health is a powerful motivator for change.

P469

# A REVIEW OF THE MATERNITY CARE RECEIVED BY WOMEN PRESENTING WITH DECREASED FETAL MOVEMENTS

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**Background:** Decreased fetal movements (DFM) are associated with adverse outcomes and the Australian and New Zealand Stillbirth Alliance (ANZSA) have developed clinical practice guidelines for the management of women who report DFM.

**Method:** A medical records review of women presenting to Barwon Health, Victoria, with DFM between 1 April 2010 and May 31 2012 was conducted to describe the characteristics of women presenting with DFM, the alignment of practice with ANZSA guidelines and birth outcomes.

**Results:** The records of 146 women presenting with DFM were reviewed. Thirty (20.5%) re-presented with DFM between 2–6 times (median 2). Women presented between 28–41 weeks (median 36 weeks). Complications were present in 88 (60.3%) women, including 11 (7.5%) with gestational diabetes, 42 (28.1%) with a BMI  $\geq 30$ , 11 (7.5%) with hypertension disorder and 8 fetuses (5.5%) were growth restricted.

Management was consistent with the guidelines for CTG which was performed for 141 (96.6%) and ultrasound biometry for 89 (61%)

(recommended only if concerned after clinical assessment). The median time to review was 105 minutes however, 26 (17%) women were seen more than 2 hours after arrival (less than 120 minutes recommended) and fundal height was measured in only 81 (55.5%) women. Investigations were more likely to be conducted were the woman re-presented with DFM. There were four fetal deaths detected on initial presentation with DFM. Other outcomes included preterm births (n = 3), Agars  $<7$  at 5 minutes (n = 11) and admissions to SCN (n = 13).

**Conclusions:** In this high risk population increased awareness of the ANZSA guidelines may increase alignment of practice with the guidelines and improve pregnancy outcomes.

P470

# EXPLORING THE EXPERIENCE AND PRACTICES OF MIDWIVES DISCUSSING DECREASED FETAL MOVEMENTS WITH WOMEN IN PREGNANCY

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**Background:** Australian and New Zealand Stillbirth Alliance (ANZSA) clinical practice guidelines recommend that all women are given verbal and written information on normal fetal movements and made aware of the importance of reporting decreased fetal movements (DFM). This study explores midwives' knowledge, practice and attitudes regarding communicating to women regarding DFM following guideline implementation.

**Method:** Between August–September, 2012, a survey was distributed to midwives providing pregnancy care at Barwon Health, Victoria. Items included 5 point Likert scale for agreement with statements and 4 point Likert scale for frequency of practices, open text responses and demographic details.

**Results:** Response rate was 49% (42/85). Verbal information on normal fetal movements was consistently provided by 14 midwives (33%) and written information by 2 midwives (5%). There was considerable variation regarding the definition of DFM and the advice provided to women experiencing DFM, including to drink cold liquid (n = 22), eat (n = 10) and palpate abdomen (n = 7). Where women reporting DFM 83% (n = 35) midwives agreed women should be seen within 12 hours and 91% (n = 38) agreed women presenting with DFM should have fundal height measured. Midwives advised women concerned about DFM to wait at least 24 hours before review 'some of the time' (n = 42) and 43% (n = 18) agreed with the use of a kick chart (not recommended). Half of the responders (n = 21) were aware of internal guidelines on DFM and 5% (n = 2) for ANZSA guidelines.

**Conclusions:** Providing women with the recently released ANZSA brochure on decreased fetal movements is recommended to support verbal information. Professional development for midwifery staff may align practice with best evidence and improve pregnancy outcomes.

P471

# ACCURACY OF ENDOTRACHEAL TUBE PLACEMENT IN NEWBORN INFANTS AT DELIVERY

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**Background:** Placement of an endotracheal tube (ETT) in newborns at delivery is common neonatal practice. In premature infants, this practice is often followed by administration of surfactant prior to confirmation of ETT position by chest X-ray (CXR). The aim of this audit was to investigate the accuracy of oral and nasal endotracheal intubation of newborn infants at delivery.

**Method:** Retrospective review of all infants intubated at delivery between 2004 and 2011 at our institution. Position of ETT as per CXR report was



categorised in 'good position' = mid-tracheal, 'borderline high' = level of clavicles, 'high' = above clavicles, 'borderline low' = just above carina, or 'low' = level of carina or bronchus.

**Results:** 582 infants were included: gestational age range 22.5–41.2 weeks, birthweight range 362–4400 g. 301 infants were intubated orally, 281 infants nasally. Good position vs borderline or poor position, oral vs nasal: 42.6% vs 50.9% (95% CI 0.7078–1.002,  $p = 0.0561$ ). Good position vs borderline or poor position, oral vs nasal infants  $\leq 1000$  g: 37.1% vs 48.9% (95% CI 0.5869–0.9829,  $p = 0.0386$ ). Good or borderline position vs poor position, oral vs nasal: 60.8% vs 74% (95% CI 0.7327–0.9207,  $p = 0.0008$ ). Good or borderline position vs poor position, oral vs nasal infants  $\leq 1000$  g: 58.3% vs 72.3% (95% CI 0.6854–0.9491,  $p = 0.0123$ ).

**Conclusions:** In this cohort, less than 50% of ETT were well positioned. 32.8% of ETT were unacceptably high or low positioned. Nasal intubation achieved better ETT position than oral intubation.

P472

#### ACCIDENTAL EXTUBATION AND SUBSEQUENT TRIAL OF NON-INVASIVE VENTILATION IN THE NEONATAL INTENSIVE CARE UNIT

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**Background:** Accidental extubation (AE) occurs as an infrequent complication of mechanical ventilation. Following AE, a trial of non-invasive ventilation may be considered if a neonate is showing adequate respiratory effort. We aimed to investigate the success and failure rate of NIV management of neonates experiencing AE.

**Method:** Retrospective single-centre study. Review of AE episodes occurring in our unit during Feb 2004 and Nov 2012. Data were collected from the departmental database and medical records. Infants managed on NIV following AE were included. Re-intubation within six hours of a trial of NIV following AE was defined as NIV failure.

**Results:** 55 episodes of AE were identified. Of those, 38 infants were trialled on continuous positive airway pressure (CPAP) following AE. This was successful in 25 (69%) and failed in 13 (31%) infants. Infants who failed a trial of NIV were re-intubated between 0.45 and 5.25 hours following AE. Infants who failed on CPAP after AE were slightly more premature but had a slightly higher birth weight: median gestation at birth 26 + 4 weeks vs 27 + 3 weeks; median birth weight 960 vs 929 g. In the group of infants re-intubated immediately following AE, one infant experienced a respiratory arrest because of a difficult re-intubation.

**Conclusions:** Following neonatal AE, a trial of CPAP may be considered as a treatment option.

P473

#### THE PREVALENCE OF POTENTIAL RISK FACTORS FOR RECURRENT MISCARRIAGE: A RETROSPECTIVE CLINICAL AUDIT OF THE RECURRENT MISCARRIAGE CLINIC AT THE ROYAL WOMEN'S HOSPITAL, 2003–2009

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**Background:** Recurrent miscarriage affects 1% of all couples and is defined as the loss of three or more consecutive pregnancies prior to 20 weeks gestation. There is limited Australian data regarding the prevalence of risk factors for recurrent miscarriage.

**Method:** A retrospective clinical audit of women first attending the Recurrent Miscarriage Clinic (RMC) at The Royal Women's Hospital between January 2003 and December 2009 was undertaken. Only women who had 3 or more consecutive miscarriages with the same partner were

included. The results of relevant investigations were retrieved from medical records.

**Results:** A total of 589 women met the eligibility criteria. Women presenting to the RMC were older than the general antenatal population attending the hospital (mean age  $34.4 \pm 6.0$  vs  $30.9 \pm 2.1$  yrs,  $p < 0.001$ ) and 115 women (19.5%) were aged over 40 years at the first consultation. An abnormal parental karyotype was identified in 5.3% of women and 2.6% of male partners. Lupus anticoagulant was detected in 2.6% of women, while 3.6% had anticardiolipin antibodies of at least moderate titres. The Factor V Leiden mutation was identified in 4.1%, while 3.3% were heterozygous for the Prothrombin gene mutation. No investigation abnormalities were found in 149 women (25.3%).

**Conclusions:** The prevalence of abnormalities associated with recurrent miscarriage is similar in this cohort to that in the published literature. No abnormal investigations were found in 25% of women.

P474

#### NEONATAL ABSTINENCE SYNDROME MANAGEMENT IN A SOUTH AUSTRALIAN TERTIARY SPECIAL CARE NURSERY

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**Background:** Illicit drug use in pregnancy is a significant clinical problem with short- and long-term consequences for the unborn child. Further, the incidence of maternal drug misuse is increasing. Infants with neonatal abstinence syndrome (NAS) may require prolonged admission and treatment with significant implications for clinical workload.

**Method:** A prospective audit of all consecutive NAS admissions to the Women's and Children's Hospital Adelaide from August 2009 to July 2011. Clinical characteristics are presented as median (range) or n (%).

**Results:** Infants with NAS accounted for 32/1652 SCBU admissions (2%). Maternal age was 31 (23–42) and parity 3 (1–9). Self-reported drug use agreed with urine toxicology in 19 (59%) of cases. Methadone users comprised 23 (72%) of subjects with a median daily dose of 70 mg (30–120 mg). Time to first Finnegan's score was 5 hrs (1–69) with 16 (58%) infants requiring treatment at 37 hrs (13–264). For infants treated for NAS, duration of treatment was 37 days (10–95) and total admission 15 days (7–34). No relationship was observed between maternal methadone dose and length of treatment. Eleven infants (68%) continued treatment after discharge.

**Conclusions:** In our population, the incidence of NAS infants requiring treatment did not reflect maternal methadone use. Further, most women had exposures to more than one illicit drug confounding the severity of withdrawal. Due to the poorly understood pathophysiology of NAS, optimal evaluation and treatment for this disorder continues to evolve.

P475

#### THE ESSENTIAL ROLE OF THE NEONATAL RESEARCH CO-ORDINATOR – NEED FOR POSITION SECURITY

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**Background:** Investigator led clinical research underpins high quality care yet research nurse positions are often undervalued. With the volume and scope of neonatal research increasing the implications for management within a local neonatal unit are increasingly complex and costly. We describe the research profile of a tertiary neonatal unit in 2 time periods: 2000–2005 and 2006–2011.

**Method:** Records of research conducted between 2000 and 2011 in the neonatal unit of the Women's and Children's Hospital were reviewed.

**Results:** 54 studies were conducted with 1442 infants enrolled in primary studies and 284 in follow-up studies. The average (range) number of studies in any one year has increased from 10 (7–17) in 2000–2005 to 15 (13–18) in 2006–2011. Only 67% of the studies were funded. The overall participation rate of families in primary studies improved over time (72% vs. 76% for 2000–2005 vs. 2006–2011). 259 (18%) infants were enrolled in 2 to 3 studies and 16 (1%) in 4 to 6 studies. In 2003, management of neurodevelopmental follow-up assessments for trials was added to the Coordinator's role. This expanded the workload with need to manage both in-hospital and follow-up trial assessments.

**Conclusions:** While the majority of families willingly participate in research and investigators continue to drive research initiatives, up to one third of projects remain unfunded creating a financial burden on local units. Experienced research nurses, with appropriate job security, are essential for the continuing high quality conduct of research in neonatal settings. Provision for research nurse positions in tertiary unit operational budgets, needs to be supported.

#### P476

### DEVELOPING A HARMONISED REPRODUCTIVE HEALTH REGISTRY TO IMPROVE DATA FOR REPRODUCTIVE, CHILD AND MATERNAL HEALTH; DEVELOPMENT OF CRITERION BASED DATASETS

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**Background:** Population-based health registries are essential to monitor progress towards Millennium Development Goals 4 & 5 (reduce child mortality and improve maternal health). While various datasets on reproductive health have been developed, they are not harmonised with one another nor with the health initiatives they intend to evaluate, resulting in ineffectiveness and inefficiencies in data collection and usage.

**Method:** We are undertaking an international project through the Norwegian Institute of Public Health in collaboration with the Department of Reproductive Health and Research, WHO, and funded by Norad. This project aims to develop a framework to assist in developing datasets and IT solutions which are region specific. We will present part of this framework which includes development of indicators to enable monitoring of the implementation of WHO Essential Interventions for pregnancy, childbirth and newborn care. For each intervention, four performance indicators are identified: a process indicator and outcome indicator for both screening and treatment/prevention. A global working group of experts has been established to take this work forward prior to wider consultation.

**Results:** Indicators and data points to effectively measure the WHO Essential interventions have been identified for further testing and piloting.

**Conclusions:** Development of a framework to identify datasets to address maternal and newborn priority interventions seems feasible. It is envisaged that this project will promote the development of harmonised reproductive health registries ultimately combining with innovative e-Health and m-Health solutions for open-source data collection and management.

#### P477

### MANUAL VENTILATION OF NEONATES: ADJUSTMENT TIME OF PEAK INFLATION PRESSURE DIFFER SIGNIFICANTLY BETWEEN DEVICES

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**Background:** International resuscitation guidelines recommend self-inflating bags (SIB), flow-inflating bags and T-piece resuscitators (TPR) for manual ventilation of neonates without giving preference for one over the other. Aim of the study was to compare these devices regarding the time needed to adjust to predefined positive inflation pressures (PIP) and their reliability of providing positive end expiratory pressure (PEEP).

**Methods:** Randomized simulation study of neonatal resuscitation of a 1 kg newborn manikin. Four devices were tested: a) SIB with attached manometer, and b) SIB without manometer, c) Neopuff® Infant T-Piece Resuscitator, and d) Panda® T-piece resuscitator. Measurements included ventilation pressures (targets: PIP 20 cmH<sub>2</sub>O, PEEP 5 cmH<sub>2</sub>O) and respiratory rates (RR, target: 60/min) for each participant and device, and device specific times needed to increase PIP.

**Results:** Median PIP was statistically significantly higher than target pressure with all four devices (Table 1). Time to adjust PIP was significantly longer for both TRPs, compared to SIB. The SIB without manometer produced highest PIPs (>14% of target), SIB with manometer and both TPRs had <5% deviation from target PIP. Inter-operator variability of PIP provision was considerably greater with the SIB, but significantly decreased with manometer attached. Generated PEEP levels were significantly different between devices. All devices generated PEEP below target values, except for SIB with attached manometer.

**Conclusions:** The time to adjust to different inflation pressures was significantly longer for TPRs than SIBs. TPRs more reliably provide defined PIPs and PEEP. Whenever SIBs are used, they should be used with an attached manometer.

Table 1. Comparison of adjustment times

| Resuscitator | Deviation from the target value (%)<br>Median (IQR) | Inter-operator variability<br>Interquartile range | P-Value      |
|--------------|-----------------------------------------------------|---------------------------------------------------|--------------|
| <b>PIP</b>   |                                                     |                                                   |              |
| SIB          | 13.8% (–0.5%–70.8%)                                 | 71.3%                                             | <b>0.007</b> |
| SIBM         | 4.3% (0%–7.8%)                                      | 7.8%                                              | <b>0.001</b> |
| TPR1         | 2.0% (1.0%–2.5%)                                    | 1.5%                                              | <b>0.001</b> |
| TPR2         | 2.0% (1.0%–6.0%)                                    | 5.0%                                              | <b>0.001</b> |
| <b>PEEP</b>  |                                                     |                                                   |              |
| SIB          | –5.0% (–10.0%–1.0%)                                 | 11.0%                                             | <b>0.005</b> |
| SIBM         | 0% (–4.0%–4.0%)                                     | 8.0%                                              | 0.741        |
| TPR1         | –2.0% (–5.0%–4.0%)                                  | 9.0%                                              | 0.409        |
| TPR2         | –8.0% (–14%–3%)                                     | 11.0%                                             | <b>0.001</b> |
| <b>RR</b>    |                                                     |                                                   |              |
| SIB          | –8.3% (–42.1%–26.7%)                                | 68.8%                                             | 0.416        |
| SIBM         | –10.0% (–33.8%–21.7%)                               | 55.4%                                             | 0.689        |
| TPR1         | –20.4% (–37.1%–4.6%)                                | 32.5%                                             | <b>0.002</b> |
| TPR2         | –27.1% (–42.9%–0.8%)                                | 42.1%                                             | <b>0.021</b> |

**Legend table 1:** Relative deviation of the applied pressures (peak inflation pressure = PIP, positive end expiratory pressure = PEEP) and respiratory rates (RR) form the target values (PIP = 20 cmH<sub>2</sub>O, PEEP = 5 cmH<sub>2</sub>O and RR = 60/min) and the inter-operator variability measured by the inter-quartile range of the relative deviations for the 4 resuscitators (statistically significant deviations are printed in bold). SIB = self-inflating bag, SIBM = self-inflating bag with a manometer, TPR1 = T-piece resuscitator 1 (Neopuff®), TPR2 = T-piece resuscitator 2 (Panda®).

## P478

## EFFECT OF ENDOTRACHEAL TUBE LEAK ON CAPNOGRAPHIC MEASUREMENTS IN VENTILATED NEWBORNS: AN IN-VITRO STUDY

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**Background:** Uncuffed endotracheal tubes (ET) are commonly used for mechanically ventilated infants to protect airways, but this bears the risk of air leak around the ET. In contrast to the measurements of tidal volume and respiratory mechanics, only little is known about the effect of ET leaks on capnographic measurements. Aim was to investigate the relationship between ET leak and the CO<sub>2</sub> measuring error of the exhaled breathing gas in a neonatal lung model.

**Methods:** Two leak-free silicon bellows were used as a lung model. A constant, low flow of pure CO<sub>2</sub> was injected so that the exhaled breathing gas contained an adjustable CO<sub>2</sub>-plateau. This lung model was ventilated via a 3 mm inner diameter ET with different respiratory rates (RR) (20, 40 and 60/min) using the Leoni Plus ventilator (Heinen & Löwenstein, Germany). ET leaks (0–80%) were simulated by a needle valve. Recordings of CO<sub>2</sub> were made using a pneumotachmograph (CO2SMO+ Monitor; Novamatrix, Wallingford, USA).

**Results:** The end-expiratory CO<sub>2</sub> partial pressure (P<sub>e</sub>CO<sub>2</sub>) was reduced to zero when air leak was simulated. Provided that the exhaled CO<sub>2</sub> reached a plateau before entering of CO<sub>2</sub> washout of the sample chamber by the leak flow, the magnitude of the CO<sub>2</sub> decreased independently of RR, and only slightly with increasing ET leak. For ET leaks of up to 20% the CO<sub>2</sub> plateau error was <10%.

**Conclusion:** In newborn infants with stiff lungs and a short alveolar plateau capnographic measurements should be interpreted with caution in presence of ET leakage.

## P479

## A RETROSPECTIVE AUDIT OF NECROTIZING ENTEROCOLITIS AND SEPSIS IN PRETERM INFANTS PRE-PROBIOTICS

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**Background:** Necrotizing enterocolitis (NEC) and sepsis remain significant causes of morbidity and mortality in preterm infants. Nepean Neonatal Intensive Care Unit (NICU) is the first in Australia to adopt the routine use of probiotics in preterm infants commencing in January 2012. The beneficial effects of probiotics can be assessed with a comparison to retrospective data of this local population.

**Method:** A retrospective audit was carried out of 251 infants (less than or equal to 32 weeks and/or less than or equal to 1500 g) admitted to Nepean NICU between January 2010 and December 2011. Data was collected from the NICUS database and patient records. Primary outcomes were incidence of NEC, proven late onset sepsis (P-LOS) and death.

**Results:**

| N = 251      | Total | NEC confirmed | NEC suspected | P-LOS   | Deaths  |
|--------------|-------|---------------|---------------|---------|---------|
| Total        | 251   | 8             | 13            | 40      | 15      |
| (% of total) |       | (3.2%)        | (5.2%)        | (15.9%) | (6%)    |
| <28 weeks    | 63    | 2             | 7             | 23      | 10      |
| (% of cases) | (25%) | (25%)         | (53.8%)       | (57.5%) | (66.7%) |
| <1000 g      | 63    | 2             | 7             | 23      | 9       |
| (% of cases) | (25%) | (25%)         | (53.8%)       | (57.5%) | (60%)   |

Confirmed NEC included 4 (1.6%) surgically and 4 (1.6%) medically managed infants. 40 events of P-LOS occurred in 33 infants including

central line associated bacteraemia in 2 infants (0.8%) and ventilator associated pneumonia in 9 infants (3.6%). No deaths were attributable to NEC.

**Conclusions:** The rate of NEC in our regional area of Australia has been reported as 3.8% and the rate at Nepean NICU was 3.2%. In our cohort, 75% of confirmed NEC occurred in infants over 28 weeks and over 1000 g.

## P480

## HIFLOW COMPARED TO CPAP FROM 30 WEEKS GESTATION AT LIVERPOOL HOSPITAL

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**Background:** The use of humidified high flow nasal cannulae (HiFlow) has become widespread over the last several years with little scientific evidence. Liverpool Hospital NICU was using HiFlow from 30 weeks corrected gestation and the impression was that it was effective as well as liked by staff and parents. It was decided to perform a small RCT as a quality project to support this practice.

**Method:** In 2011 31 Babies were randomised to either CPAP or HiFlow when stable but still requiring CPAP and at least 5 days old once reaching 30–31+6 weeks corrected gestation. Daily stability and medium term outcomes were compared.

**Results:** 16 Babies were randomised to CPAP and 15 to HiFlow. Patient characteristics were similar for GA (27.9 vs 28.1), BW (1150 vs 1195), Male (60% vs 56%) although there was a trend to more chorioamnionitis (p = 0.08) and antenatal steroid use (p = 0.17) in the HiFlow Group. During the Study period the CPAP Babies had less bradycardias (p = 0.008) and a trend to increased abdominal distension (p = 0.18). There were no deaths or severe adverse outcomes in either group and there was no difference between the corrected gestation at which respiratory support was ceased (34.9 vs 33.8).

**Conclusions:** This study compared the evolving new practice of using HiFlow in stable babies from 30 weeks corrected gestation to the previous practice of CPAP, and showed no adverse outcome in the HiFlow group. Given that HiFlow is preferred by both staff and parents this practice is considered safe to continue.

## P481

## ASSESSING THE IMPACT OF THE LANCET'S STILLBIRTH SERIES

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**Background:** In April 2011, The Lancet launched its Stillbirth Series to bring attention to the global problem of stillbirth and to promote action. The series was met with unprecedented media coverage but did it result in the impact that was hoped for?

**Method:** A web-based survey of International Stillbirth Alliance member organizations was conducted to determine the response to the Series in their region and organization.

**Results:** Most of the 222 responses came from Australia, New Zealand and the Pacific region (78%), with fewer from Europe (11%), North America (5%), Asia (2%), and South America, Africa-Middle East (each 1%). Just over half of the respondents' organizations focused on parents (58%), with the remainder focusing on health professionals. The majority



of respondents (72%) were unsure whether the Lancet's Series reached all key stakeholders, with only 20% indicating that the reach was successful or highly successful. Again, most respondents were unsure (63%) whether the response to the Series in their country was successful in raising awareness of stillbirth; 17% indicated a satisfactory or highly satisfactory response. While approximately half of respondents (42%) were unsure whether their organization participated in activities surrounding the launch, approximately the same proportion of respondents (39%) reported no participation in launch activities.

**Conclusions:** Despite widespread media attention, the Lancet's Stillbirth Series has not had a substantial impact (among ISA member organisations) in reaching key stakeholders or raising awareness around stillbirth. A more comprehensive approach is needed to fully examine the impact of the Lancet's Stillbirth Series on global changes in awareness, advocacy, and policy.

#### P482

##### PARENTS' ATTITUDES TO ASPECTS OF PERINATAL TRIALS. AN ONLINE SURVEY VIA SOCIAL MEDIA

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**Background:** Government policy recommends that consumers become partners and decision makers in clinical research.<sup>1</sup> Miracle Babies Foundation is a leading Australian support and advocacy group for parents of babies in neonatal units. We undertook an online survey via its Facebook page (<http://www.facebook.com/MiracleBabiesFoundation>).

**Method:** Respondents were invited to answer 4 questions about randomized controlled trials

**Results:** There were 410 responses; 98.3% were female, 97.3% were parents, 81% had had a baby in a neonatal unit and 8.5% were nurses or midwives.

| Question                                                                                                                                                                                                  | Yes | Unsure | No  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----|
| 1. Would you lose trust in a specialist who invited your baby to join a fair test to compare two treatments (a clinical trial) because he or she did not know which treatment was better?                 | 16% | 35%    | 49% |
| 2. Do we need a parent-friendly, educational resource/simple self-taught course to explain what perinatal trials are and why they are important to improve healthy survival for sick or preterm newborns? | 87% | 8%     | 5%  |
| 3. Should hospitals report the number of babies and women enrolled in perinatal trials each year as a quality indicator?                                                                                  | 82% | 11%    | 7%  |
| 4. Should schoolchildren learn why fair tests of treatments save lives?                                                                                                                                   | 58% | 24%    | 18% |

**Conclusions:** Many parents lack understanding about trials but appreciate their importance for high quality care. Better education and visibility for perinatal trials could help equip parents to become partners and decision makers.

#### Reference

1. National Health and Medical Research Council; Consumers Health Forum of Australia. Model Framework for Consumer and Community Participation in Health and Medical Research. 2005.

#### P483

##### PRECISION AND ACCURACY OF NELLCOR AND MASIMO PULSE OXIMETERS AT LOW OXYGEN SATURATIONS IN A LAMB MODEL

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**Background:** Recommendations for oxygen treatment in the delivery room are based on an oxygen saturation (SpO<sub>2</sub>) reference range. SpO<sub>2</sub> values <70% are common in the first minutes of life. Accuracy of SpO<sub>2</sub> measurement against the 'gold standard' arterial blood (SaO<sub>2</sub>) gas co-oximetry has not been determined in this range. The aim of this study was to determine the accuracy of Nellcor and Masimo oximeters at SaO<sub>2</sub> < 70%.

**Method:** Prospective study in ventilated anaesthetised normothermic newly born lambs with a carotid artery catheter in situ. Ventilation was adjusted to achieve saturations <70%. Nellcor (Oxi-Max 600 with Max-N sensor) and Masimo (Rad 4 with LNOP sensor) sensors were applied to the right forelimb. The location of sensors, proximal or distal was alternated. A Posey wrap prevented cross talk between sensors. Arterial blood samples were collected from the carotid artery at 5 minute intervals and concurrent oximeter readings were recorded. We used Bland-Altman analysis to determine precision and accuracy of each oximeter.

**Results:** 17 lambs were studied, 165 measurements were obtained. There were 123 measurements when SaO<sub>2</sub> < 70%. The mean SpO<sub>2</sub> - SaO<sub>2</sub> (± 1.96 SD) was 17 (13)% for the Nellcor and 13 (13)% for the Masimo oximeter. In clinical practice this means that when either oximeter displays a low SpO<sub>2</sub> it is likely that the 'true' SaO<sub>2</sub> is lower.

**Conclusions:** At SaO<sub>2</sub> < 70 both monitors overestimated oxygen saturations compared to the gold standard.

#### P484

##### RETROSPECTIVE REVIEW OF BETAMETHASONE THERAPY FOR PREVENTION OF CHRONIC LUNG DISEASE AND FACILITATING EXTUBATION FOR PRETERM INFANTS

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**Background:** Bronchopulmonary dysplasia (BPD) is one of the most challenging and persistent problems of modern neonatal medicine. Betamethasone is not commonly used as a postnatal glucocorticoid to treat BPD in premature infants. We used a low dose and short course of betamethasone based on the reasoning that it would have similar efficacy but a better safety profile compared to the conventional use of dexamethasone.

**Methods:** Retrospective data collection for preterm infants who received betamethasone. Variables including gestation, age, pre and post-betamethasone Fio<sub>2</sub>, MAP (Mean airway pressure), Pco<sub>2</sub> and mode of ventilation were collected. Adverse events like hyperglycemia, hypertension and GI bleeding were recorded for the safety profile. Wilcoxon signed rank test for paired samples was used to compare the medians for various variables (Fio<sub>2</sub>, MAP, Pco<sub>2</sub>) pre and post steroid therapy.

**Results:** Five infants received betamethasone with one infant receiving the course twice. The gestational ages ranged from 23 weeks to 28 weeks. The mode of ventilation could be changed from HFOV for two infants. The Fio<sub>2</sub> values were statistically significantly different (p value 0.03) when compared with Fio<sub>2</sub> 6 hours after the first dose and trend towards borderline significance when compared with fio<sub>2</sub> 3 days after the first dose. Similarly MAP and Pco<sub>2</sub> were compared.

**Conclusions:** This study can be considered as a pilot study for the use of betamethasone and the ongoing data collection at our NICU will provide

a direction towards further use of this steroid including trends in efficacy and safety. Long-term neurodevelopmental follow-up will be planned as more data evolves.

P485

#### DEVELOPMENT OF A LARGE PNEUMATOCOELE AFTER MECONIUM ASPIRATION SYNDROME

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**Background:** Meconium aspiration syndrome (MAS) remains an important cause of significant morbidity and mortality in the term, or near-term neonate. A large retrospective review reported MAS rates of 1.8% in term infants and a mortality rate of 1.2% among those affected by MAS. The ongoing need for oxygen after severe disease requiring ventilation often prolongs the hospital stay.

**Method:** We present the case of a 40-week infant, birth weight 3765 g with severe meconium aspiration syndrome. Due to respiratory distress shortly after birth he was transferred to our neonatal intensive care unit. He deteriorated quickly needing high frequency oscillatory ventilation (HFOV). The oxygenation index increased from 19 to 38 and the patient was transferred to the pediatric intensive care unit on site for consideration of extracorporeal membrane oxygenation (ECMO). Due to clinical improvement with HFOV, ECMO was avoided and after 5 days, the neonate was extubated to CPAP. During resolution of the disease, he developed a large pneumatocoele causing mediastinal shift and increased non-invasive respiratory support requirement. Two intercostal chest drains were inserted under radiological guidance to relieve the resultant respiratory distress.

**Results:** On discharge at 36 days of age the infant was asymptomatic with a near normal chest radiograph. There were no indications radiologically of an underlying congenital lung lesion.

**Conclusions:** To the best of our knowledge this is the first reported case of such a large pneumatocoele developing after severe meconium aspiration syndrome and requiring surgical intervention.

P486

#### LIPID PEROXIDATION PRODUCTS AND ANTI-OXIDANT ENZYMES IN SICK PRETERM NEONATES AND THEIR RELATION WITH BRONCHOPULMONARY DYSPLASIA

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**Background:** Preterm neonates have inadequate antioxidant defense system which may be one of the predisposing factors to develop bronchopulmonary dysplasia (BPD).

**Methods:** This was an exploratory study in a Level III neonatal unit over 1 year period. Fifty consecutive sick neonates of  $\leq 34$  weeks gestation were enrolled and followed till discharge or death. Their baseline blood and urine levels of Malondialdehyde (MDA) and blood levels of anti-oxidant enzymes super oxide dismutase (SOD) and Catalase were compared in the babies who developed BPD vs those who did not.

**Results:** The mean birth weight and gestational age were  $1416 \pm 219$  grams and  $29.9 \pm 2.5$  weeks respectively. Base line blood MDA ( $4.6 \pm 1.2$  vs  $3.9 \pm 1.2$  nMol/ml) was higher ( $p = 0.032$ ) and SOD ( $p = 0.02$ ) ( $26.6 \pm 28.2$  vs  $46 \pm 21.6$  U/ml) and Catalase ( $p = 0.00$ ) were significantly lower ( $0.8 \pm 0.8$  vs  $1.8 \pm 1.2$  nMol/g protein) in babies with gestation  $< 30$  wks ( $n = 24$ ) as compared to  $\geq 30$  wks babies ( $n = 26$ ) respectively. In babies who developed BPD ( $n = 17$ ), blood ( $5 \pm 0.5$  vs  $3.8 \pm 1.2$ ) and urine MDA levels ( $9.4 \pm 5.1$  vs  $5.6 \pm 1.8$ ) were significantly higher ( $p = 0.00$ ) and SOD ( $19.1 \pm 14.4$  vs  $43.9 \pm 26.4$ ) and Catalase levels ( $0.6 \pm 0.3$  vs  $2 \pm 1.1$ ) were significantly ( $p = 0.00$ ) lower than those who did not develop BPD ( $n = 21$ ).

**Conclusion:** Baseline oxidant levels are higher and anti-oxidant enzymes are lower in  $< 30$  weeks gestation babies and in those who developed BPD.

P487

#### PATTERNS OF RESPIRATORY OUTCOMES IN HIGH FLOW NASAL CANNULA (HFNC) TREATED NEONATES IN A TERTIARY CARE NICU

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**Background:** HFNC is an emerging alternate, non-invasive respiratory support for premature infants. However its safety and efficacy including the effect on chronic lung disease (CLD) rates needs to be determined.

**Method:** This retrospective study, aims to examine the respiratory outcomes (CLD, need for home oxygen) in infants  $< 32$  weeks gestation needing invasive or non-invasive ventilation support at birth, treated subsequently with HFNC (3–6 L/min of blended humidified O<sub>2</sub>/air) as a weaning modality from CPAP versus no HFNC, between 2008–11. CLD was defined as need for any O<sub>2</sub> or respiratory support at 36 weeks PMA. **Results:** 689 infants (HFNC 222 vs. controls 467) were included in the study period with the HFNC group being younger, lighter and needing longer duration of invasive ventilation/CPAP. The median gestation of starting high flow was 33 weeks (IQR 31.1–36.4) with median duration of 253 hours (IQR 138–463). CLD rates were three times (105/222 [48%] vs. 75/467 [16.1%]) and need for home O<sub>2</sub> seven times (62 [28%] vs. 21 [4.6%]) higher in the HFNC group. HFNC group had longer duration of O<sub>2</sub> therapy and higher proportion received postnatal steroids (27 [12.3%] vs. 11 [2.4%]). In multivariate logistic model HFNC, lower gestation, longer duration of ventilation and sepsis were independent risk factor for CLD and need for home O<sub>2</sub>. HFNC increased CLD by 4 times and home O<sub>2</sub> therapy by 6 times.

**Conclusions:** Infants treated with HFNC in our experience seem to be sicker than the age matched cohort and larger prospective trials are required to examine its association with increased CLD and need for home O<sub>2</sub>.

P488

#### IMPACT OF HIGH FLOW NASAL CANNULA (HFNC) ON RESPIRATORY OUTCOMES IN A TERTIARY CARE NICU – A RETROSPECTIVE EPOCH BASED ANALYSIS

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**Background:** Neonatal respiratory therapy is changing with introduction of gentle ventilation and newer non-invasive support like HFNC. However its safety and efficacy including the effect on chronic lung disease (CLD) rates are not well examined.

**Method:** This retrospective study, examines the respiratory outcomes (CLD, need for home O<sub>2</sub>) in infants  $< 32$  weeks gestation needing invasive or non-invasive ventilation support at birth in two epochs; epoch I – pre HFNC (2004–2007) & epoch II – HFNC (2008–2011). HFNC was first introduced in our unit for weaning off CPAP from 2008 onwards. CLD was defined as need for any O<sub>2</sub> or respiratory support at 36 weeks PMA. **Results:** 1367 infants (pre-HFNC 597 vs. HFNC 788), mean (SD) gestation, 28.1 (2.2) vs. 28.5 (2.3) weeks were enrolled. The median gestation of starting HFNC was 33 weeks (IQR 31.1–36.4) with median duration of 253 hours (IQR 138–463). Duration of ventilation (median, IQR) was shorter in epoch II (IMV: 24 {0–142} vs. 10 {0–96} hrs.  $p = 0.001$ ; CPAP: 132 {26–740} vs. 101 {10.5–528.5} hrs.  $p = 0.001$ ). CLD rates have reduced by 6% in epoch II (170 {29%} vs. 183 {23%};  $p = 0.01$ ) whereas need for home O<sub>2</sub> was not different (71 {12%} vs. 84 {11%},  $p = 0.4$ ). In logistic regression analysis HFNC did not influence the CLD rates.

**Conclusions:** Overall CLD rates in our unit have decreased between the two epochs. However introduction of HFNC seems to have minimal effect on respiratory outcomes.

## P489

EVALUATING THE USE OF END-TIDAL CO<sub>2</sub> IN THE MANAGEMENT OF NEONATAL RESPIRATORY DISEASES

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**Background:** Varying respiratory diseases and small tidal volumes in mechanical ventilation of neonates are challenges to the measurements of end-tidal CO<sub>2</sub> (P<sub>ET</sub>CO<sub>2</sub>). Improvement of sensor technology in sensor speed and gas volume in mainstream capnography poses an opportunity to re-evaluate its clinical application as an approximation of arterial CO<sub>2</sub>. **Methods:** Simultaneous mainstream P<sub>ET</sub>CO<sub>2</sub> measurements (Capnotrue AMP) and P<sub>a</sub>CO<sub>2</sub> from arterial samples from indwelling catheters or P<sub>cap</sub>CO<sub>2</sub> from capillary blood samples were compared for agreement.

**Results:** A total of 78 paired measurements (53 arterial and 25 capillary) were analysed from 23 ventilated neonates. The median gestation was 32.2 (24–41) weeks and median birth weight 1560 (680–3945) g. The mean P<sub>a</sub>CO<sub>2</sub>–P<sub>ET</sub>CO<sub>2</sub> difference was 2.3 (SD 3.86) and it correlated (r = 0.94) over an Altman plot range of 35–78 mmHg. The coefficient of determination R<sup>2</sup> was 0.89. Respiratory disease diagnosis of respiratory distress syndrome, meconium aspiration or peri-operative ventilation did not post an additional influence. The range of P<sub>ET</sub>CO<sub>2</sub> obtained during the 23 capillary blood sample comparisons was not different to that of the 53 arterial samples comparisons (mean 46 v 47 mmHg). For the 23 capillary samples, the mean P<sub>cap</sub>CO<sub>2</sub>–P<sub>ET</sub>CO<sub>2</sub> difference was 4.8 (SD 7.0); r = 0.79.

**Conclusion:** Mainstream capnography has the potential to provide non-invasive PaCO<sub>2</sub> estimation in neonatal mechanical ventilation. The End-tidal CO<sub>2</sub> readings trend to be 2 mmHg lower than that of arterial and 5 mmHg lower when compared with capillary blood samples. Further studies will be required to establish its clinical utility over capillary blood samples.

## P490

## IS THE HEAT-MOISTURE EXCHANGER (HME) USEFUL WHEN VENTILATING INFANTS? – ANIMAL EXPERIMENT

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**Background:** Heat-moisture exchanger (HME) is a simple and effective way of maintaining temperature and humidity of inspired gas in chronically intubated patients. HME may be useful for short-term usage in the resuscitation of newborn infants.

**Objectives:** To study immediate effects of HME on temperature and humidity of inspired gas as well as its influence on dead space and ventilation efficiency in intubated rats.

**Method:** Two adult rats (wt 400–500 g) were anesthetized and tracheostomized with 2 mmID Portex® tracheal tube. Then the rats were ventilated by Humming II® ventilator (Flow 5 L/min, Pres 8/3 cmH<sub>2</sub>O, RR 90/min, Ti 0.25 sec). Temperature (T), relative and absolute humidity (RH and AH), end-tidal CO<sub>2</sub> (ETCO<sub>2</sub>), proximal airway pressure (Paw) and flow (V') were measured as well as arterial blood gas analysis with and without an HME (Humid-vent mini, Gibeck®; dead space ~2.4 mL).

**Results:** T, RH and AH all increased from 30 degrees, 78%, 23 mg/L without an HME to 33 degrees, 90%, 33 mg/L with an HME, respectively.

PaCO<sub>2</sub> increased from 47 mmHg to 50 mmHg with an HME attached. Similarly, V' swing increased from 35 mL/sec to 50 mL/sec.

**Conclusions:** There were obvious immediate beneficial effects of HME on temperature and humidity maintenance with a minimal dead space effect such as CO<sub>2</sub> accumulation and change in ventilation pattern. HME seemed to be useful in the resuscitation and for short-term use in the NICU.

## P491

## IS THE HEAT-MOISTURE EXCHANGER (HME) USEFUL WHEN VENTILATING INFANTS? – CLINICAL STUDY

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**Background:** Heat-moisture exchanger (HME) is a simple and effective way of maintaining temperature and humidity of inspired gas in chronically intubated patients. HME may be useful for short-term usage in the resuscitation of newborn infants.

**Objectives:** To study immediate effects of HME on temperature and humidity of inspired gas in intubated infants.

**Method:** Subjects were 7 ventilated infants (5 in closed and 2 in open incubators) in the NICU at Tokai University Hospital. Infants were manually ventilated with a flow-inflating bag through an endotracheal tube (2.5–4.0 mmID Portex® tracheal tube). Temperature (T), relative and absolute humidity (RH and AH) of inspired gas was measured and compared before and after attaching an HME (Humid-vent mini®, Gibeck). Influence of %leak [(V<sub>ti</sub>–V<sub>te</sub>)/V<sub>tix</sub>100] around the ETT was also assessed.

**Results:** T, RH and AH all increased with an HME attached at the proximal airway. Mean increases were, 1.7 degrees in T, 13.5% in RH and 6.4 mg/L in AH. Two cases with %leak >90% showed obviously low temperature and humidity before attaching an HME. However, they still showed similar temperature and humidity increase (to those subjects with less leak) after attaching an HME.

**Conclusions:** HME may be useful in the resuscitation of infants in the DR as well as for short-term usage in the NICU. It seemed still efficacious in patients with large ETT leaks.

## P492

## INSPIRED HUMIDITY AND TEMPERATURE WITH THE SOPHIE VENTILATOR DURING CONVENTIONAL VENTILATION

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**Background:** Currently there is no information on the Sophie ventilator with an inbuilt humidifier system in regard to delivered inspired humidity (IH) and endotracheal tube temperature (ETTT).

**Methods:** A closed ventilator circuit within a humidicrib using an artificial lung immersed in 37°C water was used. Ventilator settings were 25/5 (peak/end expiratory pressure) x60 (respiratory time) and inspiratory time of 0.35 in air. Humidicrib temperature (32°C, 34.5°C, and 37°C), and inspired temperature offset (39–2°C, 39–1°C and 40–3°C) were modified for their respective readings. Sets of five readings, five minutes apart, of IH and ETTT for each change in humidicrib temperature and inspired temperature offset were taken.

**Results:** A temperature offset of 39–1°C produced significantly higher absolute IH (AIH: Table). ETTT increased with an increase in humidicrib temperature (Table).

**Conclusions:** The IH was dependent on temperature offset, inspired temperature setting and humidicrib temperature. All AIH measurements met the minimum USA limit of 30 mgH<sub>2</sub>O/L gas. At 39–1°C, AIH measurements met the USA and UK (33 mgH<sub>2</sub>O/L gas) minimum standards. ETTT was dependent on humidicrib temperature.



Table.

| Humidicrib temp      | 32°C                     |                           | 34.5°C                  |              | 37°C                    |              | General Linear model <sup>d</sup> |          |
|----------------------|--------------------------|---------------------------|-------------------------|--------------|-------------------------|--------------|-----------------------------------|----------|
| Inspired temp offset | AIH <sup>a</sup> (n = 5) | ETTT <sup>b</sup> (n = 5) | AIH (n = 5)             | ETTT (n = 5) | AIH (n = 5)             | ETTT (n = 5) | AIH Sig                           | ETTT Sig |
| 39–2°C               | 32.1 ± 0.1 <sup>c</sup>  | 33.0 ± 0.1                | 33.1 ± 0.1              | 35.4 ± 0.1   | 31.6 ± 0.3              | 37.3 ± 0.0   | <0.001                            | <0.001   |
| 39–1°C               | 33.8 ± 0.1 <sup>e</sup>  | 33.5 ± 0.1                | 35.2 ± 0.1 <sup>e</sup> | 35.3 ± 0.0   | 34.6 ± 0.4 <sup>f</sup> | 37.3 ± 0.1   | <0.001                            | <0.001   |
| 40–3°C               | 32.5 ± 0.4               | 33.0 ± 0.1                | 33.5 ± 0.4              | 35.3 ± 0.0   | 34.0 ± 1.0              | 37.3 ± 0.1   | <0.001                            | <0.001   |

<sup>a</sup>AIH:mgH<sub>2</sub>O/L, <sup>b</sup>ETTT: °C, <sup>c</sup>mean ± SE, <sup>d</sup>Bonferroni adjustment, <sup>e</sup>Significantly higher than 39–2°C and 40–3°C p < 0.01, <sup>f</sup>Significantly higher than 39–2°C p < 0.01.

## P493

# IS INSURE TECHNIQUE FOLLOWED BY CPAP: A FEASIBLE THERAPEUTIC OPTION FOR PRETERM INFANTS WITH RESPIRATORY DISTRESS SYNDROME DURING RETRIEVAL

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**Background:** To identify clinical characteristics of a retrieval cohort of mechanically ventilated (MV) preterm infants with RDS who were favorably extubated (FE) shortly after admission. These infants may be suitable for INSURE followed by CPAP in a retrieval setting.

**Method:** Preterm MV infants with RDS retrieved between 2010–2011 with gestational age (GA) >28 wks, birth weight ≥1000 gms, and age ≤48 hrs. FE infants, defined as successful extubation within 6 hrs at destination hospital, were compared to those who were extubated after.

**Results:** A total of 223 infant, mean GA.33.1 wks (SD 1.9), mean birth weight 2239 g (SD 568) were examined. Of these, 50% received ante-natal steroids and 53% were <4 hrs of age. Half (49.7%) were receiving CPAP and remaining MV at stabilization by NETS. Thirty (16%) infants were favorably extubated at destination. FiO<sub>2</sub>(st) on stabilization by NETS correlated with admission FiO<sub>2</sub> (r = 0.79). No significant differences were noted for GA, age at call, and ventilator settings at NETS stabilisation between FE infants and those who were not. FiO<sub>2</sub>(st) after surfactant administration were significantly lower in the FE group (mean 0.25 v 0.37 respectively; p = 0.005). However, the AUC from ROC based on FiO<sub>2</sub>(st) was 0.63 (p = 0.01). However, sensitivity and specificity of FiO<sub>2</sub>(st) cutoff points (between 0.25–0.30) were low.

**Conclusion:** FiO<sub>2</sub>(st) after surfactant administration, though significantly lower in the FE infants, has only a weak predictive value. Therefore, it would not likely justify a clinical trial using FiO<sub>2</sub>(st) as eligibility criteria for a randomized trial of INSURE during retrieval.